

WDW-10

Global Clock-Neutral Relational Descent and Universality Closure

A three-clock descent theorem over the Branch-A Wheeler–DeWitt realization

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Exact scope. The theorem is relative to the transversely germified retained source lineage of the accepted Branch-A, compact-hyperbolic, metric–real-scalar, rank-one, no-spectator WDW-9 target. It constructs three local function-valued clock presentations and their clock-neutral descent. It does not assert one global monotonic clock, full superspace, singularity resolution, Standard-Model coupling, topology change, or uniqueness after changing any non-clock coordinate.

Global clock-neutral relational descent

Abstract

The accepted WDW-9 endpoint supplies one complete local Wheeler–DeWitt realization on the function-valued matter profile gauge $\phi(\cdot) = \tau\mathbf{1}$. It does not supply a plural clock atlas, transition maps, a native observable–record category, or a descent theorem. This paper constructs those missing objects on the exact fixed Branch-A target.

At the compact hyperbolic source anchor

$$z_\star = (h_\star, \pi_\star; 0, \epsilon_\star \sqrt{h_\star}), \quad K_{\star ij} = c_{\epsilon_\star} h_{\star ij}, \quad c_{\epsilon_\star} = \sqrt{1 + \kappa \epsilon_\star^2 / 6} > 0,$$

we retain the inherited scalar profile and construct two intrinsic metric profiles,

$$\nu_{h_\star}(h) = \frac{1}{3} \log \frac{d\mu_h}{d\mu_{h_\star}}, \quad R^{(3)}(h).$$

Their lapse responses at the anchor are respectively

$$N \mapsto c_{\epsilon_\star} N, \quad N \mapsto 4c_{\epsilon_\star} (3 - \Delta_{h_\star}) N.$$

Multiplication by $c_{\epsilon_\star}$ is invertible and $3 - \Delta_{h_\star}$ is a positive elliptic isomorphism on the closed hyperbolic carrier. Parameter-dependent inversion on fixed quantitative flow boxes therefore gives residual-copy-free scalar, volume, and curvature clock neighborhoods with a common triple overlap. The scalar clock is field-valued; the volume clock is order-zero metric-valued; and the curvature clock has an order-two metric principal symbol. Their quotient profiles cannot be related by one same-neighborhood scalar reparameterization.

The final theorem retains the orbit-complete common-source and enriched-recovery structures while separating the connected proper-spatial quotient X_{10}^{pf} from the guarded WDW-3 refoliation groupoid and explicitly denies that the latter supplies an off-shell quantum action. It first pulls each clock neighborhood O back to the saturated raw preimage $\tilde{O} = \pi_{\text{pr}}^{-1}(O)$ and localizes the actual WDW-6 successor source. Separate lemmas prove ideal invariance for local differential/counterterm generators and finite proper actions. One complete WDW-5/6 support is retained: all labels, target sorts, counterterm columns, graph topology, and the literal common domain. Each clock grammar is defined by conjugating that support through its independently formed orbit-complete flow-box/proper-action chart $\Psi_{\alpha, O}(z, g) = g \cdot s_O(\kappa_{\alpha, O}(z))$. The full proper-arrow coordinate is retained; restriction to $g = e$ is not used as a carrier map. The primal, common-domain, and continuous-dual maps obey the proved evaluation-naturality square; the source kernel, its evaluation image, and the full transposed kernel are never identified. The inverse chart and label markings remain explicit. Thus $\mathbf{G}_{\alpha\beta} = \mathbf{\Gamma}_\alpha \mathbf{\Gamma}_\beta^{-1}$ is positively constructed without treating W5.12 or W6.15 as an existence theorem for refoliation arrows.

The new intrinsic-clock quantum gate is proved directly. At the hyperbolic anchor, $dT_R = 12dT_\nu$ and both covectors are nonzero pure trace. W6R.S12 retains their regulator-free finite-rank Hessian trace, and the accepted DeWitt tensor is strictly negative on that line. With the normalized lapses this gives $a_{\nu, \star} \neq 0$ and $a_{R, \star} = 12a_{\nu, \star} \neq 0$. Quantitative shrinking keeps both coefficients bounded away from zero. Compact-open finite-word estimates then prove graph-smooth Cauchy restriction and first-normal differentiation for both intrinsic charts while retaining g as a tangential variable.

The accepted rank-one physical quotient is mapped isometrically onto the line generated by the localized authenticated joint-kernel doublet. The renormalized Hamiltonian member is reapplied with its complete normalized-lapse label along each independently constructed chart-normal line. The three lines are proved to coincide on the retained homogeneous source interval $\mathcal{E}_\star$; no off-line equality or transverse connection is claimed. On germ overlaps, physical and enriched graph transitions are

$$T_{\alpha\beta} = \text{Cond}_\alpha \text{Cond}_\beta^{-1}, \quad \mathbf{G}_{\alpha\beta, O} = \mathbf{\Gamma}_{\alpha, O} \mathbf{\Gamma}_{\beta, O}^{-1}.$$

They preserve the physical form, source carrier, successor graph topology and common domain, evaluation naturality, total label-transporting grammar, target sorts, counterterms, all three kernel levels, represented algebra, records, continuation, first failures, boundary data, and nonclaims. The connection intertwiner includes the derivative of the base map, the forced clock Jacobian, and the companion term independently derived in both charts. Source factorization proves the complete triple cocycle and removes higher defects without presupposing a global object.

The compatible local profiles first form an explicit clock-descent object: a lawful-refinement class of the three-chart cover, readout relations, overlap coherence, orientations, anchors, continuations, and first failures. Replacing the distinguished-clock coordinate of the local category by that object defines a separate clock-neutral physical recipient before global descent. The remaining compatible local packages descend on the transversely germified retained lineage to a nonempty clock-neutral realization canonically equivalent to the common physical anchor with its source-history, record, continuation, and failure sheaves. Its form is positive, its admitted physical-record probe is separating, and every loop transport is identity after the maximal complete-probe-neutral presentation quotient. For arbitrary same-target atlases that pass the printed local

antecedents, the cross-atlas transitions are constructed before comparison. Their underlying source covers have a common fibre-product cover; the two clock data pulled back to that cover remain distinct and are compared by mutually inverse transition-supported roofs. They therefore descend to the same neutral object without any claim that distinct clock profiles possess a literal common clock-data refinement. The enriched marked scalar restriction is first identified with the complete enriched scalar local object; applying the conservative WDW-9 reduct then recovers the accepted WDW-9 endpoint coordinatewise. The reduct retains every predecessor label, grammar, domain, counterterm, kernel, record, failure, and nonclaim coordinate while forgetting only successor-added atlas structure as such. The enriched recovery category is defined independently with explicit atlas, incidence, transition, coherence, holonomy, source/label marking, and gluing coordinates; its arrows, composition, restrictions, underlying-object functor, comparison-marked objects, and object/arrow reduct are all typed before descent. The global recovery marking is formed only after the enriched scalar-presentation isomorphism is proved in the neutral recipient. Its clock component is a roof whose left leg is literal restriction refinement and whose right leg is the proved $t_{\Phi\alpha}$ clock equivalence with inverse $t_{\alpha\Phi}$, so it neither chooses a global clock nor allows the reduct to export an unrelated local atlas member. The lawful gluing quotient is a singleton because empty-word point evaluations and the independent source/label markings exclude every hidden graph centralizer while the normalized authenticated section removes phase freedom. The final status is

$$\mathfrak{R}_{10}^{\text{global}} \in \text{EnrPhysRec}_{10}(X_{10}^{\text{reg}}), \quad \text{For}_{10}(\mathfrak{R}_{10}^{\text{global}}) = \mathfrak{R}_{10}^{\text{und}} \in \text{NeuPhysRec}_{10}(X_{10}^{\text{reg}}),$$

$$\text{Res}_{9,\Phi}(\mathfrak{R}_{10}^{\text{global}}) \cong \mathfrak{R}_9^*, \quad |\text{Glue}_{10}^{\text{law}}/\Gamma_{10}^{\text{law}}| = 1, \quad G_{10} = \text{PASS-GLOBAL}.$$

No clock-only domain extension beyond the inherited scalar-chart locus is claimed; the stronger PASS-GLOBAL-EXTENDED label is therefore not issued.

Principal result. The accepted WDW-9 local theorem is conservatively recovered from a positively populated, genuinely plural, clock-neutral descent object on the declared regular clockable locus. The three local presentations are not obtained by relabeling one clock, every transition is constructed before descent, and every same-target gluing lies in one lawful equivalence orbit.

Result map

Layer	Positive construction	Terminal result
Accepted source	Final WDW-9 endpoint and complete non-clock target	No atlas output is imported
Local clocks	Scalar profile, intrinsic volume ratio, and intrinsic scalar curvature	Three occupied charts; all pairwise and triple overlaps nonempty
Local quantum data	One saturated raw-localized WDW-6 support; three conjugated chart grammars and inverse source markings; exact evaluation naturality; retained finite-rank DeWitt noncharacteristicity; source-side graph-smooth Cauchy doublets; separate source/evaluation/transposed kernels; chart-normal lines; records	Three complete objects of PhysRec_{10}
Transitions	Differentiable base maps, $T_{\alpha\beta}$ and common-source $\mathbf{G}_{\alpha\beta,O} = (G_{\alpha\beta,O}, \mu_{\alpha\beta})$	Label/grammar/domain/source-marking preservation, companion/Jacobian intertwining, exact enriched cocycle
Global object	Compatible-family descent with authenticated global section; explicit enriched record and underlying-object functor	Positive, separating, clock-neutral realization in EnrPhysRec_{10} with underlying NeuPhysRec_{10} object and explicit ClockDesc_{10} coordinate
Universality	Common source covers, transition-supported comparison roofs, and common-anchor mapping theorem	Atlas independence and multiple-choice closure
Containment	Typed marked-enriched restriction and arrow followed by object/arrow Red_9	Exact WDW-9 recovery without erasing successor structure
Exhaustion	Ordered grammar, complete probe, and evaluation/source-marking rigidity	One lawful gluing orbit; $G_{10} = \text{PASS-GLOBAL}$

Locked dependency order

$$W10.1 \rightarrow (W10.2, W10.3) \rightarrow (W10.4, W10.5, W10.6) \rightarrow W10.7 \rightarrow W10.8 \rightarrow W10.9 \rightarrow \cdots \rightarrow W10.24.$$

Local population precedes transition maps; transitions precede coherence; coherence precedes descent; descent precedes holonomy and uniqueness; recovery and exhaustion occur last. In particular W10.16 does not populate W10.5 or W10.6, W10.5–W10.6 make no use of W10.9, and W10.20 does not define W10.10.

Status boundary

The mathematical arguments are contained in the manuscript. Reproducibility checks confirm source lineage, typed incidences, dependency order, cross-references, and build cleanliness; they do not replace proof of the elliptic inverse, graph-domain, unitarity, descent, holonomy, universality, or imported WDW-1–9 theorems.

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1. Accepted handoff, theorem object, and no-backflow seal

1.1 Authorized predecessor

The sole WDW-10 predecessor is the final WDW-9 publication endpoint. Its thirteen-role joint object, twenty overlap equalities, physical line, represented algebra, scalar-profile clock, complete record, continuation, WDW-8 recovery object, failure ledger, nonclaims, assignment typing, and $G_9 = \text{PASS}$ status are imported literally.

Definition 1.1 (Accepted WDW-9 realization). Write

$$\mathfrak{R}_9^* = (\mathcal{X}_9, \mathcal{C}_9, \mathcal{D}_9, \widehat{\mathcal{C}}_9, \mathcal{H}_{\text{phys}}, \langle \cdot, \cdot \rangle_{\text{phys}}, \rho_{\text{phys}}, \mathfrak{C}_\Phi, \mathfrak{R}_{\text{rec}}, \mathfrak{R}_{\text{cont}}, \mathfrak{F}_{\leq 9}, \mathfrak{N}_{\leq 9}, \text{Cert}_9)$$

for the accepted source-attached Branch-A package. Here $\mathcal{X}_9$ is the regular proper- redescription/refoliation history-germ base, $\mathcal{C}_9$ the classical joint constraint surface, $\mathcal{D}_9$ the accepted common graph domain, $\widehat{\mathcal{C}}_9$ the anomaly-closed quantum constraint family, $\mathcal{H}_{\text{phys}}$ the nonzero positive rank-one quotient, and $\rho_{\text{phys}}(\mathbb{C}I) = \mathbb{C}I$ the represented physical algebra. The last five entries retain the scalar clock, record, continuation, failures/nonclaims, and complete certificate.

The construction uses the WDW-4A anchor

$$z_\star = (h_\star, \pi_\star; \phi_\star, p_{\phi,\star}) = \left(h_\star, -\frac{c_{\epsilon_\star} \sqrt{h_\star}}{\kappa} h_\star^{-1}; 0, \epsilon_\star \sqrt{h_\star} \right), \quad \text{Ric}(h_\star) = -2h_\star, \quad R(h_\star) = -6, \quad (1)$$

where $\epsilon_\star > 0$ and $c_{\epsilon_\star} = \sqrt{1 + \kappa \epsilon_\star^2 / 6}$. WDW-4A proves that (1) satisfies the complete Einstein–Klein–Gordon constraints, has no generalized Killing initial data, lies in a split regular constraint chart, and develops to the retained source history $[\gamma_\star]_{\text{src}}$.

Definition 1.2 (Exact handoff). The WDW-10 input is

$$I_{9 \rightarrow 10} = (\mathfrak{R}_9^*, \text{Roles}_9, \text{Edges}_9, \mathfrak{F}_{\leq 9}, \mathfrak{N}_{\leq 9}, \text{ClockAtlasSlot} = \emptyset, \text{Authorize}_{10} = \text{true}). \quad (2)$$

No intrinsic-volume chart, intrinsic-curvature chart, transition, cocycle, descent object, atlas comparison, or gluing class is an entry of (2). The empty slot is an input statement: WDW-9 does not prepopulate successor atlas data. It is not a WDW-9 signature coordinate requiring every successfully constructed WDW-10 object to retain an empty atlas after extension.

1.2 Frozen target and regularity scale

Let Σ be the fixed compact connected oriented hyperbolic three-manifold of WDW-3/4. Fix $s_0 > 7/2 + 2$. At each Sobolev stage $s \geq s_0$, use the WDW-5 metric–scalar chart

$$\mathcal{Q}^s = \text{Met}_+^s(\Sigma) \times H^s(\Sigma; \mathbb{R}), \quad \mathcal{P}^s = T^* \mathcal{Q}^s$$

with the inherited density conventions, and the WDW-2 connected proper-spatial quotient. Scalar curvature maps H^s metrics to H^{s-2} ; the volume-ratio and scalar profiles map to H^s . All inverses below are compatible under the accepted scale restrictions, so their inverse limit gives the smooth-tame WDW-5 neighborhood. The manuscript writes the inverse-limit objects unless one Sobolev stage is needed to display an elliptic domain.

Every non-clock standing-bearing coordinate is frozen: carrier and topology, Branch-A source, orientation, connected proper groupoid, regular HDA stratum, configuration calculus, regulator and counterterms, complete common graph domain, anomaly target, normalized joint solution, positive physical form, represented algebra, source-attached WDW-8 recovery subtype, compact-boundaryless status, no-spectator requirement, record lineage, and every first-failure/nonclaim coordinate.

Definition 1.3 (Clock-neutral base). Let $\widetilde{X}_{10}^{\text{raw}}$ be the corresponding raw source-attached regular history/Cauchy-presentation locus and let $\pi_{\text{pr}} : \widetilde{X}_{10}^{\text{raw}} \rightarrow X_{10}^{\text{pr}}$ denote only its connected proper-spatial quotient map. The accepted guarded WDW-3 refoliation arrows form a groupoid

$$\mathcal{R}_{10}^{\text{rf}} \rightrightarrows X_{10}^{\text{pr}}.$$

Write $\mathfrak{X}_{10} = [X_{10}^{\text{pr}} / \mathcal{R}_{10}^{\text{rf}}]$ for its classical groupoid presentation. This notation is not a coarse orbit set on which arbitrary off-shell successor functions are evaluated, and it does not supply a quantum action of $\mathcal{R}_{10}^{\text{rf}}$ on the WDW-6 carrier. The off-shell successor grammar is retained on one proper-localized WDW-6 support; only connected proper actions are imported there. Guarded WDW-3 arrows govern classical source-history and record

comparison, while a quantum lift would require a separate positive construction and is nowhere assumed below. Only proper-basic observables and the authenticated joint kernel descend when their own invariance has been proved. For site notation set $X_{10}^{\text{base}} = X_{10}^{\text{pr}}$ and retain refoliation as arrows. The base is neither $\mathcal{H}_{\text{phys}}$ nor a clock-value interval. The former authenticates quantum fibres; the latter are local readouts on proper-spatial presentations of this base.

The data (1) have spatially constant scalar momentum, constant mean curvature, and hyperbolic metric. Their accepted WDW-4A development therefore contains a connected homogeneous source interval $\mathcal{C}_\star$ about $z_\star$ on which

$$h(u) = a(u)^2 h_\star, \quad K(u) = k(u)h(u), \quad \phi(u), \quad p_\phi(u)/\sqrt{h(u)}, \quad R(h(u)) \quad \text{are spatially constant.}$$

Shrink this interval once so that every inherited WDW-9 guard and all three response inverses below remain valid. This is a restriction of the inherited source history, not a new solution or an alternative continuation.

For each clock profile C_α , let $\mathcal{S}_{\alpha,\star}$ be the local transverse plaque-parameter slice supplied by the split characteristic chart, and choose once and for all a radius pair $(r_\alpha, \sigma_\alpha) > 0$ such that: the exact response family differs from its anchor value by less than $(2\|L_{\alpha,\star}^{-1}\|)^{-1}$ in operator norm; the guarded plaque map is injective on $(\mathcal{S}_{\alpha,\star} \cap B_{r_\alpha}(z_\star)) \times B_{\sigma_\alpha}(0)$; the parameter-dependent inverse sends the profile ball back into $B_{\sigma_\alpha}(0)$; and every frozen source, successor-graph, record, and continuation guard persists there. Let U_α be the connected source-attached quotient image of this certified product neighborhood in X_{10}^{pr} . The restricted guarded refoliation arrows remain attached to its classical source-history record but are not promoted to off-shell quantum carrier maps. Thus U_α is a deliberately restricted residual-copy-free inverse-function neighborhood, not a maximal anchor component. Define

$$X_{10}^{\text{reg}} = \text{Germ}_{\mathcal{C}_\star}(U_\Phi \cup U_\nu \cup U_R). \quad (3)$$

The underlying retained events of (3) are the points of $\mathcal{C}_\star$ in the proper-spatial quotient; its structure object consists of successor-source germs on the displayed full configuration neighborhoods. Hence transverse functional derivatives, operator graphs, and the profile-separation witnesses remain present even though the eventual physical continuation is the accepted source lineage. Guarded refoliation identifies presentation standing through arrows, not by assigning one scalar value to every off-shell function. Two representatives define the same germ only when they agree on some source-attached open neighborhood of $\mathcal{C}_\star$. The boundary $\partial_{\text{clk}} X_{10}^{\text{reg}}$ is the first loss of any local or overlap condition, including exit from one of the fixed certified radius pairs. Non-clock WDW-9 failures remain outside every U_α .

Theorem 1.4 (W10.1 — exact handoff and fixed target). *Equation (2) is a nonempty, source-attached handoff. Equations (1)–(3) define the exact WDW-10 transversely germified lineage target without an atlas output in any antecedent. Every WDW-9 non-clock, failure, boundary, recovery, and nonclaim coordinate is retained. Failure of any later WDW-10 gate leaves $G_9 = \text{PASS}$ unchanged.*

Proof. Nonemptiness and every fixed coordinate are coordinates of the accepted WDW-9 joint occupant. The source-lineage declaration identifies the imported endpoint and governing WDW-10 contract. Inspection of (2) shows that the atlas slot is empty. The definitions of X_{10}^{pr} , $\mathcal{R}_{10}^{\text{f}}$, and X_{10}^{reg} use only predecessor base objects, accepted WDW-3 arrows, and candidate-local pass conditions; none refers to a WDW-10 transition, descent, or final orbit. The dependency is therefore one way. WDW-10 does not alter the predecessor tuple, so a later failure changes only the extension coordinate. $\square$

No-backflow seal. A chart change cannot repair an anomaly, restore a lost common domain, erase a failed prefix, create a missing source, or convert a WDW-8 recovery failure into success. A construction that changes source, matter, gauge, topology, boundary, operator family, physical form, or use target is assigned the out-of-scope disposition before atlas testing.

1.3 Claim and nonclaims

The theorem is global only relative to the transversely germified retained lineage (3). It does not assert a global monotonic scalar, a preferred foliation, an external Schrödinger parameter, or a chart through a singular or non-clock failure. It does not add Standard-Model fields, fermions, local gauge constraints, asymptotic time, boundary charges, topology change, a second physical state, an ambient Fock carrier, a Born rule, or a decoherence theorem. It also does not claim the optional strict point-set extension $(U_\nu \cup U_R) \setminus U_\Phi \neq \emptyset$ away from $\mathcal{C}_\star$. The endpoint label will distinguish this from the stronger extension claim.

2. Clock site and native physical observable–record category

2.1 The source-attached clock site

An object of ClockSite_{10} is a source-attached germ neighborhood $\text{Germ}_{\mathcal{C}_*}(U)$ in X_{10}^{pr} that is stable under connected proper redescription and temporal restriction and on which all retained WDW-9 coordinates are defined. Its underlying events lie on $\mathcal{C}_*$, while its source sheaf retains full transverse germs on U . A morphism $V \rightarrow U$ is a germ restriction followed, when needed, by an accepted WDW-2 proper presentation arrow or a guarded WDW-3 arrow in $\mathcal{R}_{10}^{\text{rf}}$. This is a classical site statement: a WDW-3 arrow transports source-history and record coordinates but does not automatically act on the off-shell WDW-6 carrier. A quantum graph component is present only when separately constructed. No coarse refoliation-orbit scalar is presumed. A family $\{U_i \rightarrow U\}_{i \in I}$ covers when the images are jointly surjective on retained quotient-history points of $\mathcal{C}_*$ and their neighborhoods are cofinal for the retained source germs. Only set-sized families are used.

Finite pullbacks are literal source pullbacks:

$$U_{i_0 \cdots i_k} = U_{i_0} \times_{X_{10}^{\text{pr}}} \cdots \times_{X_{10}^{\text{pr}}} U_{i_k}. \quad (4)$$

A point of (4) is one quotient source event together with all indicated chart certifications. Thus equality of numerical clock ranges never creates an overlap. Empty, singular, domain-losing, or theorem-object- changing pullbacks are retained as typed failures.

Definition 2.1 (Lawful refinement). A refinement $r : \mathcal{A}' \rightarrow \mathcal{A}$ assigns to each $V_j \in \mathcal{A}'$ an index $r(j)$ and a source-preserving inclusion $V_j \hookrightarrow U_{r(j)}$, is jointly surjective on the same base, and preserves every complete failure coordinate. The fibre-product family $\{U_i \times_X V_j\}_{i,j}$ is a common refinement of the *underlying source covers* of two lawful atlases on the same fixed target. It does not identify their clock profiles or by itself define a common refinement of their clock-descent data.

Theorem 2.2 (W10.2 — clock-neutral base and site). *ClockSite₁₀ is a Grothendieck site with typed finite pullbacks and the cover topology above. The underlying source covers of lawful same-target atlases admit the common refinement just displayed. This theorem defines neither the three new chart objects nor their cover; their positive population is deferred to W10.4–W10.9.*

Proof. Identity restrictions cover. Pulling a jointly surjective germ family back along $V \rightarrow U$ gives a jointly surjective family on V , because membership is tested at the same source event. A composite of jointly surjective families is jointly surjective. The WDW-2 proper-presentation functor preserves restriction, complete descriptor labels, and source ancestry. The WDW-3 functor preserves the classical source-history coordinates only; none of the site axioms requires an off-shell quantum lift. Fibre products of regular germ neighborhoods are represented by their open intersections and carry the intersection of the finite certifications. The product cover is jointly surjective because each point is hit in both original covers. Failure coordinates are pulled back rather than discarded. $\square$

2.2 Local objects

For $U \in \text{ClockSite}_{10}$, a local physical observable–record object is

$$\mathfrak{R}_C(U) = (C, U, \mathcal{V}_C^{\text{ren}}, \mathcal{D}_C^{\text{ren}}, J_C, \mathcal{H}_C, \text{Cond}_C, \text{Conn}_C, \rho_C, \mathcal{A}_C^{\text{phys}}, \text{Rec}_C, \text{Cont}_C, \text{Fail}_C, \text{Anchor}_C, \text{Cert}_C). \quad (5)$$

The clock C is a function-valued profile with its use, anchor, orientation, and failure data. $\mathcal{V}_C^{\text{ren}}$ is a chart presentation of the localized WDW-6 graph object

$$(V_{6,C}^{\text{ren}}, \mathcal{D}_{6,C}^{\text{ren}}, \mathcal{I}_C, \{\widehat{C}_{C,j}\}_{j \in \mathcal{I}_C}, \mathcal{P}_C, \Psi_C, \widehat{Q}_C, \{\widehat{U}_{g,C}\}, \{\mathcal{R}_{a,C}\}, \rho_C, \mathbf{\Gamma}_C, \mathcal{E}_C^{\text{src}}, \ell_C),$$

including the complete WDW-5/6 labels, target sorts, counterterm columns, the full proper-arrow chart domain, reconstruction map, slice-complete primal map, proper translations and slice-change maps, the independently formed chart map $\mathbf{\Gamma}_C$, its common-source marking $\mathcal{E}_C^{\text{src}} = \mathbf{\Gamma}_C^{-1}$, and the label marking ℓ_C back to the one common label set. No guarded WDW-3 quantum action is an entry. It is not the proper-basic scalar subspace and is not the physical Hilbert line. The map J_C is the Cauchy trace on the orbit-complete chart image of the localized successor source; it retains the proper-arrow coordinate as a tangential variable. The Hilbert object is the no-spectator line generated by the authenticated trace, and conditioning is an isometry from $\mathcal{H}_{\text{phys}}$ onto that line. The connection coordinate is the typed pair

$$\text{Conn}_C = (\mathcal{N}_C, \nabla_C^{\mathcal{N}_C}; \mathcal{L}_*, \nabla_C^{\mathcal{L}_*}).$$

Here $\mathcal{N}_C$ is the independently constructed chart-normal line field on the full germ neighborhood, and $\mathcal{L}_* = T\mathcal{C}_*$ is its proved restriction to the retained source lineage. Only the $\mathcal{L}_*$ restriction is compared and descended.

No equality of the three off-line normal fields and no connection in residual-profile directions is claimed. The represented physical algebra and records are proper-basic where appropriate; the off-shell grammar remains connected-proper-equivariant, the refoliation groupoid remains classical at this stage, and continuation ends at the first failure.

For $V \subset U$, restriction applies simultaneously to the base, localized successor source, proper-arrow chart domain and reconstruction, Cauchy family, graph domain, represented algebra, record history, continuation interval, and failure object. It never resets a lineage.

2.3 Morphisms and composition

A morphism $\Theta_{\alpha\beta} : \mathfrak{R}_\beta(U_{\alpha\beta}) \rightarrow \mathfrak{R}_\alpha(U_{\alpha\beta})$ is the septuple

$$\Theta_{\alpha\beta} = (t_{\alpha\beta}, T_{\alpha\beta}, \mathbf{G}_{\alpha\beta}, A_{\alpha\beta}, R_{\alpha\beta}, K_{\alpha\beta}, F_{\alpha\beta}). \quad (6)$$

The entries are respectively the differentiable common-event base map/readout relation, unitary fibre map, enriched graph matching $\mathbf{G}_{\alpha\beta} = (G_{\alpha\beta}^V, G_{\alpha\beta}^K, G_{\alpha\beta}^D, \mu_{\alpha\beta})$, $*$ -algebra intertwiner, record isomorphism, continuation natural transformation, and failure comparison. Lawfulness requires preservation of the source, raw-localization and common-source-presentation certificate, the orbit-complete reconstruction and proper/slice-change actions, physical form, complete localized constraint graphs, total labels, target sorts, counterterms, domains, the constraint-normal certificates and their $\mathcal{L}_*$ restrictions, represented observables, record identities and values, retained order, first failures, boundary data, and every nonclaim.

Composition is coordinatewise where both maps have common source domain:

$$\Theta_{\alpha\beta} \circ \Theta_{\beta\gamma} = (t_{\alpha\beta}t_{\beta\gamma}, T_{\alpha\beta}T_{\beta\gamma}, (G_{\alpha\beta}^V G_{\beta\gamma}^V, G_{\alpha\beta}^K G_{\beta\gamma}^K, G_{\alpha\beta}^D G_{\beta\gamma}^D, \mu_{\alpha\beta} \circ \mu_{\beta\gamma}), A_{\alpha\beta}A_{\beta\gamma}, R_{\alpha\beta}R_{\beta\gamma}, K_{\alpha\beta}K_{\beta\gamma}, F_{\alpha\beta}F_{\beta\gamma}).$$

Identity maps are the seven identities. Associativity follows componentwise and is typed by the triple source pullback. An automorphism is lawful only when it fixes the complete probe introduced in Section 9.2.

Lemma 2.3 (Category laws). *The objects (5), morphisms (6), restrictions, identities, and composition form a category $\text{PhysRec}_{10}(U)$ for every U , and restriction gives a pseudofunctor $\text{PhysRec}_{10} : \text{ClockSite}_{10}^{\text{op}} \rightarrow \mathbf{Cat}$.*

Proof. Every component category is already defined: Hilbert isometries, orbit-complete source-marked bicontinuous graph isomorphisms with their proper-action and slice-change coordinates, $*$ -algebra maps, record-object isomorphisms, and continuation/failure arrows. The common-source and fixed-target conditions are equalizer conditions on their product. Equalizers are closed under identities and composition. Restricting twice equals direct restriction on every component. Connected proper raw-presentation transport is functorial up to the accepted WDW-2 presentation 2-isomorphism. Guarded WDW-3 arrows remain confined to the classical record component unless an independent quantum matching is supplied. The conservative pseudofunctor language records those presentation choices; Section 5 proves literal coherence for the populated datum from its common-source chart maps. $\square$

Definition 2.4 (Fixed-target presentation congruence). For two same-domain objects or arrows x, y , write $x \sim_{10}^{\text{pres}} y$ only when the complete probe $\mathcal{P}_{10}$ has the same value on them. This definition does not assert that the relation is maximal or that every kernel-pair pair is connected by a presentation arrow; those facts are proved in W10.23.

2.4 Clock-descent and clock-neutral physical recipients

The first coordinate in (5) makes $\text{PhysRec}_{10}(U)$ a category of *single-clock presentations*. It is therefore the correct local source category but cannot be the recipient of a realization with no preferred global readout. We define that recipient before constructing any global object.

A clock predescent datum on U is

$$\mathbf{c} = (\{j_a : U_a \rightarrow U\}_{a \in A}, \{C_a\}_{a \in A}, \{t_{ab}\}, \{\kappa_{abc}\}, \text{Orient}, \text{Anchor}, \text{Fail}^{\text{clk}}), \quad (5a)$$

where the first coordinate is a set-sized lawful cover; each C_a is a function-valued clock profile with its local use, orientation, anchor, continuation, and first failure; t_{ab} is the differentiable common-event readout relation on U_{ab} ; and κ_{abc} records the identity, inverse, and triple compatibility data on the literal source pullbacks. All failure prefixes and empty intersections are retained.

Let $\text{PreClockDesc}_{10}(U)$ have the data (5a) as objects. Its 1-arrows are componentwise source-preserving maps of covers, profiles, readout relations, coherence, anchors, orientations, and failure prefixes. A transition-supported

natural isomorphism between two such arrows is a family of the declared invertible readout arrows on a common source refinement, commuting with every anchor, orientation, continuation, coherence, and failure square. Vertical and horizontal pasting use the identity, inverse, and cocycle data κ .

A lawful refinement is a 1-arrow that is jointly surjective on the same base and changes no complete clock coordinate: its local profiles and readout data are literal restrictions, not replacements by a different clock. Define

$$\text{ClockDesc}_{10}(U) := \text{PreClockDesc}_{10}(U)[(\text{lawful refinements})^{-1}].$$

Concretely an arrow in $\text{ClockDesc}_{10}(U)$ is represented by a roof

$$c \xleftarrow{r} c'' \xrightarrow{u} d, \quad r \text{ a lawful refinement,}$$

and two roofs agree when, after pullback to a common lawful refinement of their respective source data, their right legs are joined by a transition-supported natural isomorphism. Identities are identity roofs. Composition is formed by the source fibre product of the middle covers. The induced middle clock data are pulled back separately; literal identifications use only lawful-refinement legs, while any genuine clock change remains in the transition-supported right leg. The cover axioms and failure preservation make the resulting left leg a lawful refinement. Associativity follows after the canonical triple-pullback comparison and the κ -pasting law. A 1-arrow with a two-sided inverse up to the displayed natural isomorphisms therefore becomes an isomorphism in $\text{ClockDesc}_{10}(U)$; a genuine clock transition is not renamed a refinement. Restriction intersects every chart and incidence with $V \subset U$, retains typed empty intersections and failure prefixes, and applies to roofs componentwise. It therefore defines a pseudofunctor

$$\text{ClockDesc}_{10} : \text{ClockSite}_{10}^{\text{op}} \longrightarrow \mathbf{Cat}.$$

The singleton functor S_{clk} sends one profile C to the one-chart datum with identity readout and coherence. It does not select a chart from a plural datum.

An object of the clock-neutral recipient $\text{NeuPhysRec}_{10}(U)$ is

$$\mathfrak{R}_{\mathfrak{C}}^{\text{neu}}(U) = (\mathfrak{C}, U, \mathcal{V}^{\text{ren}}, \mathcal{D}^{\text{ren}}, J, \mathcal{H}, \text{Cond}, \text{Conn}, \rho, \mathcal{A}^{\text{phys}}, \text{Rec}, \text{Cont}, \text{Fail}, \text{Anchor}, \text{Cert}), \quad (5b)$$

where $\mathfrak{C} \in \text{ClockDesc}_{10}(U)$ replaces the distinguished profile C and every other coordinate has the same fixed target and type as the corresponding non-clock coordinate in (5). A morphism is a ClockDesc_{10} arrow together with the six non-readout components of (6), commuting with the clock-descent anchor, record, continuation, and failure squares. Identities, composition, and restriction are componentwise. Thus $\text{NeuPhysRec}_{10} : \text{ClockSite}_{10}^{\text{op}} \rightarrow \mathbf{Cat}$ is a pseudofunctor.

There is a canonical singleton embedding

$$S_{\text{neu}} : \text{PhysRec}_{10}(U) \longrightarrow \text{NeuPhysRec}_{10}(U), \quad \mathfrak{R}_C \longmapsto \mathfrak{R}_{S_{\text{clk}}(C)}^{\text{neu}}, \quad (5c)$$

which retains every non-clock coordinate and sends the readout component of (6) to the induced singleton clock-descent arrow. It is pseudonatural under restriction and conservative on the complete non-clock signature. A plural clock-descent object may be isomorphic to a singleton presentation after restricting to a scalar chart, but (5c) supplies no inverse and no globally preferred member.

Lemma 2.5 (Clock-descent and neutral-recipient laws). *The roof construction above is a category, restriction makes ClockDesc_{10} a pseudofunctor, the objects (5b) and their arrows form NeuPhysRec_{10} , and (5c) is a pseudonatural singleton functor conservative on every non-clock standing-bearing coordinate.*

Proof. Common source fibre products exist by W10.2. They provide the common source cover for composing roofs; the two induced clock data on that cover are retained separately and compared through the already typed transition-supported right legs. A further common source refinement together with the κ -pasting law identifies the two parenthesizations, proving the localized category laws. Identity, inverse, and horizontal/vertical composition of natural isomorphisms follow componentwise from the pre-descent coherence. Restriction preserves lawful refinement, roof equivalence, natural isomorphisms, identities, and composition. The non-clock coordinates in (5b) form the same product and equalizer categories used for PhysRec_{10} ; replacing the first factor by ClockDesc_{10} therefore preserves identities, composition, and restrictions. The singleton functor is identity on every non-clock coordinate, so it preserves and reflects equality there. Every claim is made before descent of the populated WDW-10 datum. $\square$

2.5 Enriched neutral descent records, predecessor signature, and conservative reduct

Neither the local category $\text{PhysRec}_{10}(U)$ nor the neutral recipient $\text{NeuPhysRec}_{10}(U)$ contains the chosen atlas presentation and its descent certificates. We therefore define the required successor category rather than treating those coordinates as prose attached to (5b). An object of $\text{EnrPhysRec}_{10}(U)$ is the dependent record

$$\mathfrak{E} = (\mathfrak{R}, \mathcal{A}, \mathcal{I}, \Theta, \text{Coh}, \text{Hol}, \mathcal{E}^{\text{src}}, \ell, \text{GlueCert}), \quad (6a)$$

with the following typed coordinates.

- $\mathfrak{R} \in \text{NeuPhysRec}_{10}(U)$ is the underlying clock-neutral physical-record object.
- $\mathcal{A} = \{(j_a : U_a \rightarrow U, \mathfrak{R}_a, C_a)\}_{a \in A}$ is a set-sized lawful cover with $\mathfrak{R}_a \in \text{PhysRec}_{10}(U_a)$ and its available readout.
- $\mathcal{I}$ is the full finite Čech incidence diagram $U_{a_0 \dots a_k}$, including source, anchor, record, boundary, nonclaim, and first-failure data.
- $\Theta = \{\Theta_{ab}\}$ contains the complete morphisms (6), including their (G^V, G^K, G^D, μ) components, on every ordered double overlap.
- Coh records identities, inverses, carrier and label cocycles, and the accepted proper-presentation 2-cells and higher comparisons on their literal domains; Hol records the complete carrier-and-label action of a generating loop basis.
- $\mathcal{E}^{\text{src}} = \{\mathcal{E}_a^{\text{src}}\}$ and $\ell = \{\ell_a\}$ are the inverse common-source and common-label markings for every member of $\mathcal{A}$.
- GlueCert contains the quotient maps, representative-independence squares, descended structure maps, and authenticated compatible section, or the typed first failure of one of them.

The record (6a) is defined for typed candidate data before any coordinate is certified as passing. Thus its definition neither assumes effective descent nor promotes a certificate to a physical observable.

An arrow

$$\Xi = (f, \sigma, \{f_a\}_{a \in A}, f_{\mathcal{I}}, f_{\Theta}, f_{\text{Coh}}, f_{\text{Hol}}, f_{\mathcal{E}}, f_{\ell}, f_{\text{Glue}}) : \mathfrak{E} \longrightarrow \mathfrak{E}' \quad (6b)$$

consists of $f : \mathfrak{R} \rightarrow \mathfrak{R}'$ in $\text{NeuPhysRec}_{10}(U)$, an atlas map $\sigma : A \rightarrow A'$, local PhysRec_{10} arrows $f_a : \mathfrak{R}_a \rightarrow \mathfrak{R}'_{\sigma(a)}$ over the declared base maps, and component maps on the remaining displayed coordinates. They must commute with every face, transition, marking, coherence, holonomy, and gluing square. Identities use identity atlas maps and identity component arrows. Composition uses $\sigma'\sigma$, $f'_{\sigma(a)}f_a$, and componentwise pasting on the remaining coordinates. The commutative-square conditions are stable under this composition, which is associative because every component category is associative.

For $j : V \hookrightarrow U$, define $j^{*, \text{enr}} \mathfrak{E}$ by intersecting every U_a and every incidence domain with V , restricting $\mathfrak{R}$, the local objects, arrows, markings, certificate data, and failure prefixes, and retaining typed empty intersections. Apply the same operation to (6b). Restriction respects identities and composition, with the accepted proper-presentation comparison on iterated pullback. Hence

$$\text{EnrPhysRec}_{10} : \text{ClockSite}_{10}^{\text{op}} \longrightarrow \mathbf{Cat}$$

is a pseudofunctor. Forgetting every coordinate of (6a) except $\mathfrak{R}$, and every coordinate of (6b) except f , defines a pseudonatural functor

$$\text{For}_{10} : \text{EnrPhysRec}_{10} \longrightarrow \text{NeuPhysRec}_{10}. \quad (6c)$$

For $U \subset U_{\Phi}$, an object of $\text{EnrPhysRec}_{10}^{\Phi\text{-mark}}(U)$ is $(\mathfrak{E}; m_{\Phi})$, where $\mathfrak{E} \in \text{EnrPhysRec}_{10}(U)$ and

$$m_{\Phi} = (a_{\Phi}, \chi_{\Phi}), \quad \chi_{\Phi} : \mathfrak{R} \xrightarrow{\cong} S_{\text{neu}}(\mathfrak{R}_{a_{\Phi}}) \quad \text{in } \text{NeuPhysRec}_{10}(U),$$

a_{Φ} selects an atlas member with the inherited scalar-presentation type, and χ_{Φ} certifies that it presents the underlying object. A marked arrow Ξ is an arrow (6b) with $\sigma(a_{\Phi}) = a'_{\Phi}$ and

$$\chi'_{\Phi} f = S_{\text{neu}}(f_{a_{\Phi}}) \chi_{\Phi}. \quad (6c')$$

Forgetting (a_{Φ}, χ_{Φ}) defines $\text{U}_{\text{mark}} : \text{EnrPhysRec}_{10}^{\Phi\text{-mark}}(U) \rightarrow \text{EnrPhysRec}_{10}(U)$. The singleton-enrichment functor Sing_{Φ} sends a local scalar object to (6a) with underlying object $S_{\text{neu}}(\mathfrak{R}_{\Phi})$, the identity cover, identity incidence/-transition/coherence/holonomy/gluing coordinates, and its already constructed inverse source and label markings, with $\chi_{\Phi} = \text{id}$. Thus a local marked lift and a full restricted global presentation inhabit the same declared category without being identified.

For $U \subset U_{\Phi}$, let $\text{Cert}_9(U)$ be the category whose objects are restrictions of complete WDW-9 certificate packages on U and whose arrows preserve the full WDW-9 signature: all thirteen roles, twenty overlaps, source lineage, response realization, assignment typing, operator grammar, complete constraint labels, graph domain, counterterms, joint-kernel data, record, continuation, failures, boundary data, nonclaims, and G_9 . Thus Cert_9 is not the physical Hilbert-line category and does not forget any predecessor standing-bearing coordinate.

Define the conservative predecessor reduct on objects by extracting the marked scalar local component of (6a) and projecting that component to its complete WDW-9 signature; the comparison χ_{Φ} prevents this operation from

exporting an unrelated atlas member instead of a presentation of $\mathfrak{R}$. On an arrow (6b), extract the mark-preserving local arrow f_Φ and project it to the same predecessor coordinates:

$$\text{Red}_9 : \text{EnrPhysRec}_{10}^{\Phi\text{-mark}}(U) \longrightarrow \text{Cert}_9(U), \quad \text{Red}_9(\Xi) = f_\Phi|_{\text{sig}_9}. \quad (6d)$$

The action is exact:

Coordinate block	Location in (6a)	Action of Red_9
Thirteen roles, twenty overlaps, WDW-9 grammar, labels, domain, counterterms, kernels, records, failures, and nonclaims	Marked scalar local component	Retain componentwise
WDW-10-only local chart certificates, including $\mathcal{P}$, Ψ , $\hat{Q}$, proper translations, and slice-change maps	Marked scalar local component	Forget only successor additions
Extra clocks, incidence, transitions, coherence, holonomy, source/label markings, and gluing certificate	$\mathcal{A}, \mathcal{I}, \Theta$; Coh, Hol; $\mathcal{E}^{\text{src}}, \ell$; GlueCert	Forget as successor signature
Underlying clock-neutral object	$\mathfrak{R}$	Export through the marked singleton presentation certified by $\chi_\Phi : \mathfrak{R} \cong S_{\text{neu}}(\mathfrak{R}_{a_\Phi})$; an unlinked local member is not in the reduct domain

Lemma 2.6 (Enriched-category laws and conservative-reduct functoriality). *The records (6a), arrows (6b), identities, composition, and enriched restrictions form the pseudofunctor EnrPhysRec_{10} ; (6c) is a functor; and (6d) preserves identities and composition. It commutes with restriction:*

$$\text{Red}_9(j^{*,\text{enf}}(\mathfrak{E}; m_\Phi)) = j^* \text{Red}_9(\mathfrak{E}; m_\Phi) \quad (6e)$$

for every $j : V \hookrightarrow U \subset U_\Phi$, with the same equality on arrows.

Proof. The construction following (6b) proves the category and restriction laws componentwise, while (6c) is the first-coordinate projection from the now explicitly displayed dependent record. A marked identity has identity scalar component and satisfies (6c'), while the marked component of Ξ/Ξ is $f'_\Phi f_\Phi$; projecting to the WDW-9 signature therefore preserves identities and composition. Pasting the two marking squares proves (6c') for the composite. Enriched restriction intersects the marked scalar domain and restricts each one of its predecessor coordinates, including χ_Φ . Extracting that component before or after restriction gives the same tuple and the same restricted arrow, proving (6e). The retained-coordinate row prevents an accidental reduction to the physical line, while the other rows show that every forgotten entry is an actual coordinate of the printed domain of (6d). $\square$

Remark 2.7 (Recovery polarity). The empty slot in the authorization tuple (2) says that WDW-9 did not supply an atlas. It is not a demand that every lawful WDW-10 successor object continue to have an empty atlas. Exact predecessor recovery is therefore an isomorphism after (6d), not an isomorphism between objects with different signatures. No map from a bare WDW-9 certificate is allowed to manufacture the successor-only atlas coordinates.

Theorem 2.8 (W10.3 — local, clock-descent, neutral, enriched, and recovery categories). *The pseudofunctors PhysRec_{10} , ClockDesc_{10} , NeuPhysRec_{10} , and EnrPhysRec_{10} , the singleton functor S_{neu} , the forgetful functors For_{10} and U_{mark} , the predecessor category Cert_9 , the comparison-marked enriched category $\text{EnrPhysRec}_{10}^{\Phi\text{-mark}}$, the singleton enrichment Sing_Φ , and the object-and-arrow reduct Red_9 are fixed before descent or recovery. Their restrictions and comparison laws are typed by (6b)–(6e). These categories retain all physical, atlas, domain, record, continuation, failure, boundary, and nonclaim coordinates required by their respective signatures.*

Proof. The first category lemma supplies PhysRec_{10} ; (5a)–(5c) supply the clock-descent and neutral recipients and the singleton embedding without selecting a global clock. Equations (6a)–(6c) explicitly supply the added object and arrow coordinates, laws, restrictions, and underlying-object functor. The marking square (6c'), U_{mark} , and Sing_Φ supply the two recovery domains, and the functoriality lemma supplies Red_9 on objects and arrows with restriction naturality. No stack, global-object, effective-descent, or recovery assertion was used. Membership of the populated datum follows later from W10.9–W10.13. $\square$

Remark 2.9. The stack property is not stipulated here. Section 6 proves the exact effective descent theorem needed for the populated atlas and then proves the arbitrary-atlas same-target statement from common-anchor factorization.

3. Positive population of the three-clock atlas

3.1 The three profile mechanisms

The scalar profile is inherited:

$$C_\Phi(h, \phi) = \phi, \quad C_\Phi = \tau \mathbf{1}.$$

It carries the complete WDW-9 clock-use package, not merely the function ϕ . The two new profiles are defined on the same source-attached metric neighborhood by

$$C_\nu(h; h_\star) = \frac{1}{3} \log \left(\frac{d\mu_h}{d\mu_{h_\star}} \right), \quad C_\nu = s \mathbf{1}, \quad (7)$$

$$C_R(h) = R^{(3)}(h), \quad C_R = r \mathbf{1}. \quad (8)$$

The reference density in (7) is an anchor coordinate: a proper redescription transports both h and $h_\star$, so $C_\nu(g^*h; g^*h_\star) = g^*C_\nu(h; h_\star)$. Equation (8) is natural. Therefore both conditions descend to profile equations after the WDW-2 proper-spatial quotient; neither uses a preferred coordinate volume form.

At (1), $(\tau, s, r) = (0, 0, -6)$. The three readouts are labels of constant profile slices. They are not scalar functionals proposed as slices of the infinite-dimensional lapse distribution.

3.2 Normal response operators

With the frozen WDW-3 convention $K_{ij} = \frac{1}{2} \mathcal{L}_n h_{ij}$, a lapse N gives

$$\delta_N h_{ij} = 2N K_{ij}, \quad \delta_N \phi = N \frac{p\phi}{\sqrt{h}}. \quad (9)$$

Consequently the three profile responses at the anchor are

$$L_{\Phi, \star} N = \epsilon_\star N, \quad (10)$$

$$L_{\nu, \star} N = \frac{1}{3} N \operatorname{tr}_{h_\star} K_\star = c_{\epsilon_\star} N, \quad (11)$$

$$L_{R, \star} N = DR_{h_\star}[2N K_\star] = 4c_{\epsilon_\star} (3 - \Delta_{h_\star}) N. \quad (12)$$

Here $\Delta = D^i D_i$ has nonpositive spectrum. For completeness, the scalar-curvature variation used in (12) is

$$DR_h[q] = -\langle \operatorname{Ric}(h), q \rangle_h + D^i D^j q_{ij} - \Delta_h \operatorname{tr}_h q. \quad (13)$$

Substituting $q = 2c_{\epsilon_\star} N h_\star$ and $\operatorname{Ric}(h_\star) = -2h_\star$ gives (12) term by term.

Lemma 3.1 (Intrinsic-volume response inverse). *$L_{\nu, \star} : H^s(\Sigma) \rightarrow H^s(\Sigma)$ is an isomorphism with inverse $N \mapsto c_{\epsilon_\star}^{-1} N$. On a sufficiently small source-attached neighborhood on which $|\operatorname{tr}_h K| \geq 3c_{\epsilon_\star}/2$, the exact response*

$$L_{\nu, z} N = \frac{1}{3} N \operatorname{tr}_h K$$

has a smooth compatible inverse family.

Proof. The anchor assertion follows from $c_{\epsilon_\star} > 0$. The displayed lower bound makes the exact multiplier and its reciprocal bounded on every H^s stage. Smooth dependence follows from Sobolev multiplication and the smooth inversion map on functions bounded away from zero. Stage restrictions commute with pointwise inversion. $\square$

Lemma 3.2 (Intrinsic-curvature elliptic inverse). *For every $s \geq s_0$, $L_{R, \star} : H^s(\Sigma) \rightarrow H^{s-2}(\Sigma)$ is an isomorphism. After shrinking the source-attached regular chart, the exact family*

$$L_{R, z} N = -2N \langle \operatorname{Ric}(h), K \rangle_h + 2D^i D^j (N K_{ij}) - 2\Delta_h (N \operatorname{tr}_h K) \quad (14)$$

is a smooth family of elliptic isomorphisms with compatible inverses.

Proof. For real N ,

$$\langle N, (3 - \Delta)N \rangle_{L^2} = 3\|N\|_{L^2}^2 + \|DN\|_{L^2}^2.$$

Thus the self-adjoint elliptic operator $3 - \Delta$ has trivial kernel and closed range. Its index is zero, so it is onto and elliptic regularity gives the stated Sobolev isomorphisms. Formula (14) is (13) applied to $2NK$. Its principal symbol at the anchor is $-4c_{\epsilon_*}\Delta$, hence uniformly elliptic. Ellipticity and bounded invertibility persist under a sufficiently small H^s coefficient perturbation; the resolvent identity gives smooth parameter dependence. The elliptic estimates are compatible on the WDW-5 scale, so the inverses define a smooth-tame inverse-limit family. $\square$

Tangential shifts add Lie derivatives of the profile. They are exactly the WDW-2 proper-spatial arrows and are removed only after their source and boundary guards pass. The inverse statements concern the normal lapse block on that quotient; no improper transformation is absorbed.

3.3 Guarded profile slices

Let $\mathcal{S}_{\alpha,*}$ be the local transverse plaque-parameter slice fixed in Section 1, and let $\mathcal{P}_z(N)$ denote the WDW-3 guarded characteristic plaque with $z \in \mathcal{S}_{\alpha,*}$ and normal descriptor N , its source history held fixed. For $\alpha \in \{\Phi, \nu, R\}$ define

$$\mathcal{F}_\alpha(z, N, c) = C_\alpha(\mathcal{P}_z(N)) - c\mathbf{1}.$$

Equations (10)–(12) are the derivatives in N at the anchor. The parameter-dependent implicit-function theorem therefore gives unique small lapse profiles $N_\alpha(z, c)$ solving $\mathcal{F}_\alpha = 0$. The WDW-4A no-KID theorem removes normal isotropy, while the WDW-2 connected proper quotient removes only the retained spatial presentation arrows.

Proposition 3.3 (Unique local intersections). *There are nonempty connected source-attached open sets U_Φ, U_ν, U_R containing the anchor event such that every guarded characteristic plaque in U_α meets the corresponding constant-profile slice exactly once modulo connected proper redescription. These are the fixed certified radius neighborhoods of Section 1, not maximal continuations. The first failures are respectively:*

- *loss of scalar timelikeness/oriented speed, inherited WDW-9 scalar-clock data, or exit from its certified flow box;*
- *vanishing of $\text{tr } K$, loss of the volume-response inverse, source/record/domain data, or exit from the certified volume flow box;*
- *loss of ellipticity/invertibility of (14), source/record/domain data, or exit from the certified curvature flow box.*

Every non-clock predecessor failure is copied literally into all three failure objects.

Proof. For Φ this is the imported WDW-4A/7 profile flow box, restricted to a certified sub-box. For ν and R , the two inverse lemmas give the required split derivative. Choose the radius pairs so the Neumann estimate

$$\|L_{\alpha,z} - L_{\alpha,*}\| < (2\|L_{\alpha,*}^{-1}\|)^{-1}$$

holds, the parameter inverse maps the profile ball into the lapse ball, and the plaque map from the transverse slice–lapse product is injective. The quantitative inverse/implicit-function theorem then gives exactly one small solution $N_\alpha(z, c) \in B_{\sigma_\alpha}(0)$ for every admitted (z, c) . Compatible radius choices on the Sobolev scale give the inverse-limit result. This proves residual-copy freedom inside the declared neighborhood. It makes no inference from local invertibility to injectivity on a maximal component and makes no global-injectivity claim. Exiting the certified box is therefore a typed first failure rather than an invitation to continue the chart by assertion. $\square$

3.4 Chart-normal Cauchy seeds

Choose one convention for every analytic normal coordinate. Put

$$Z_\alpha = \int_\Sigma w_\alpha d\mu_{h_*} > 0, \quad \langle f \rangle_{w_\alpha} = Z_\alpha^{-1} \int_\Sigma w_\alpha f d\mu_{h_*}, \quad T_\alpha = \langle C_\alpha - c_{\alpha,*} \rangle_{w_\alpha}, \quad (15a)$$

where $w_\Phi = \epsilon_*^{-1}$, $w_\nu = c_{\epsilon_*}^{-1}$, and $w_R = (12c_{\epsilon_*})^{-1}$. Thus the harmless constant weights are retained as chart certificates but cancel from the normalized average. Define

$$\chi_\alpha = C_\alpha - c_{\alpha,*} - T_\alpha \mathbf{1}, \quad \langle \chi_\alpha \rangle_{w_\alpha} = 0. \quad (15b)$$

Equations (15a)–(15b), rather than an unnormalized weighted integral, are used throughout the Cauchy normal form, companion operator, connection, and overlap Jacobians. For the exact response-normal lapse

$$N_\alpha(z) = L_{\alpha,z}^{-1} \mathbf{1},$$

one has identically

$$dT_\alpha(\mathfrak{X}_z(N_\alpha(z))) = \langle L_{\alpha,z} N_\alpha(z) \rangle_{w_\alpha} = 1, \quad d\chi_\alpha(\mathfrak{X}_z(N_\alpha(z))) = 0. \quad (15c)$$

Hence $C_\alpha = c_{\alpha,\star} + T_\alpha \mathbf{1} + \chi_\alpha$ is a literal decomposition and ∂_{T_α} is represented by the same exact lapse inverse at every point. The function-valued physical slice is $\chi_\alpha = 0$; the normalized mean coordinate is not promoted to a clock.

For the moment, choose a local WDW-2 proper slice over the certified flow box and retain the associated proper-arrow coordinate g . For a WDW-6 successor-source element F , write its orbit-complete raw chart representative as $\widehat{F}_\alpha(T_\alpha, \chi_\alpha, g)$. Formally set

$$J_{\alpha,x}^{\text{src}} \widehat{F}_\alpha = \left(\widehat{F}_\alpha|_{T_\alpha=t, \chi_\alpha=0}, \partial_{T_\alpha} \widehat{F}_\alpha|_{T_\alpha=t, \chi_\alpha=0} \right), \quad (15d)$$

where g and all non-profile tangential source coordinates remain in both components; $\chi_\alpha = 0$ is the already declared function-valued physical gauge. Section 4 constructs every $\widehat{F}_\alpha$ by a slice-complete pullback from one common proper-localized WDW-6 support, proves (15d) continuous and graph-smooth, and proves that another proper slice changes it only by the accepted WDW-2 orbit-coordinate presentation map. The one-slice evaluation $g = e$ is never used as the source presentation. No refoliation lift or physical-line map is used to type the trace. The full doublet is retained because the WDW constraint is second order in the chart-normal direction. At the anchor its first component on the authenticated solution $F_{A,\mu}^{(0)}$ is one, so $Y_{\alpha,\star} = J_{\alpha,\star}^{\text{src}} F_{A,\mu}^{(0)} \neq 0$ for all three charts. Nonvanishing persists on smaller anchor components.

Theorem 3.4 (W10.4 — inherited scalar-chart certificate). *The complete WDW-9 scalar-profile realization restricts to a predecessor certificate*

$$\mathfrak{R}_\Phi^9(U_\Phi) \in \text{Cert}_9(U_\Phi)$$

with every WDW-9 coordinate unchanged. This theorem does not prepopulate the WDW-10-only atlas, common-source markings, or gluing certificates.

Proof. Use the WDW-9 clock/conditional and record/recovery coordinates, the WDW-7 Cauchy doublet and generated line, and their inherited restrictions. All thirteen roles and twenty incidences remain attached, together with the predecessor grammar, labels, graph domain, counterterms, and joint-kernel data. No formula in this section reconstructs, approximates, or enriches that object. $\square$

Theorem 3.5 (W10.5 — intrinsic-volume profile-clock population). *The profile (7) has a full lapse-response inverse, locally unique guarded intersections, an authenticated nonzero formal Cauchy seed, and the explicit certified-neighborhood first-failure boundary above. These analytic clock-population data are constructed without assuming conditioning, graph-domain transport, a physical form, an algebra representation, a connection, or complete local-object membership. In particular, the prior fact that total volume changes along the scalar-clock history is not promoted to clock standing.*

Proof. The response and certified local-intersection results were proved above. The authenticated WDW-6 solution has value one at the anchor, so its formal doublet is nonzero there and remains nonzero after shrinking. No assertion in this proof invokes W10.9 or a later theorem. The complete successor-source trace, conditioning, connection, record, and standing conclusion are downstream obligations. $\square$

Theorem 3.6 (W10.6 — intrinsic-curvature profile-clock population). *The profile (8) has the quotient elliptic inverse (14), locally unique guarded intersections, an authenticated nonzero formal Cauchy seed, and an explicit certified-neighborhood first-failure boundary. These analytic clock-population data are constructed without assuming any later local quantum object or transition.*

Proof. The elliptic inverse, certified guarded intersection, and nonzero formal seed were constructed without using descent or W10.9. Curvature is natural under spatial redescription and the profile slice is defined on the proper quotient. Homogeneous monotonicity is nowhere used as the inverse theorem. The proof stops here; the successor-source trace and all quantum coordinates are downstream obligations. $\square$

3.5 Genuine plurality

Lemma 3.7 (Pairwise differential separation). *On every sufficiently small common WDW-5 configuration neighborhood, no pair among C_Φ, C_ν, C_R is related by a locally invertible scalar reparameterization preserving the complete clock-use package.*

Proof. The differential of C_Φ acts on the scalar field, while those of C_ν and C_R act on the metric. A pure scalar variation and a pure metric variation therefore rule out reparameterizations involving C_Φ .

For the two metric profiles, take conformal variations $q = 2uh_*$. Equations (7), (12), and (13) give

$$DC_\nu[q] = u, \quad DC_R[q] = 4(3 - \Delta)u.$$

If $C_R = f(C_\nu)$ on a neighborhood, differentiation at the anchor would make $4(3 - \Delta)$ multiplication by the scalar $f'(0)$. Constants force $f'(0) = 12$, while any nonconstant Laplace eigenfunction gives $4(3 + \lambda) \neq 12$. This is impossible. Nonconstant conformal variations are not connected proper spatial arrows: a pure conformal Lie derivative would generate a conformal Killing field, and the compact hyperbolic anchor has none. The argument therefore survives the proper quotient. $\square$

Theorem 3.8 (W10.7 — genuine clock plurality). *The finite seed atlas contains three pairwise profile-distinct clock mechanisms. In particular at least two are not same-neighborhood reparameterizations. Their numerical relation along the homogeneous source line is presentation coincidence on that line and has no neighborhood-finality force.*

Proof. Apply the separation lemma and retain the distinct anchors, differential orders, normal responses, and failure objects in the clock registry. The Physical Clock Pair Normal-Form theorem classifies these as quotient-profile or anchor divergence within the enlarged atlas-use target, not as equivalent skins. $\square$

3.6 Occupied cover and incidence

All three certified neighborhoods contain the retained interval $\mathcal{C}_*$. Their pairwise and triple germ intersections are therefore nonempty. Shrink their fixed radii simultaneously so the WDW-6 common graph domain, represented algebra, WDW-8 source lineage, compact-boundaryless coordinate, and first-failure records are common on each intersection.

Theorem 3.9 (W10.8 — germ cover and overlap population). *The positively occupied analytic/source cover of (3) is*

$$\mathcal{A}_{10}^0 = \{\text{Germ}_{\mathcal{C}_*} U_\Phi, \text{Germ}_{\mathcal{C}_*} U_\nu, \text{Germ}_{\mathcal{C}_*} U_R\}.$$

Every pairwise pullback and $U_{\Phi\nu R}$ is nonempty and contains the authenticated common source lineage with its full transverse germ. The profile, source, record/continuation, boundary, and nonclaim incidences are actual pullbacks over X_{10}^{base} . Graph-domain and quantum-object overlap maps are not asserted here; they are constructed in W10.9–W10.10. The first-failure boundary is the union of the retained local failures and the first loss of a required common incidence.

Proof. Covering is the definition (3), but population is not: W10.4–W10.6 exhibit the inherited scalar certificate and independently formed analytic clock data in the two new members. The anchor event lies in every certified chart by (1), (7), and (8), and the homogeneous interval was shrunk so all pass conditions persist. All three germ representatives therefore contain $\mathcal{C}_*$, and their finite intersections are source-attached neighborhoods of that lineage. Each pullback is formed over the same quotient source event and germ, not the clock ranges. Shrinking only inside the fixed certified radius pairs preserves the retained history prefix and yields the printed boundary. No W10.9 object is used. $\square$

4. Slice-complete common-source local realization

4.1 Saturated raw localization

Write $\Phi_6^{\text{tr}} = \Phi_{5A}^{\text{tr}}$ for the Branch-A smooth trace-core source and $\mathcal{G}_6^{\text{ren}}$ for the registered WDW-6 successor grammar. For every source-attached open neighborhood $O \subset X_{10}^{\text{Pr}}$ define its saturated raw preimage and raw restriction ideal by

$$\tilde{O} = \pi_{\text{pr}}^{-1}(O), \quad \mathcal{I}_{\text{raw}}(\tilde{O}) = \{F \in \Phi_6^{\text{tr}} : F|_{\tilde{O}} = 0\}, \quad \Phi_{6,O}^{\text{tr}} = \Phi_6^{\text{tr}} / \mathcal{I}_{\text{raw}}(\tilde{O}). \quad (16)$$

The restriction in (16) is evaluated on the raw source where every $F \in \Phi_6^{\text{tr}}$ is defined. It is not a restriction of a non-invariant function to quotient O , and no inverse of global restriction is used.

Lemma 4.1 (Local-differential ideal invariance). *Let A be a registered local differential, regulated Hamiltonian, constraint, counterterm-insertion, or typed coefficient/target constructor in $\mathcal{G}_6^{\text{ren}}$. Then*

$$A\mathcal{I}_{\text{raw}}(\tilde{O}) \subset \mathcal{I}_{\text{raw}}(\tilde{O}).$$

Proof. The listed WDW-6 primitives are finite-order Bastiani differential operators in the configuration argument, with smooth local, separated, finite-rank, or regular-kernel coefficients. Spatial smoothing and counterterm data occur in those coefficients and typed target constructors; they do not replace the configuration argument of F . If F vanishes on the raw open set $\tilde{O}$, every finite configuration derivative of F vanishes there. The product rule gives $(AF)|_{\tilde{O}} = 0$. $\square$

Lemma 4.2 (Finite proper-action ideal invariance). *For every connected proper-redescription arrow g whose source and range meet the raw locus, let $(U_g F)(q) = F(g^{-1}q)$. Then*

$$U_g \mathcal{I}_{\text{raw}}(\tilde{O}) \subset \mathcal{I}_{\text{raw}}(\tilde{O}),$$

and the same holds for every finite composable word of such actions and inverses.

Proof. Saturation is the exact premise: $q \in \tilde{O}$ implies $g^{-1}q \in \tilde{O}$ whenever the arrow is defined, because $\pi_{\text{pr}}(g^{-1}q) = \pi_{\text{pr}}(q) \in O$. Thus $F|_{\tilde{O}} = 0$ gives $(U_g F)(q) = 0$. Induction proves the finite-word statement. Unlike the preceding lemma, this proof explicitly handles evaluation at another raw presentation point. $\square$

The two lemmas induce every type-correct localized word and connected proper action:

$$\widehat{C}_{O,i}^{\text{src}}[F]_{\tilde{O}} = [\widehat{C}_i^{\text{src}} F]_{\tilde{O}}, \quad U_{g,O}[F]_{\tilde{O}} = [U_g F]_{\tilde{O}}. \quad (17)$$

They prove no corresponding statement for a finite WDW-3 refoliation arrow, and none is used below.

4.2 One common localized WDW-6 support

WDW-5 defines the branch-appropriate complete label set as the disjoint union

$$\mathfrak{J}_6 = \coprod_{t \in \{D, G, J, H\}_A} \mathfrak{J}_{6,t}, \quad i_5 = (A, t, \sigma; \mathcal{K}_A, \epsilon, \nabla_A^{\text{conf}}, \mathfrak{o}_A, \mathfrak{o}_{\text{dens}}, \rho_A^{\text{pol}}), \quad i = (i_5, \mathfrak{c}_i, \text{Sort}(i_5)). \quad (18)$$

Here σ contains the lapse, shift, or internal smearing appropriate to t ; the four slots D, G, J, H are retained, with the accepted typed-empty internal slots on Branch A. The remaining W5 entries retain the regulator, smoothing/pivot, configuration connection, ordering, density, and polarization. The W6-enriched label attaches the accepted counterterm column $\mathfrak{c}_i$ and the target sort definitionally determined by i_5 ; neither is a free added coordinate.

For a connected proper presentation arrow g , W5.3 and the already populated proper action give the complete matching tuple to which W5.12/W6.15 apply. Hence

$$U_{g,O} \widehat{C}_{O,i}^{\text{src}} = \widehat{C}_{O,\mu_g(i)}^{\text{src}} U_{g,O}, \quad U_{g,O}^\times \widehat{C}_{O,i}^\times = \widehat{C}_{O,\mu_g(i)}^\times U_{g,O}^\times, \quad \mu_{g^{-1}} = \mu_g^{-1}, \quad \mu_{g \circ h} = \mu_g \circ \mu_h. \quad (19)$$

If i is Hamiltonian with lapse N , the lapse entry of $\mu_g(i)$ is $g_* N$. Equation (19) is restricted to the accepted proper groupoid. W5.12 and W6.15 are conditional complete-matching theorems; they are not cited as existence theorems for arbitrary WDW-3 arrows.

For a compact $K \subseteq \tilde{O}$, derivative degree m , type-correct finite word $\omega = (i_1, \dots, i_k)$, and continuous probe λ on the target sort of $\widehat{C}_\omega F$, define

$$p_{K,m,\omega,\lambda}^O([F]) = \sup_{q \in K, 0 \leq j \leq m} \left| \lambda \left(D^j (\widehat{C}_\omega^{\text{src}} F)(q) \right) \right|.$$

The restriction quotient makes this independent of F . Including the empty word and point evaluations makes the family Hausdorff. Equation (19) reindexes the proper-presentation family bijectively. Write $V_6^{\text{ren}}(O)$ for the completion and $\mathcal{D}_6^{\text{ren}}(O)$ for the localized literal WDW-6 common graph domain in $\mathcal{K}_6^{\text{ren}}(O) = ((V_6^{\text{ren}}(O))')^\times$. Let $\iota_{6,O} : V_6^{\text{ren}}(O) \rightarrow \mathcal{D}_6^{\text{ren}}(O)$ be the W6R.S15 evaluation embedding. Define

$$\mathbf{V}_6(O) = \left(V_6^{\text{ren}}(O), \mathcal{K}_6^{\text{ren}}(O), \mathcal{D}_6^{\text{ren}}(O), \mathfrak{J}_6, \{\widehat{C}_{O,i}^{\text{src}}, \widehat{C}_{O,i}^\times\}_{i \in \mathfrak{J}_6}, \text{Sort}_6, \text{CT}_6, \text{Anom}_6, \{U_{g,O}, \mu_g\}_{g \in \mathcal{G}_{\text{pr}}} \right). \quad (20)$$

This is the single off-shell support used by all three clocks on every common representative neighborhood.

4.3 Slice-complete chart presentations formed by conjugation

Let $\kappa_{\alpha,O} : Z_{\alpha,O} \rightarrow O$ be the certified proper-quotient flow-box chart and choose a WDW-2 local proper slice $s_O : O \rightarrow \tilde{O}$. Let $\mathcal{G}_O^{\text{pr}}$ be the local connected proper-action coordinate supplied by that slice theorem. The orbit-complete chart domain and reconstruction map are

$$\mathcal{P}_{\alpha,O} := \{(z, g) : z \in Z_{\alpha,O}, g \in \mathcal{G}_O^{\text{pr}}, g \cdot s_O(\kappa_{\alpha,O}(z)) \text{ is defined}\}, \quad \Psi_{\alpha,O}(z, g) := g \cdot s_O(\kappa_{\alpha,O}(z)).$$

After the certified shrink already allowed in W10.8, the accepted WDW-2 free-regular slice theorem makes $\Psi_{\alpha,O} : \mathcal{P}_{\alpha,O} \rightarrow \tilde{O}$ a smooth proper-action trivialization: every raw point of the saturated preimage occurs, and freeness makes its pair (z, g) unique. Define the slice-complete primal map on the raw restriction quotient by

$$\widehat{Q}_{\alpha,O}[F](z, g) := F(\Psi_{\alpha,O}(z, g)), \quad \widehat{\Phi}_{\alpha,O}^{\text{tr}} := \widehat{Q}_{\alpha,O}(\Phi_{6,O}^{\text{tr}}).$$

It is well defined because $\Psi_{\alpha,O}$ takes values in $\tilde{O}$. It is injective because $\Psi_{\alpha,O}$ is onto: $\widehat{Q}_{\alpha,O}[F] = 0$ implies $F|_{\tilde{O}} = 0$ and hence $[F] = 0$. On the displayed algebraic range its inverse is the explicit map

$$\Psi_{\alpha,O}^{-1}(q) = \left(\kappa_{\alpha,O}^{-1}(\pi_{\text{pr}} q), g_{s,q} \right), \quad q = g_{s,q} \cdot s_O(\pi_{\text{pr}} q), \quad \widehat{Q}_{\alpha,O}^{-1} f = [f \circ \Psi_{\alpha,O}^{-1}]_{\tilde{O}}.$$

The former one-slice restriction is only the projection $f \mapsto f(z, e_z)$ where the identity arrow is defined; it is not used as a carrier map.

For $f = \widehat{Q}_{\alpha,O}[F]$, a compact $\widehat{K} \in \mathcal{P}_{\alpha,O}$, and the same m, ω, λ used before (20), define before conjugating any operator

$$\widehat{p}_{\widehat{K}, m, \omega, \lambda}^{\alpha, O}(f) := \sup_{(z, g) \in \widehat{K}, 0 \leq j \leq m} \left| \lambda \left(D^j (\widehat{Q}_{\alpha,O} \widehat{C}_{O, \omega}^{\text{src}} F)(z, g) \right) \right|.$$

Injectivity makes this independent of the displayed F . The empty word includes point evaluations at every (z, g) . Write $V_{6, \alpha}^{\text{ren}}(O)$ for the Hausdorff completion in this family. If $\widehat{Q}'_{\alpha, O} : V'_{6, \alpha}(O) \rightarrow (V_{6, \alpha}^{\text{ren}}(O))'$ is its continuous transpose, let $\Gamma_{\alpha, O}^{\mathcal{K}} = (\widehat{Q}'_{\alpha, O})^\times$ be the induced map on the WDW-6 continuous-dual carrier, $\Gamma_{\alpha, O}^{\mathcal{D}}$ its exact common-domain restriction, and $\rho_{\alpha, O} : \mathcal{I}_6 \rightarrow \mathcal{I}_{\alpha, O}$ the literal coordinate recoding of the same abstract label. Let $\iota_{\alpha, O} : V_{6, \alpha}^{\text{ren}}(O) \rightarrow \mathcal{K}_{6, \alpha}^{\text{ren}}(O)$ be the chart evaluation embedding. Define, rather than assume,

$$\widehat{C}_{\alpha, \rho_{\alpha, O}(i)}^{\text{src}} := \widehat{Q}_{\alpha, O} \widehat{C}_{O, i}^{\text{src}} \widehat{Q}_{\alpha, O}^{-1}, \quad \widehat{C}_{\alpha, \rho_{\alpha, O}(i)}^\times := \Gamma_{\alpha, O}^{\mathcal{K}} \widehat{C}_{O, i}^\times (\Gamma_{\alpha, O}^{\mathcal{D}})^{-1}, \quad \mathbf{\Gamma}_{\alpha, O} = (\widehat{Q}_{\alpha, O}, \Gamma_{\alpha, O}^{\mathcal{K}}, \Gamma_{\alpha, O}^{\mathcal{D}}, \rho_{\alpha, O}).$$

Thus

$$\widehat{Q}_{\alpha, O} \widehat{C}_{O, i}^{\text{src}} = \widehat{C}_{\alpha, \rho_{\alpha, O}(i)}^{\text{src}} \widehat{Q}_{\alpha, O}, \quad \Gamma_{\alpha, O}^{\mathcal{K}} \widehat{C}_{O, i}^\times = \widehat{C}_{\alpha, \rho_{\alpha, O}(i)}^\times \Gamma_{\alpha, O}^{\mathcal{D}}$$

are conjugation identities. Proper action is retained explicitly: for every composable proper arrow h ,

$$(\widehat{U}_{h, \alpha, O} f)(z, g) := f(z, h^{-1}g), \quad \widehat{Q}_{\alpha, O} U_{h, O} = \widehat{U}_{h, \alpha, O} \widehat{Q}_{\alpha, O},$$

with the evident groupoid source restrictions. If a second slice has $s'_O(x) = a_O(x) \cdot s_O(x)$, then

$$\mathcal{R}_{\alpha, O}(z, g') = (z, g' a_O(\kappa_{\alpha, O}(z))), \quad \widehat{Q}_{\alpha, O}^{s'} = \mathcal{R}_{\alpha, O}^* \widehat{Q}_{\alpha, O}^s.$$

The chart carries the independent common-source markings

$$\mathcal{E}_{\alpha, O}^{\text{src}} := \mathbf{\Gamma}_{\alpha, O}^{-1}, \quad \ell_{\alpha, O} := \rho_{\alpha, O}^{-1}.$$

These maps are constructed locally before any transition or descent object.

Lemma 4.3 (Slice-complete common-source chart-presentation theorem). *For every α and $O \subset U_\alpha$, $\mathbf{\Gamma}_{\alpha, O}$ is a bicontinuous presentation isomorphism from (20) onto a chart graph object $\mathbf{V}_{6, \alpha}(O)$, maps $\mathcal{D}_6^{\text{ren}}(O)$ onto the literal domain $\mathcal{D}_{6, \alpha}^{\text{ren}}(O)$, and transports every finite word, proper action, target sort, counterterm, and anomaly coordinate. Its inverse reconstructs the entire saturated raw source, not merely the chosen slice. Changing s_O acts by the explicit $\mathcal{R}_{\alpha, O}^*$ matching above and the already populated proper law (19), not by a WDW-3 refoliation lift.*

Proof. Well-definedness and injectivity were proved before completion. The WDW-2 slice theorem supplies the smooth inverse $\Psi_{\alpha, O}^{-1}$ on the declared free-regular neighborhood. On each compact cap, the chain rule and finite Faà di Bruno formula bound a pulled-back chart seminorm by finitely many seminorms of (20) on its compact image; applying the same argument to $\Psi_{\alpha, O}^{-1}$ gives the reverse estimate. Because the empty-word family tests every (z, g) , no nonzero proper-orbit variation lies in the kernel. Completion therefore gives the bicontinuous carrier maps. The domain and every chart operator were defined by exact image and conjugation, so their squares and typed targets are literal. The displayed formula for $\widehat{U}_h$ proves proper-action conjugation. The slice-change formula follows from $g' \cdot s'_O(x) = g' a_O(x) \cdot s_O(x)$; inverse and triple slice changes compose by groupoid associativity. No statement about a finite normal-deformation arrow enters the proof. $\square$

Lemma 4.4 (Evaluation naturality and exact source–domain square). *For every chart and localized source element $F \in V_6^{\text{ren}}(O)$,*

$$\Gamma_{\alpha,O}^{\mathcal{D}} \iota_{6,O} F = \Gamma_{\alpha,O}^{\mathcal{K}} \iota_{6,O} F = \iota_{\alpha,O} \widehat{Q}_{\alpha,O} F. \quad (20a)$$

Moreover, for every $i \in \mathfrak{I}_6$,

$$\widehat{C}_{\alpha,\rho_{\alpha,O}(i)}^{\times} \iota_{\alpha,O} \widehat{Q}_{\alpha,O} F = \iota_{\alpha,O} \widehat{C}_{\alpha,\rho_{\alpha,O}(i)}^{\text{src}} \widehat{Q}_{\alpha,O} F.$$

Thus source equations, their evaluation images, and arbitrary transposed-kernel elements remain three distinct typed levels.

Proof. W6R.S15 gives $\mathcal{D}_6^{\text{ren}}(O) = \mathcal{K}_6^{\text{ren}}(O)$ and its chart counterpart, so the evaluation image lies in the literal common domain. For every $\Lambda \in V_{6,\alpha}^{\text{ren}}(O)'$,

$$((\widehat{Q}'_{\alpha,O})^{\times} \iota_{6,O} F)(\Lambda) = \iota_{6,O} F(\widehat{Q}'_{\alpha,O} \Lambda) = (\widehat{Q}'_{\alpha,O} \Lambda)(F) = \Lambda(\widehat{Q}_{\alpha,O} F) = (\iota_{\alpha,O} \widehat{Q}_{\alpha,O} F)(\Lambda).$$

This proves (20a); the $\mathcal{D}$ equality is the restriction of the same carrier map to the exact domain. The final square follows from the W6R.S15 extension identity $\widehat{C}_{\alpha,i}^{\times} \iota_{6,O} = \iota_{6,O} \widehat{C}_{\alpha,i}^{\text{src}}$ and the two conjugation identities above. $\square$

Theorem 4.5 (Constraint-native carrier exhaustion). *Within the populated fixed-target WDW-6 comparison class, the lawful local off-shell carrier is the one common support $[\mathbf{V}_6(O)]_{\sim_6}$ together with its chart presentations. A fixed-lapse grammar on a proper-basic scalar source is excluded because it freezes a standing-bearing descriptor contradicted by (19). A label-erased quotient is excluded because it loses Reference and fixed-domain identity. Added bundle or sheaf data have no standing unless definitionally induced from (20). AASC does not create a refoliation action, a chart map, or an occupant.*

Proof. WDW-5/6 positively populate (20), while the flow box and the WDW-2 proper-action trivialization positively populate each slice-complete $\Gamma_{\alpha,O}$. WDW-3 proves that a fixed $\widehat{H}[N]$ is not basic when a presentation changes; the proper covariance law (19) retains the full descriptor family. Hence fixed-label and label-erased candidates fail Admissibility, Standing, or Reference before comparison with the survivor. AASC supplies only the exhaustive same-scope disposition. $\square$

4.4 Intrinsic-clock noncharacteristicity

Let $V_{\star} = \text{Vol}(h_{\star})$ and $\text{Av}_{\star} f = V_{\star}^{-1} \int_{\Sigma} f d\mu_{h_{\star}}$. For a metric variation q , the normalized coordinates (15a) have anchor differentials

$$dT_{\nu,\star}[q] = \frac{1}{6} \text{Av}_{\star}(\text{tr}_{h_{\star}} q), \quad dT_{R,\star}[q] = 2 \text{Av}_{\star}(\text{tr}_{h_{\star}} q) = 12 dT_{\nu,\star}[q].$$

The second equality uses (13), $\text{Ric}(h_{\star}) = -2h_{\star}$, and integration of the divergence terms on compact boundaryless Σ . Both covectors lie on the nonzero parallel pure-trace line. For the accepted inverse DeWitt tensor,

$$\mathcal{G}_{ijkl}(h_{\star}) h_{\star}^{ij} h_{\star}^{kl} = -\frac{3}{2\sqrt{h_{\star}}} \neq 0.$$

This calculation must be made in the accepted *renormalized* operator, not in the obstructed unrenormalized heat family. The two covectors are smooth L^2 covector-densities, so a chart cylinder $f(T_{\alpha}, \chi_{\alpha})$ has a rank-one clock Hessian in the W6R.S12 finite-separated channel. W6R.S12 retains the ordinary regulator-free trace of that channel,

$$\mathfrak{p}_{\star} := 2\kappa \text{Tr}_{L^2}(M_{\mathbf{1}_{\mathbb{K}_A}}(h_{\star})(dT_{\nu,\star} \otimes dT_{\nu,\star})) < 0.$$

The inequality is the displayed pure-trace DeWitt contraction. Gate-II local counterterms replace the nonpositive local heat-coefficient column; W6R.6 and W6R.S12 explicitly keep finite-rank and separated Hessian channels in the regular finite part, so no counterterm coefficient enters $\mathfrak{p}_{\star}$. Because this retained second-configuration-differential block is linear in the lapse and quadratic in the clock covector, while $N_{\nu,\star} = c_{\epsilon_{\star}}^{-1} \mathbf{1}$ and $N_{R,\star} = (12c_{\epsilon_{\star}})^{-1} \mathbf{1}$, its two clock-normal coefficients satisfy

$$a_{\nu,\star} = c_{\epsilon_{\star}}^{-1} \mathfrak{p}_{\star} \neq 0, \quad a_{R,\star} = 12a_{\nu,\star} \neq 0.$$

The scalar coefficient $a_{\Phi,\star} \neq 0$ is the accepted W7.4 coefficient on the inherited scalar chart.

Lemma 4.6 (Uniform intrinsic-clock noncharacteristicity). *After shrinking only inside the fixed certified radius pairs, each coefficient a_{α} is smooth and obeys*

$$|a_{\alpha}(z) - a_{\alpha,\star}| < \frac{1}{2} |a_{\alpha,\star}|, \quad |a_{\alpha}(z)| \geq \frac{1}{2} |a_{\alpha,\star}| > 0$$

on U_{α} . Hence multiplication by a_{α}^{-1} is continuous in every admitted graph seminorm.

Proof. W6R.S12 and W6R.S12a make the retained finite-separated trace Bastiani smooth in the configuration, kinetic multiplier, lapse, and smooth L^2 covector coordinates. The response inverse theorems make $z \mapsto N_\alpha(z) = L_{\alpha,z}^{-1} \mathbf{1}$ smooth. The displayed regulator-free anchor calculations give nonzero values. Continuity supplies the quantitative half-value neighborhood; shrinking the already fixed boxes to it preserves every prior guard. Smooth reciprocal calculus on functions bounded away from zero gives the last assertion. $\square$

4.5 Cauchy traces and companion factorization

Define the source trace first, and only then its evaluation-image presentation:

$$\begin{aligned} J_{\alpha,x}^{\text{src}}(\widehat{Q}_{\alpha,O}F) &= \left(\widehat{F}_\alpha|_{T_\alpha=t, \chi_\alpha=0}, \partial_{T_\alpha} \widehat{F}_\alpha|_{T_\alpha=t, \chi_\alpha=0} \right), & \widehat{F}_\alpha &= \widehat{Q}_{\alpha,O}F, \\ J_{\alpha,x}^{\text{ev}}(\Gamma_{\alpha,O}^{\mathcal{D}} \iota_{6,O}F) &:= J_{\alpha,x}^{\text{src}}(\widehat{Q}_{\alpha,O}F). \end{aligned} \quad (21)$$

Equation (20a) makes the second line single-valued. The function-valued physical gauge sets $\chi_\alpha = 0$, while the full proper-arrow coordinate g and every other non-profile tangential source coordinate remain in both components. In particular, (21) does not evaluate at $g = e$ and does not collapse a non-basic source to one proper slice. Write $J_{\alpha,x} = (J_{\alpha,x}^{\text{src}}, J_{\alpha,x}^{\text{ev}})$ for this typed pair; its two entries do not share the full transposed carrier as a domain.

For a proper-slice change $s'_O = a_O \cdot s_O$, let $\dot{\mathcal{R}}_{a,\alpha,t}^*$ denote the T_α derivative of the smooth family $\mathcal{R}_{a,O}^*$ at the trace surface, holding its argument fixed. Differentiating the already proved identity $\widehat{F}_\alpha^{s'} = \mathcal{R}_{a,O}^* \widehat{F}_\alpha^s$ gives the complete Cauchy presentation map

$$\mathcal{J}_{a,\alpha,t} := \begin{pmatrix} \mathcal{R}_{a,O}^* & 0 \\ \dot{\mathcal{R}}_{a,\alpha,t}^* & \mathcal{R}_{a,O}^* \end{pmatrix}, \quad J_{\alpha,t}^{\text{src},s'} \widehat{F}_\alpha^{s'} = \mathcal{J}_{a,\alpha,t} J_{\alpha,t}^{\text{src},s} \widehat{F}_\alpha^s. \quad (21a)$$

This triangular map is invertible; its inverse is obtained from a^{-1} and the ordinary derivative-of-inverse formula. For successive slice changes, the product rule gives $\mathcal{J}_{b,a \cdot s} \mathcal{J}_{a,s} = \mathcal{J}_{ba,s}$.

Lemma 4.7 (Proper-slice Cauchy naturality). *The map (21a) is bicontinuous in every declared Cauchy graph seminorm, transports the authenticated source-solution doublet and its generated line, and obeys inverse, triple, and quadruple proper-presentation coherence. It is a presentation map and not a WDW-3 normal-deformation lift.*

Proof. Smoothness of a_O , $\mathcal{R}_{a,O}$, and the local proper action makes both diagonal entries and $\dot{\mathcal{R}}_{a,\alpha,t}^*$ finite sums of registered proper-action and chart differential words with smooth coefficients. The compact-cap graph estimates therefore prove continuity, and the same argument for a^{-1} proves bicontinuity. Equation (21a) applies to every source element, hence to the localized authenticated source solution and its span. The displayed product rule and WDW-2 groupoid associativity give the higher comparisons. $\square$

Lemma 4.8 (Three-clock graph-smooth Cauchy trace). *For $\alpha \in \{\Phi, \nu, R\}$, restriction and first T_α derivative in (21) are continuous on the declared chart germ source, smooth in the readout, and bounded in every finite-word graph seminorm by finitely many seminorms of the common support (20). They therefore restrict continuously to the localized authenticated source-solution orbit and its evaluation image. No extension to an arbitrary element of $\mathcal{K}_6^{\text{ren}}(O)$ or $\mathcal{Z}_6^\times(O)$ is asserted. No higher normal derivative is required after the constraint is reapplied. Its change under a different proper slice is exactly (21a).*

Proof. For Φ this is W7.2–W7.4 restricted to the common support. For ν and R , composition with the smooth inverse flow box and the full proper-action trivialization, restriction to a closed cap of $\mathcal{P}_{\alpha,O} \cap \{\chi_\alpha = 0\}$, and application of the smooth vector field ∂_{T_α} at fixed g give the same compact-open estimates. Conjugating a finite WDW-6 word through $\widehat{Q}_{\alpha,O}$ produces a finite sum of chart differential and proper-translation words with smooth coefficients. The chain rule and finite Faà di Bruno formula therefore bound every requested derivative by finitely many inherited word seminorms on the compact image. The preceding lemma permits division by a_α uniformly. Reapplying the second-order constraint expresses $\partial_{T_\alpha}^2$ through the value, first normal derivative, and tangential words, proving the final assertion exactly as in W7.4. $\square$

Let $\mathfrak{X}_z(N)$ denote the WDW-3 normal characteristic vector at z with lapse N . The exact response inverse supplies

$$N_\alpha(z) = L_{\alpha,z}^{-1} \mathbf{1}, \quad X_\alpha^{\text{nor}}(z) = \mathfrak{X}_z(N_\alpha(z)), \quad \mathcal{N}_{\alpha,z} = \text{span}\{X_\alpha^{\text{nor}}(z)\}.$$

The flow box gives $D\kappa_\alpha(\partial_{T_\alpha}) = X_\alpha^{\text{nor}}$, $dT_\alpha(X_\alpha^{\text{nor}}) = 1$, and $d\chi_\alpha(X_\alpha^{\text{nor}}) = 0$.

Let $\bar{i}_\alpha^{\text{nor}} \in \mathfrak{J}_6$ be the common-source Hamiltonian label whose lapse entry is N_α and whose other coordinates are the imported WDW-6 regulator, density, ordering, connection, polarization, counterterm, and target entries. Put $i_\alpha^{\text{nor}} = \rho_\alpha(\bar{i}_\alpha^{\text{nor}}) \in \mathfrak{J}_\alpha$. Its chart presentation has the proved noncharacteristic form

$$\hat{\mathcal{H}}_{\alpha,O}^{\text{ren}} = -\hbar^2 a_\alpha \partial_{T_\alpha}^2 + B_\alpha \partial_{T_\alpha} + \Theta_\alpha, \quad C_\alpha^{\text{cmp}} = \hbar^{-2} a_\alpha^{-1} \Theta_\alpha, \quad D_\alpha^{\text{cmp}} = \hbar^{-2} a_\alpha^{-1} B_\alpha. \quad (22a)$$

On the localized source joint kernel and, by (20a), its evaluation image,

$$\partial_{T_\alpha} J_\alpha^{\text{src}} F_\alpha = A_\alpha^{\text{cmp}} J_\alpha^{\text{src}} F_\alpha, \quad A_\alpha^{\text{cmp}} = \begin{pmatrix} 0 & I \\ C_\alpha^{\text{cmp}} & D_\alpha^{\text{cmp}} \end{pmatrix}. \quad (22b)$$

Direct division of the constraint fixes (22b) before the connection is defined. Every tangential word and counterterm remains in B_α or Θ_α .

Lemma 4.9 (Common-normal core lemma). *On $\mathcal{C}_\star$ the independently constructed normal lines coincide:*

$$\mathcal{N}_{\Phi,z} = \mathcal{N}_{\nu,z} = \mathcal{N}_{R,z} = \mathcal{L}_{\star,z} := \text{span}\{\mathfrak{X}_z(\mathbf{1})\}, \quad z \in \mathcal{C}_\star. \quad (23)$$

Off $\mathcal{C}_\star$ no equality is asserted.

Proof. Along the homogeneous source interval, $K = kh$, the scalar velocity is spatially constant, and intrinsic scalar curvature is constant. Each exact response maps the constant-lapse subspace to the constant-profile subspace with nonzero coefficient. Since the full response is invertible, the unique solution of $L_{\alpha,z} N = \mathbf{1}$ is spatially constant. Thus every $X_\alpha^{\text{nor}}(z)$ is a nonzero multiple of $\mathfrak{X}_z(\mathbf{1})$. Symmetry and uniqueness on the full lapse space prove the claim. $\square$

4.6 Joint kernel, physical line, and local endpoint

Define separately the source joint kernel, its evaluation image, and the full transposed joint kernel:

$$\mathcal{Z}_6^{\text{src}}(O) = \bigcap_{i \in \mathfrak{J}_6} \ker \hat{C}_{O,i}^{\text{src}} \subset V_6^{\text{ren}}(O), \quad \mathcal{Z}_6^{\text{ev}}(O) = \iota_{6,O} \mathcal{Z}_6^{\text{src}}(O), \quad \mathcal{Z}_6^\times(O) = \bigcap_{i \in \mathfrak{J}_6} \ker \hat{C}_{O,i}^\times.$$

The WDW-6 extension identity $\hat{C}_{O,i}^\times \iota_{6,O} F = \iota_{6,O} \hat{C}_{O,i}^{\text{src}} F$ proves $\mathcal{Z}_6^{\text{ev}}(O) \subseteq \mathcal{Z}_6^\times(O)$; it does not identify the two kernels. Their chart images are

$$\mathcal{Z}_{6,\alpha}^{\text{src}}(O) := \hat{Q}_{\alpha,O} \mathcal{Z}_6^{\text{src}}(O), \quad \mathcal{Z}_{6,\alpha}^\times(O) := \Gamma_{\alpha,O}^\mathcal{K} \mathcal{Z}_6^\times(O), \quad \mathcal{Z}_{6,\alpha}^{\text{ev}}(O) := \Gamma_{\alpha,O}^\mathcal{D} \mathcal{Z}_6^{\text{ev}}(O) = \iota_{\alpha,O} \mathcal{Z}_{6,\alpha}^{\text{src}}(O).$$

The two conjugation identities and bijectivity of $\rho_{\alpha,O}$ prove

$$\mathcal{Z}_{6,\alpha}^{\text{src}}(O) = \bigcap_{j \in \mathfrak{J}_{\alpha,O}} \ker \hat{C}_{\alpha,j}^{\text{src}}, \quad \mathcal{Z}_{6,\alpha}^\times(O) = \bigcap_{j \in \mathfrak{J}_{\alpha,O}} \ker \hat{C}_{\alpha,j}^\times.$$

This is chart transport from one already populated kernel, not a refoliation-covariance theorem.

Lemma 4.10 (Localized authenticated source-kernel lemma). *Let $F_{A,\mu}^{(0)}$ be the accepted normalized WDW-6 source solution and let O be a source-attached representative neighborhood containing the anchor. Its restriction class*

$$F_{A,\mu,O}^{(0)} := [F_{A,\mu}^{(0)}]_{\tilde{O}}$$

belongs to $\mathcal{Z}_6^{\text{src}}(O)$, has $F_{A,\mu,O}^{(0)}(q_\star) = 1$, and therefore is nonzero. Consequently $\iota_{6,O} F_{A,\mu,O}^{(0)} \in \mathcal{Z}_6^{\text{ev}}(O) \subseteq \mathcal{Z}_6^\times(O)$.

Proof. The accepted source solution obeys $\hat{C}_i^{\text{src}} F_{A,\mu}^{(0)} = 0$ for every complete label. Equation (17) therefore gives $\hat{C}_{O,i}^{\text{src}} F_{A,\mu,O}^{(0)} = 0$ in the raw restriction quotient, and continuity preserves that equality in $V_6^{\text{ren}}(O)$. Since $q_\star \in \tilde{O}$, restriction preserves the accepted normalization value one, so the class is not zero. The last inclusion is exactly the W6R.S15 extension identity, not equality of the source and transposed kernels. $\square$

Define

$$\begin{aligned} Y_{\alpha,x} &= J_{\alpha,x}^{\text{src}}(\hat{Q}_{\alpha,O} F_{A,\mu,O}^{(0)}) \\ &= J_{\alpha,x}^{\text{ev}} \Gamma_{\alpha,O}^\mathcal{D} \iota_{6,O} F_{A,\mu,O}^{(0)} \neq 0, \end{aligned} \quad \text{Cond}_{\alpha,x}[u] = I_6([u]) Y_{\alpha,x}, \quad \mathcal{H}_{\alpha,x} = \mathbb{C} Y_{\alpha,x}. \quad (24)$$

The value component at the anchor is the accepted normalization $F_{A,\mu,O}^{(0)}(q_\star) = 1$, so the displayed Cauchy doublet is nonzero; continuity supplies the already certified local nonvanishing interval. Transport the physical form and represented scalar algebra through $\text{Cond}_{\alpha,x}$. On the chart-normal line set

$$\nabla_{\alpha,X_\alpha^{\text{nor}}}^{\mathcal{N}_\alpha} = \partial_{T_\alpha} - A_\alpha^{\text{cmp}}. \quad (25)$$

Equation (22b) makes (25) preserve the generated line. Its restriction to (23) is $\nabla_{\alpha|\star}^{\mathcal{L}_\star}$. No transverse connection is claimed.

The local record is the complete source record at the certified profile intersection with the chart readout retained separately. The first-failure object is

$$\text{Fail}_\alpha = \min \left\{ \begin{array}{l} F_{\text{radius}}, F_{\text{profile}}, F_{\text{intersection}}, F_{\text{raw ideal}}, F_{\text{orbit presentation}}, F_{\text{label}}, \\ F_{\text{source marking}}, F_{\text{noncharacteristic}}, F_{\text{germ}}, F_{\text{source}}, F_J, \\ F_{\text{grammar}}, F_{\text{common domain}}, F_{\text{joint kernel}}, F_{\text{normal core}}, \\ F_{\text{form}}, F_{\text{record}}, F_{\text{continuation}}, F_{\leq 9} \end{array} \right\}. \quad (26)$$

The minimum is lineage order, and localization never resets a failed prefix.

Theorem 4.11 (W10.9 — slice-complete common-source local quantum realization and constraint-normal physical line). *For each $\alpha \in \{\Phi, \nu, R\}$, equations (16)–(26) construct*

$$\mathfrak{R}_\alpha(U_\alpha) = (C_\alpha, U_\alpha, \mathbf{V}_{6,\alpha}, \mathcal{D}_{6,\alpha}^{\text{ren}}, J_\alpha, \mathcal{H}_\alpha, \text{Cond}_\alpha, \text{Conn}_\alpha, \rho_\alpha, \mathcal{A}_\alpha^{\text{phys}}, \text{Rec}_\alpha, \text{Cont}_\alpha, \text{Fail}_\alpha, \text{Anchor}_\alpha, \text{Cert}_\alpha) \in \text{PhysRec}_{10}(U_\alpha).$$

Every chart is a bicontinuous, proper-orbit-complete conjugated presentation of one common localized WDW-6 support and retains the total label action, typed targets and counterterms, literal common domain, graph-continuous and proper-slice-natural Cauchy trace, source, evaluation, and transposed joint kernels, authenticated physical line, noncharacteristic normalized companion, constraint-derived normal connection, the evaluation-naturality square, common-normal restriction, records, failures, and nonclaims. No quantum implementation of a WDW-3 refoliation arrow is asserted or required. For the scalar chart, the independently defined singleton enrichment gives $\mathfrak{R}_\Phi^{10,\text{loc},\text{mark}} := \text{Sing}_\Phi(\mathfrak{R}_\Phi(U_\Phi)) \in \text{EnrPhysRec}_{10}^{\Phi\text{-mark}}(U_\Phi)$ and a canonical local comparison

$$\epsilon_\Phi^{\text{loc}} : \text{Red}_9(\mathfrak{R}_\Phi^{10,\text{loc},\text{mark}}) \xrightarrow{\cong} \mathfrak{R}_\Phi^9(U_\Phi). \quad (26a)$$

Proof. W10.4–W10.6 supply the inherited scalar package and independently constructed analytic profile data; W10.8 supplies the germ cover. The ideal-invariance lemmas localize the actual source. Equation (20) is the accepted common WDW-6 support; the chart-presentation theorem constructs each local grammar by conjugation. The retained finite-rank DeWitt calculation proves the two new clock-normal coefficients nonzero before division, the evaluation-naturality lemma types the authenticated source orbit, and the three-clock trace lemma supplies the required graph estimates without extending pointwise operations to arbitrary dual elements. Equations (22)–(25) then construct the companion, physical line, and connection. All antecedents precede W10.10 and global descent. For $\alpha = \Phi$, the construction retains the imported WDW-9 coordinates literally and adds only the successor presentation certificates required by (5). The functor Sing_Φ supplies the identity atlas, identity incidence, transition, coherence, holonomy, and gluing coordinates required by (6a), so the marked lift is an actual object of the printed reduct domain rather than a bare pair. Applying Red_9 deletes precisely those added coordinates and leaves the imported tuple componentwise fixed. The identity maps on its thirteen roles, twenty overlaps, repairs, failures, boundary data, and nonclaims therefore assemble to (26a). This is a comparison in Cert_9 , not an inverse that reconstructs the enriched scalar object from bare WDW-9 data. $\square$

The WDW-7 exclusion of scalar-valued clock substitutes remains intact. T_ν and T_R are analytic normal coordinates; the physical gauges are the function-valued equations (7) and (8). This construction does not claim a quantum refoliation lift, a many-fingered connection, a common star envelope in transverse directions, equality of the three normal fields off $\mathcal{C}_\star$, or the operator-flat Tomonaga–Schwinger extension left open by W7.14.

5. Common-source transitions, complete preservation, and coherence

5.1 Actual overlaps and normal-line base maps

For $\alpha, \beta \in \{\Phi, \nu, R\}$ let $O_{\alpha\beta} \subset U_\alpha \cap U_\beta$ be a source-attached representative neighborhood of $\mathcal{C}_*$; two choices represent the same germ after a further shrink. Both chart packages are restrictions of the same localized object $\mathbf{V}_6(O_{\alpha\beta})$. Their base/readout relation is

$$b_{\alpha\beta} = \kappa_\alpha^{-1} \kappa_\beta, \quad t_{\alpha\beta} : c_\beta(x) \mapsto c_\alpha(x), \quad x \in \mathcal{C}_*. \quad (27)$$

It is the coordinate change at one common source event, not a WDW-3 quantum evolution map and not equality of clock values.

Choose $X_*(z) = \mathfrak{X}_z(\mathbf{1})$ on $\mathcal{L}_*$. In the normalized convention put

$$\lambda_\alpha(z) = dT_\alpha(X_*(z)) = \langle L_{\alpha,z} \mathbf{1} \rangle_{w_\alpha} \neq 0, \quad X_\alpha^{\text{nor}} = \lambda_\alpha^{-1} X_*, \quad j_{\alpha\beta} = \frac{\lambda_\alpha}{\lambda_\beta}. \quad (28)$$

At the anchor $(\lambda_\Phi, \lambda_\nu, \lambda_R) = (\epsilon_*, c_{\epsilon_*}, 12c_{\epsilon_*})$. The common-normal lemma gives

$$Db_{\alpha\beta}(X_\beta^{\text{nor}}) = j_{\alpha\beta} X_\alpha^{\text{nor}}$$

on $\mathcal{C}_*$. Off that lineage residual-profile components may occur and no line-intertwining claim is made.

5.2 Independent physical maps and common-source graph maps

Define on the actual overlap

$$\begin{aligned} T_{\alpha\beta,x} &= \text{Cond}_{\alpha,b_{\alpha\beta}x} \text{Cond}_{\beta,x}^{-1}, \\ G_{\alpha\beta,O}^V &= \widehat{Q}_{\alpha,O} \widehat{Q}_{\beta,O}^{-1}, \\ G_{\alpha\beta,O}^\mathcal{K} &= \Gamma_{\alpha,O}^\mathcal{K} (\Gamma_{\beta,O}^\mathcal{K})^{-1}, \quad G_{\alpha\beta,O}^\mathcal{D} = \Gamma_{\alpha,O}^\mathcal{D} (\Gamma_{\beta,O}^\mathcal{D})^{-1}, \\ \mu_{\alpha\beta,O} &= \rho_{\alpha,O} \rho_{\beta,O}^{-1}, \quad \mathbf{G}_{\alpha\beta,O} = (G_{\alpha\beta,O}^V, G_{\alpha\beta,O}^\mathcal{K}, G_{\alpha\beta,O}^\mathcal{D}, \mu_{\alpha\beta,O}), \quad O = O_{\alpha\beta}. \end{aligned} \quad (29)$$

The first inverse is used only on the no-spectator physical range. Every other inverse is the inverse of an independently formed chart presentation. In particular,

$$\widehat{Q}_{\alpha,O}^{-1} G_{\alpha\beta,O}^V = \widehat{Q}_{\beta,O}^{-1}, \quad \mathcal{E}_{\alpha,O}^{\text{src}} \mathbf{G}_{\alpha\beta,O} = \mathcal{E}_{\beta,O}^{\text{src}}, \quad \ell_{\alpha,O} \mu_{\alpha\beta,O} = \ell_{\beta,O}.$$

These common-source marking squares precede descent and will supply rigidity.

For every $i \in \mathfrak{I}_{\beta,O}$ and every type-correct finite word ω ,

$$\begin{aligned} G_{\alpha\beta,O}^V \widehat{C}_{\beta,i}^{\text{src}} &= \widehat{C}_{\alpha,\mu_{\alpha\beta}(i)}^{\text{src}} G_{\alpha\beta,O}^V, \\ G_{\alpha\beta,O}^\mathcal{K} \widehat{C}_{\beta,i}^\times &= \widehat{C}_{\alpha,\mu_{\alpha\beta}(i)}^\times G_{\alpha\beta,O}^\mathcal{D}, \\ G_{\alpha\beta,O}^\mathcal{K} \widehat{C}_{\beta,\omega}^\times &= \widehat{C}_{\alpha,\mu_{\alpha\beta}^*(\omega)}^\times G_{\alpha\beta,O}^\mathcal{D}. \end{aligned} \quad (30)$$

This follows by substituting the source and transposed conjugation definitions through the corresponding common operators indexed by $\ell_\beta(i)$. Target sorts and counterterm/anomaly columns are the two coordinate presentations of that same abstract label. Equation (20a) also gives the transition naturality square

$$G_{\alpha\beta,O}^\mathcal{D} \iota_{\beta,O} = \iota_{\alpha,O} G_{\alpha\beta,O}^V.$$

Together with the physical algebra, common record, continuation, and failure maps, these give

$$\Theta_{\alpha\beta} = (b_{\alpha\beta}, T_{\alpha\beta}, \mathbf{G}_{\alpha\beta}, A_{\alpha\beta}, R_{\alpha\beta}, K_{\alpha\beta}, F_{\alpha\beta}).$$

Theorem 5.1 (W10.10 — common-source transition population). *Every ordered pair of seed charts carries the displayed complete germ map on its actual overlap. The physical component is formed from independently populated conditioning maps. The graph and label components factor through the independently constructed common support and markings. Before descent,*

$$\Theta_{\alpha\alpha} = \text{id}, \quad \mathbf{G}_{\beta\alpha} = \mathbf{G}_{\alpha\beta}^{-1} = ((G_{\alpha\beta}^V)^{-1}, (G_{\alpha\beta}^\mathcal{K})^{-1}, (G_{\alpha\beta}^\mathcal{D})^{-1}, \mu_{\alpha\beta}^{-1}), \quad \Theta_{\beta\alpha} = \Theta_{\alpha\beta}^{-1}.$$

Proof. W10.9 supplies exact physical ranges, the one common graph support, and all three chart presentations. Cancellation of adjacent conditioning maps gives the physical inverse. Cancellation of adjacent $\widehat{Q}$, Γ , and ρ maps in (29) gives the source, carrier, domain, and label inverses. Base maps are inverse coordinate presentations of one source event; records, continuation, and failures are defined on the source pullback. No cocycle, global object, or refoliation action is used. $\square$

5.3 Complete domain, grammar, and joint-kernel preservation

Lemma 5.2 (Complete common-source graph-domain preservation). *For every overlap representative O ,*

$$\langle T_{\alpha\beta}u, T_{\alpha\beta}v \rangle_\alpha = \langle u, v \rangle_\beta, \quad (31)$$

$$G_{\alpha\beta, O}^{\mathcal{D}} \mathcal{D}_{6, \beta}^{\text{ren}}(O) = \mathcal{D}_{6, \alpha}^{\text{ren}}(O). \quad (32)$$

Moreover the compatible $G_{\alpha\beta, O}$ pair is bicontinuous for the complete graph topologies, (30) transports every constraint kind, typed target, product, commutator, regulator, counterterm, and anomaly identity, and

$$G_{\alpha\beta, O}^{\mathcal{K}} \mathcal{Z}_{6, \beta}^\times(O) = \mathcal{Z}_{6, \alpha}^\times(O), \quad G_{\alpha\beta, O}^V \mathcal{Z}_{6, \beta}^{\text{src}}(O) = \mathcal{Z}_{6, \alpha}^{\text{src}}(O), \quad G_{\alpha\beta, O}^{\mathcal{D}} \mathcal{Z}_{6, \beta}^{\text{ev}}(O) = \mathcal{Z}_{6, \alpha}^{\text{ev}}(O).$$

Proof. Equation (31) follows because both conditioning maps push forward the same accepted WDW-6 physical form. Equation (32) is immediate from the two exact domain images in (29). Bicontinuity follows from bicontinuity of the chart maps and their inverses. Equation (30) is the source/transpose double-conjugation identity. Finally, the source kernels are $\widehat{Q}$ -images of the same $\mathcal{Z}_6^{\text{src}}$, the transposed kernels are $\Gamma^{\mathcal{K}}$ -images of the same $\mathcal{Z}_6^\times$, and the authenticated evaluation kernels are $\Gamma^{\mathcal{D}}$ -images of the same $\mathcal{Z}_6^{\text{ev}}$. Substituting (29) gives both equalities in both directions. No label-surjectivity argument is used to manufacture a missing kernel transport. $\square$

5.4 Normal labels and companion intertwining

Let $i_\alpha^{\text{nor}} = \rho_\alpha(\bar{i}_\alpha^{\text{nor}})$ be the normalized Hamiltonian chart label defined before (22a). On $\mathcal{C}_\star$ the common-source label markings obey

$$\ell_\beta(i_\beta^{\text{nor}}) = \text{Scale}_{j_{\alpha\beta}}(\ell_\alpha(i_\alpha^{\text{nor}})). \quad (33)$$

Indeed $N_\beta = \lambda_\beta^{-1} \mathbf{1} = j_{\alpha\beta} \lambda_\alpha^{-1} \mathbf{1} = j_{\alpha\beta} N_\alpha$ there. Equation (33) is an equality between two members of the one common label family, not the output of an unconstructed label map.

Differentiating the independently constructed identity $T_{\alpha\beta}Y_\beta = Y_\alpha \circ b_{\alpha\beta}$ along X_β^{nor} and using the two normalized constraint equations yields the typed generated-line identity

$$j_{\alpha\beta} A_\alpha^{\text{cmp}} Y_\alpha = (D_{X_\beta^{\text{nor}}} T_{\alpha\beta}) Y_\beta + T_{\alpha\beta} A_\beta^{\text{cmp}} Y_\beta. \quad (34)$$

Here every term is evaluated at the corresponding overlap point. The derivative term is forced by the base-dependent transition. Equation (34) does not promote $T_{\alpha\beta}$ to an unconstructed endomorphism of the full Cauchy carrier. Consequently, on the common-normal core and for generated-line sections,

$$\nabla_{\alpha, D b_{\alpha\beta} X_\beta^{\text{nor}}}^{\mathcal{L}_\star} (T_{\alpha\beta} \xi) = T_{\alpha\beta} \nabla_{\beta, X_\beta^{\text{nor}}}^{\mathcal{L}_\star} \xi, \quad \nabla_{\alpha, X_\alpha^{\text{nor}}}^{\mathcal{L}_\star} (T_{\alpha\beta} \xi) = j_{\alpha\beta}^{-1} T_{\alpha\beta} \nabla_{\beta, X_\beta^{\text{nor}}}^{\mathcal{L}_\star} \xi. \quad (35)$$

Theorem 5.3 (W10.11 — complete preservation and normal intertwining). *Every $\Theta_{\alpha\beta}$ preserves separately: the physical form; source and graph bicontinuity; exact common domains; evaluation-embedding naturality; total label-transporting covariance for all four constraint kinds; typed targets, counterterms, and anomaly data; the source, authenticated evaluation-image, and transposed joint kernels; the common-source markings; the normalized lapse relation; the generated-line companion identity and common-normal connection; the represented physical algebra; records; continuation; failures; boundary coordinates; and nonclaims. No off-core normal equality, scalar refoliation descent of the off-shell grammar, quantum refoliation lift, or transverse connection is asserted.*

Proof. The graph-domain lemma proves the carrier, domain, grammar, and joint-kernel coordinates; the transition naturality square following (30) proves compatibility of the primal and common-domain levels. The marking squares following (29) prove common-source fidelity. Equation (33) proves the normalized descriptor relation before (34)–(35) prove the companion and connection laws. The physical-algebra identity follows from the transported representations. Common-source evaluation and accepted history restriction prove the record and continuation squares; ordered failures are retained rather than reset. $\square$

5.5 Triple and higher coherence

On a triple overlap represented by $O \subset U_\alpha \cap U_\beta \cap U_\gamma$,

$$\begin{aligned} b_{\alpha\beta} b_{\beta\gamma} &= b_{\alpha\gamma}, & G_{\alpha\beta, O}^V G_{\beta\gamma, O}^V &= G_{\alpha\gamma, O}^V, \\ G_{\alpha\beta, O} G_{\beta\gamma, O} &= G_{\alpha\gamma, O}, & \mu_{\alpha\beta} \circ \mu_{\beta\gamma} &= \mu_{\alpha\gamma}, \\ \mathbf{G}_{\alpha\beta, O} \mathbf{G}_{\beta\gamma, O} &= \mathbf{G}_{\alpha\gamma, O}. \end{aligned} \quad (36)$$

These are literal cancellations of the middle κ_β , $\hat{Q}_\beta$, Γ_β , and ρ_β maps. The proper-arrow coordinate is part of each $\hat{Q}_\beta$ presentation and is therefore cancelled, not discarded. The same equations hold for finite-word labels and typed targets. The Jacobians obey $j_{\alpha\gamma} = j_{\alpha\beta}j_{\beta\gamma}$; differentiating the physical cocycle gives the affine chain rule for (34). Joint-kernel transport is path independent because all transposed chart kernels are images of the same $Z_6^\times(O)$, all source kernels are images of the same $Z_6^{\text{src}}(O)$, and all authenticated evaluation kernels are images of the same $Z_6^{\text{ev}}(O)$.

A different local proper slice inserts the explicit $\mathcal{R}_a^*$ orbit-coordinate map and the accepted WDW-2 proper-presentation natural isomorphism into each Γ_α . The two composites on a -dependent Cauchy doublets are transported by the triangular $\mathcal{J}_{a,\alpha}$ of (21a), whose off-diagonal derivative obeys the product rule. Both composites on a quadruple pullback agree by ordinary groupoid associativity and that product rule. This is the only presentation 2-cell used; no WDW-3 quantum 2-cell is asserted.

Theorem 5.4 (W10.12 — common-source triple and quadruple coherence). *On every occupied triple germ overlap,*

$$\Theta_{\alpha\beta}\Theta_{\beta\gamma} = \Theta_{\alpha\gamma}$$

as a complete PhysRec₁₀ morphism. Carrier maps, total labels, target sorts, finite words, common domains, joint kernels, source markings, normalized labels, companion terms, records, continuation, failures, boundaries, and nonclaims all compose. Proper-slice presentation and Cauchy-doublet isomorphisms obey quadruple coherence; hence no probe-visible higher obstruction precedes descent.

Proof. Conditioning maps cancel on their exact ranges. The common-source factorizations in (29) give both the G and μ equations in (36), so every typed operator, marking, and joint-kernel square composes. Base maps and Jacobians compose by the chain rule; differentiation gives the generated-line affine composition of (34). Common-source evaluation and ordered history/failure transport prove the remaining coordinates. The accepted WDW-2 proper groupoid gives the final presentation-coherence statement. $\square$

Anti-circularity check. W10.5 and W10.6 end with analytic clock data. W10.9 constructs the intrinsic noncharacteristic coefficients, trace estimates, and three chart presentations from one common WDW-6 support. W10.10 factors transitions through those already formed local maps; W10.11 verifies their structures. Neither coherence, a quantum refoliation action, nor $\mathfrak{R}_{10}^{\text{global}}$ appears in a local or transition antecedent.

6. Positive descent, obstruction, and the gluing orbit

6.1 Effective descent of the finite datum

Let $\bigsqcup_\alpha \mathcal{H}_\alpha$ be the disjoint union over the three chart domains. On overlap fibres impose the relation

$$(\beta, x, v) \sim (\alpha, x, T_{\alpha\beta,x}v). \quad (37)$$

The cocycle theorem makes (37) an equivalence relation. Define

$$\mathcal{H}_{10}^{\text{gl}} = \left(\bigsqcup_\alpha \mathcal{H}_\alpha \right) / \sim. \quad (38)$$

The local forms, enriched graph objects, algebra actions, records, continuation arrows, failures, boundary data, and nonclaims are quotiented by their corresponding components of $\Theta_{\alpha\beta}$. W10.11 makes each result independent of the local representative.

In particular, on every representative germ neighborhood O the global source and common domain are formed independently of the physical quotient:

$$\begin{aligned} V_{10}^{\text{gl}}(O) &= \left(\bigsqcup_\alpha V_{6,\alpha}^{\text{ren}}(O) \right) / ((\beta, F) \sim (\alpha, G_{\alpha\beta,O}^V F)), \\ \mathcal{D}_{10}^{\text{gl}}(O) &= \left(\bigsqcup_\alpha \mathcal{D}_{6,\alpha}^{\text{ren}}(O) \right) / ((\beta, \Lambda) \sim (\alpha, G_{\alpha\beta,O}^\mathcal{D} \Lambda)). \end{aligned}$$

The label family is glued simultaneously by $(\beta, i) \sim (\alpha, \mu_{\alpha\beta}(i))$. The enriched cocycle makes both quotients well defined; the restriction squares assemble them into a global connected-proper-equivariant germ-domain package.

Equation (30) descends every registered typed operator word and counterterm as a label-indexed family, not as one fixed-label scalar endomorphism. Likewise, (35) descends the constraint-normal connection exactly along $\mathcal{L}_\star = T\mathcal{C}_\star$; no off-core or transverse connection is introduced by descent. The transition naturality square descends the chart evaluation embeddings to one map $\iota_{10,O} : V_{10}^{\text{gl}}(O) \rightarrow \mathcal{D}_{10}^{\text{gl}}(O)$, and source, evaluation, and full transposed joint kernels descend as three separate subobjects. The source markings give canonical inverse maps from these quotients to the one common localized $V_6^{\text{ren}}(O)$, $\mathcal{D}_6^{\text{ren}}(O)$, and $\mathcal{I}_6$. Thus the quotient is a descent presentation of the already populated support, not a new off-shell carrier.

There is an explicit anchor trivialization

$$Q : \mathcal{H}_{10}^{\text{gl}} \longrightarrow X_{10}^{\text{reg}} \times \mathcal{H}_{\text{phys}}, \quad Q[\alpha, x, \text{Cond}_{\alpha,x} u] = (x, u). \quad (39)$$

It is well defined because (29) preserves u , and it is bijective because every local fibre is exactly a conditioning range. The inverse sends (x, u) to the class of any chart containing x .

Independently of the Hilbert and graph quotients, the populated clock profiles and the readout components of the already constructed transitions define

$$\mathfrak{C}_{10}^{\text{gl}} := \left[(\mathcal{A}_{10}^0, \{C_\alpha\}_\alpha, \{t_{\alpha\beta}\}_{\alpha,\beta}, \{\kappa_{\alpha\beta\gamma}\}, \text{Orient}_{10}, \text{Anchor}_{10}, \text{Fail}_{10}^{\text{clk}}) \right]_{\text{ref}} \in \text{ClockDesc}_{10}(X_{10}^{\text{reg}}). \quad (38a)$$

Here $[\]_{\text{ref}}$ is the lawful-refinement localization defined before W10.3. Identities, inverses, and the triple compatibility of the $t_{\alpha\beta}$ are the readout components of W10.10–W10.12. Equation (38a) retains all three profiles, their anchors, orientations, continuations, and first failures; it is not a function-valued global clock and contains no selected atlas member. For each α , let $\mathcal{B}_\alpha = \{U_\beta \cap U_\alpha \rightarrow U_\alpha\}_\beta$. Its literal restriction map to $j_\alpha^* \mathfrak{C}_{10}^{\text{gl}}$ is a lawful refinement, while the component family $t_{\alpha\beta}$ is a 1-arrow from $\mathcal{B}_\alpha$ to $\text{S}_{\text{clk}}(C_\alpha)$. Its inverse 1-arrow selects the $\beta = \alpha$ member. The unit is identity, and the counit on the β member is $t_{\beta\alpha}$; the inverse and cocycle laws make both triangle composites identity. The transition-supported natural isomorphisms and cocycle of W10.12 therefore give

$$\zeta_\alpha : j_\alpha^* \mathfrak{C}_{10}^{\text{gl}} \xrightarrow{\cong} \text{S}_{\text{clk}}(C_\alpha) \quad \text{in } \text{ClockDesc}_{10}(U_\alpha). \quad (38b)$$

These comparisons are coherent on overlaps. They give local singleton presentations of the neutral clock-descent object without selecting any one of them globally.

Definition 6.1 (Underlying and enriched global realizations). $\mathfrak{R}_{10}^{\text{und}}$ has first coordinate $\mathfrak{C}_{10}^{\text{gl}}$ from (38a), physical bundle (38), and the connected-proper-equivariant germ-domain package. On each O its coordinates are the source $V_{10}^{\text{gl}}(O)$, evaluation map $\iota_{10,O}$, literal domain $\mathcal{D}_{10}^{\text{gl}}(O)$, label family $\mathcal{I}_{10}^{\text{gl}}(O)$, and typed operator family $\widehat{\mathcal{C}}_{10}^{\text{gl}}(O)$. It also contains the descended typed operator grammar, $\mathcal{L}_\star$ constraint-normal connection, form, represented algebra, record, continuation, failure, anchor, and certificate structures. Equivalently it is the object transported through (39) from the common physical anchor and source-history/record sheaves on X_{10}^{reg} , with the clock coordinate kept as (38a). Thus

$$\mathfrak{R}_{10}^{\text{und}} \in \text{NeuPhysRec}_{10}(X_{10}^{\text{reg}}).$$

On every U_α , (38b) together with the corresponding representative maps for (37)–(39) defines a neutral-category isomorphism

$$\xi_\alpha : j_\alpha^* \mathfrak{R}_{10}^{\text{und}} \xrightarrow{\cong} \text{S}_{\text{neu}}(\mathfrak{R}_\alpha) \quad \text{in } \text{NeuPhysRec}_{10}(U_\alpha). \quad (39a)$$

Its clock component is ζ_α ; its graph, physical, algebra, record, continuation, failure, and connection components are the already proved $G_{\alpha\beta}$, $T_{\alpha\beta}$, and corresponding maps of W10.10–W10.12. Their cocycles make the family $\{\xi_\alpha\}$ compatible. Define the enriched object

$$\mathfrak{R}_{10}^{\text{global}} := (\mathfrak{R}_{10}^{\text{und}}, \mathcal{A}_{10}^0, \mathcal{I}_{10}^0, \Theta_{10}^0, \text{Coh}_{10}^0, \text{Hol}_{10}^{\text{raw}}, \{\mathcal{E}^{\text{src}}\}_\alpha, \{\ell_\alpha\}_\alpha, \text{GlueCert}_{10}) \in \text{EnrPhysRec}_{10}(X_{10}^{\text{reg}}), \quad (39b)$$

where the coordinates are respectively the W10.8 cover/incidence data, W10.10 transitions, W10.12 coherence, the raw generating-loop words formed from those transitions, W10.9 inverse source/label markings, and the quotient, independence, and section certificates (37)–(40). The raw loop coordinate is present before, and is only classified by, W10.15. Thus $\text{For}_{10}(\mathfrak{R}_{10}^{\text{global}}) = \mathfrak{R}_{10}^{\text{und}}$ by (6c). The scalar chart is not marked inside the unmarked global object. The atlas of available readouts, common-source markings, label recodings, and successor gluing certificates are literal coordinates of (39b); ordinary domain restriction does not erase them. Predecessor recovery is performed later in $\text{EnrPhysRec}_{10}^{\Phi\text{-mark}}$ by the typed reduct (6d).

Theorem 6.2 (W10.13 — positive clock-neutral global descent). *The finite local datum has effective descent $\mathfrak{R}_{10}^{\text{und}}$ in NeuPhysRec_{10} , and the complete populated datum assembles to the object $\mathfrak{R}_{10}^{\text{global}}$ in EnrPhysRec_{10} displayed in (39b). Its underlying object is nonempty and carries the complete connected-proper-equivariant source/graph package, the three separately typed kernel levels, the explicit clock-descent coordinate (38a), and all physical descended structures. The authenticated family*

$$x \longmapsto [\alpha, x, Y_{\alpha,x}] \quad (40)$$

is a global nowhere-zero compatible section.

Proof. Equivalence of (37) and of the enriched graph/label relations follows from identities, inverses, and W10.12. W10.11 proves that every structural coordinate is constant on equivalence classes. Equation (38a) supplies the required clock-neutral first coordinate, while (37)–(39) supply every non-clock coordinate of (5b). Hence $\mathfrak{R}_{10}^{\text{und}}$ is an actual NeuPhysRec_{10} object rather than either a formal stack label or an ill-typed single-clock object. Equations (38b) and (39a) supply its compatible local singleton presentations. W10.8, W10.10, W10.12, and the quotient construction supply every coordinate of (39b), so W10.3 types the assembled record in EnrPhysRec_{10} . For (40), $T_{\alpha\beta}Y_\beta = Y_\alpha$ by (29), so local values glue. They are nonzero by W10.9. Hence both the underlying descent object and its enriched record are positively populated. $\square$

6.2 Global form, algebra, record, and neutrality

For $[\alpha, x, v], [\alpha, x, w]$ in one global fibre, set

$$\langle [v], [w] \rangle_{10, x} = \langle v, w \rangle_{\alpha, x}. \quad (41)$$

Unitarity makes this well defined and positive. The global represented algebra is $\rho_{10}(a)[\alpha, x, v] = [\alpha, x, \rho_\alpha(a)v]$. The complete record is the common source record with the atlas of available readouts attached; continuation and failure descend similarly.

Proposition 6.3 (Global nondegeneracy). *The descended form has zero radical, the global section (40) has unit norm, and the global physical–record probe separates the admitted object. More precisely, a difference of fibre vectors is detected by the transported physical form and canonical coordinate; a difference of retained source events, histories, record germs, continuations, or failures is detected by the corresponding complete record coordinate.*

Proof. Under (39), (41) is the positive WDW-6 form, so its radical is zero and (40) corresponds to the normalized authenticated physical class. A rank-one fibre has one physical ray, but unequal vectors remain separated by the form/canonical coordinate. The base is the source-attached quotient-history locus with complete record, continuation, and failure data retained; its equality is exactly equality under those admitted coordinates. Therefore the product probe separates the declared object without claiming a larger observable algebra. $\square$

Theorem 6.4 (W10.14 — global nondegeneracy and separation). *For $_{10}(\mathfrak{R}_{10}^{\text{global}}) = \mathfrak{R}_{10}^{\text{und}}$ has a positive physical form, well-defined complete connected-proper-equivariant successor graph/domain and label action, descended typed operator grammar and counterterms, an $\mathcal{L}_\star$ constraint-normal connection, represented algebra, nonzero physical section, complete record, continuation, and irreversible failure object. Its admitted physical–record probe is separating at the frozen target.*

Proof. Apply the preceding proposition and the structural descent in W10.13. Separation is relative to the exact rank-one physical and source-record theorem object; no unconstructed state or observable is added. $\square$

Clock neutrality is now literal: (38a) is a refinement class of compatible local profiles, while (38)–(41) contain no globally preferred readout. A local clock is used to present a fibre and its valid domain, while (39) shows that the physical element is independent of that use. The representative atlas, its three profiles, relations, and failures remain visible both in $\mathfrak{C}_{10}^{\text{gl}}$ and in the enriched coordinates; neutrality does not erase real local plurality.

6.3 Obstructions and holonomy

Let $N(\mathcal{A}_{10}^0)$ be the nerve of the occupied cover and let $\text{Aut}_{\mathcal{P}_{10}}$ be the groupoid of local automorphisms preserving the complete probe. Define

$$\Omega_{\alpha\beta\gamma} = \Theta_{\alpha\gamma}^{-1} \Theta_{\alpha\beta} \Theta_{\beta\gamma}, \quad (42)$$

$$\text{Hol}(\gamma) = \Theta_{\alpha_0\alpha_1} \cdots \Theta_{\alpha_{m-1}\alpha_m}, \quad \alpha_m = \alpha_0. \quad (43)$$

All ordered pairwise and triple intersections are occupied, so the nerve is the full two-simplex on three vertices. Contractibility alone will not be used.

Lemma 6.5 (Direct defect computation). *Every triple defect (42) and every loop transport (43) acts identically on the physical anchor, carrier, complete label action, typed graph, algebra, records, continuation, failures, boundary data, and nonclaims.*

Proof. For the Hilbert component, substitute (29) around the word. Every adjacent conditioning inverse cancels on its exact generated range, leaving the identity. The common-source factorizations cancel both Γ and ρ around the word; equation (30), common-source record evaluation, and functorial continuation then give the same cancellation

in all other components. The failure coordinate is an ordered retained source coordinate, so a loop returns the same coordinate without resetting it. This is a calculation on actual intersections; no good-cover or cohomology shortcut is invoked. $\square$

A change of local frame $Y_\alpha \mapsto e^{i\theta_\alpha} Y_\alpha$ also changes the displayed conditioning map and transitions by the corresponding coboundary. It leaves (39)–(41), the normalized source class, records, and all complete-probe values fixed, so it is presentation skin. A phase inserted only around a loop would change the anchored section or the holonomy coordinate and is not an admissible frame change.

Theorem 6.6 (W10.15 — obstruction and holonomy classification). *For the finite seed atlas, every existence defect is trivial and every loop holonomy is either literal identity or a complete-probe-neutral frame coboundary. No physical holonomy or theorem-object-changing repair is present. The obstruction/holonomy ledger is therefore fully classified as TRIVIAL or PRESENTATION – SKIN.*

Proof. The direct defect lemma gives literal identity in the source-anchored trivialization. The only changes between local displayed frames are the coboundaries just classified. Any automorphism changing a form, carrier, label action, graph, observable, record, continuation, failure, source normalization, boundary, or nonclaim is detected by $\mathcal{P}_{10}$ and is not in the lawful presentation groupoid. This exhausts the actual finite-loop basis and its composites. $\square$

6.4 Common-source rigidity and uniqueness of the gluing

Lemma 6.7 (Evaluation and source-marking rigidity). *Let A be a bicontinuous automorphism of one localized common graph support (20). If A fixes every empty-word point-evaluation probe on $\Phi_{6,O}^{\text{tr}}$, every abstract label under the common label marking, and every registered operator square, then A is the identity on the completed carrier, continuous dual, and literal common domain. On the generated physical line, an isometry fixing*

$$Y_\alpha = \text{Cond}_\alpha(I_6^{-1}1)$$

is also the identity.

Proof. Empty-word evaluations separate the raw restriction quotient by construction of $\mathcal{I}_{\text{raw}}(\tilde{O})$. Equivalently, in every chart they test every (z, g) because $\Psi_{\alpha,O}$ is onto; evaluation at $g = e$ alone is not used. Hence $AF = F$ on the dense initial non-basic graph source. Bicontinuity extends this equality to its completion. The continuous-dual map is then the transpose identity, while exact-domain preservation and the registered operator squares make the domain restriction the identity. The common label marking is a bijection to $\mathfrak{I}_6$, so a label automorphism fixing it is identity. Finally every vector on the physical line is cY_α ; a complex-linear isometry fixing Y_α fixes the whole line. Thus neither a hidden graph centralizer nor a phase survives the complete probe. $\square$

Let $\tilde{\Theta}_{\alpha\beta}$ be any complete same-target transition family for the already populated local objects that passes W10.11 and preserves the independently formed common-source markings. The source-anchoring square requires

$$\tilde{T}_{\alpha\beta} \text{Cond}_\beta = \text{Cond}_\alpha.$$

Since Cond_β is onto its no-spectator range, $\tilde{T}_{\alpha\beta} = T_{\alpha\beta}$. For the graph and labels, lawfulness requires

$$\mathcal{E}_\alpha^{\text{src}} \tilde{\mathbf{G}}_{\alpha\beta} = \mathcal{E}_\beta^{\text{src}}, \quad \ell_\alpha \tilde{\mu}_{\alpha\beta} = \ell_\beta.$$

Multiplying by the inverse markings gives $\tilde{\mathbf{G}}_{\alpha\beta} = \Gamma_\alpha \Gamma_\beta^{-1} = \mathbf{G}_{\alpha\beta}$ and $\tilde{\mu}_{\alpha\beta} = \rho_\alpha \rho_\beta^{-1} = \mu_{\alpha\beta}$. Equivalently, the rigidity lemma excludes a postcomposition centralizer. The algebra, record, continuation, and failure conditions then fix the remaining components, up to probe-neutral changes of displayed proper slice.

Theorem 6.8 (W10.16 — one lawful gluing orbit). *Every same-target gluing of the populated finite datum, including its carrier and complete label action, is equivalent to (37) under the maximal lawful presentation congruence. Hence*

$$|\text{Glue}_{10}^{\text{law}}(\mathcal{A}_{10}^0)/\Gamma_{10}^{\text{law}}| = 1. \quad (44)$$

This result follows from common-anchor surjectivity, the independently constructed source markings, and evaluation rigidity, not from W6.15, the number of charts, or the rank-one dimension alone.

Proof. The preceding rigidity argument fixes every standing-bearing component carried by a lawful gluing. W10.15 removes only displayed proper-slice/frame changes that the complete probe cannot see. Thus every live gluing lies in the orbit of the constructed one. W10.13 supplies that live element, proving the quotient is a singleton rather than empty. $\square$

7. Refinement, universality, and multiple-choice closure

7.1 Refinement invariance

Let $r : \mathcal{A}' \rightarrow \mathcal{A}$ be a lawful refinement. Restrict every local object and transition to the subordinate domains. The resulting equivalence relation on the refined disjoint union maps to the old relation by inclusion. Both quotient objects have the physical trivialization (39) and the same common-source graph/label descent package.

Lemma 7.1 (Common-source-cover comparison span). *Let $\mathcal{A} = \{U_i, C_i, \dots\}$ and $\mathcal{B} = \{V_j, D_j, \dots\}$ be lawful same-target atlases whose cross-atlas transition components have been positively constructed before descent by the W10.10–W10.12 formulas and satisfy their preservation and coherence laws. Put*

$$W_{ij} = U_i \times_{X_{10}^{\text{pr}}} V_j.$$

On this common source cover form two clock data

$$\mathcal{A}_W = \{W_{ij}, C_i|_{W_{ij}}, \dots\}, \quad \mathcal{B}_W = \{W_{ij}, D_j|_{W_{ij}}, \dots\}.$$

The arrows $r_{\mathcal{A}} : \mathcal{A}_W \rightarrow \mathcal{A}$ and $r_{\mathcal{B}} : \mathcal{B}_W \rightarrow \mathcal{B}$ are literal lawful refinements. The components

$$u_{\mathcal{B}\mathcal{A},ij} = (t_{ji}, T_{ji}, \mathbf{G}_{ji}, \dots)|_{W_{ij}} : \mathcal{A}_W|_{W_{ij}} \longrightarrow \mathcal{B}_W|_{W_{ij}}$$

define a transition-supported arrow $u_{\mathcal{B}\mathcal{A}} : \mathcal{A}_W \rightarrow \mathcal{B}_W$, not a refinement. Consequently

$$\omega_{\mathcal{B}\mathcal{A}} = \left[\mathcal{A} \xleftarrow{r_{\mathcal{A}}} \mathcal{A}_W \xrightarrow{r_{\mathcal{B}} \circ u_{\mathcal{B}\mathcal{A}}} \mathcal{B} \right] \quad (44a)$$

is a clock-descent comparison roof. Reversing the transition components gives $\omega_{\mathcal{A}\mathcal{B}}$, and these roofs are mutually inverse.

Proof. W10.2 makes $\{W_{ij}\}$ jointly surjective and preserves every failed prefix. Pulling back $\mathcal{A}$ and $\mathcal{B}$ separately therefore makes $r_{\mathcal{A}}$ and $r_{\mathcal{B}}$ lawful; it does not equate $C_i|_{W_{ij}}$ with $D_j|_{W_{ij}}$. The cross-atlas formulas of W10.10–W10.11 supply t_{ji} , T_{ji} , $\mathbf{G}_{ji}$, and every record, anchor, continuation, failure, source, domain, label, and marking component on the same W_{ij} . W10.12 supplies their inverse components and the identity and cocycle 2-cells, so

$$u_{\mathcal{A}\mathcal{B}}u_{\mathcal{B}\mathcal{A}} \simeq \text{id}_{\mathcal{A}_W}, \quad u_{\mathcal{B}\mathcal{A}}u_{\mathcal{A}\mathcal{B}} \simeq \text{id}_{\mathcal{B}_W}.$$

After the literal refinement legs are inverted in ClockDesc_{10} , these are the unit and counit of the two roofs. Thus (44a) is an isomorphism without a clock datum that literally refines both atlases. $\square$

Theorem 7.2 (W10.17 — atlas refinement and comparison-span independence). *Restriction along a lawful refinement induces a canonical isomorphism*

$$\text{Desc}_{\text{NeuPhysRec}_{10}}(\mathcal{A}') \cong \text{Desc}_{\text{NeuPhysRec}_{10}}(\mathcal{A}) \quad (45)$$

preserving every underlying standing-bearing coordinate. The corresponding enriched records are related by the componentwise refinement arrows (6b); they become equivalent under the declared lawful-refinement presentation congruence, without claiming that a nonbijective atlas subdivision is an isomorphism in EnrPhysRec_{10} . Any two lawful same-target atlases with the W10.10–W10.12 cross-transition package are related by the mutually inverse comparison roofs (44a). Only their underlying source covers are commonly refined; the two pulled-back clock data remain distinct.

Proof. The clock-descent coordinates on the two sides of (45) are isomorphic by the refinement roof that is inverted in ClockDesc_{10} . Their remaining coordinates map by (39) to $X_{10}^{\text{reg}} \times \mathcal{H}_{\text{phys}}$ and to the same WDW-6 localized connected-proper-equivariant successor object, with the same complete labels, domains, source-history, grammar, target sorts, counterterms, common-normal connection, form, algebra, record, continuation, and failure structures. The maps agree on each refined local piece and hence define the canonical isomorphism. Surjectivity of the refined cover prevents loss of a base event, and failure preservation prevents loss of a failed prefix. For two atlases, the common-source-cover comparison lemma supplies separate data $\mathcal{A}_W$ and $\mathcal{B}_W$. Applying (45) to $r_{\mathcal{A}}$ and $r_{\mathcal{B}}$ and inserting $u_{\mathcal{B}\mathcal{A}}$ gives

$$\text{Desc}(\mathcal{A}) \cong \text{Desc}(\mathcal{A}_W) \xrightarrow{\text{Desc}(u_{\mathcal{B}\mathcal{A}})} \text{Desc}(\mathcal{B}_W) \cong \text{Desc}(\mathcal{B}).$$

The same transition-supported span carries the atlas, incidence, transitions, coherence, raw loop actions, inverse markings, and gluing certificate componentwise as required by (6b). Its reverse is the inverse by W10.12. This proof does not require either atlas to contain the labels Φ, ν, R . $\square$

7.2 Universal mapping property

For $\mathfrak{Z} \in \text{NeuPhysRec}_{10}(X_{10}^{\text{reg}})$, define the set of compatible local arrows

$$\text{Cone}_{\mathcal{A}}(\mathfrak{R}, \mathfrak{Z}) = \left\{ (f_{\alpha})_{\alpha} :: f_{\alpha} : \mathcal{S}_{\text{neu}}(\mathfrak{R}_{\alpha}) \rightarrow \mathfrak{Z}|_{U_{\alpha}}, \right. \\ \left. f_{\alpha} \mathcal{S}_{\text{neu}}(\Theta_{\alpha\beta}) = f_{\beta} \text{ on } U_{\alpha\beta} \right\}. \quad (46)$$

The variance is explicit: arrows go from the local/glued realization into $\mathfrak{Z}$.

Theorem 7.3 (W10.18 — clock-neutral universality). *Restriction induces a natural bijection*

$$\text{Hom}_{\text{NeuPhysRec}_{10}(X)}(\mathfrak{R}_{10}^{\text{und}}, \mathfrak{Z}) \xrightarrow{\cong} \text{Cone}_{\mathcal{A}_{10}^0}(\mathfrak{R}, \mathfrak{Z}) \quad (47)$$

The forward map is $f \mapsto (f|_{U_{\alpha}} \circ \xi_{\alpha}^{-1})_{\alpha}$. for every same-target $\mathfrak{Z}$. The inverse sends a compatible family to the unique arrow

$$f[\alpha, x, v] = f_{\alpha}(x, v).$$

The same property holds for every lawful atlas carrying that independently constructed cross-transition package. It is invariant under literal lawful refinement by (45), and it is transported between distinct same-target clock data by the comparison roofs (44a), up to the maximal lawful presentation congruence.

Proof. A global neutral arrow restricts along the canonical local singleton presentations ξ_{α} of (39a) by the map printed in (47), and its restrictions satisfy (46). Conversely, if two local representatives are related by (37), compatibility in (46), including the readout component in $\mathcal{S}_{\text{neu}}(\Theta_{\alpha\beta})$, gives the same image; hence the displayed formula is well defined. It preserves every structure because each f_{α} does so locally. The local pieces cover, so two global arrows with the same restrictions agree. This proves bijectivity and naturality. W10.17 transports the statement along literal refinements and transition-supported comparison roofs to arbitrary lawful atlases. Presentation 2-cells act trivially after the complete probe, giving the stated qualified uniqueness. $\square$

The dual mapping-in statement follows by applying the same proof to the opposite component categories, but it is not used to claim the endpoint. Equation (47), not the adjective “universal,” is the exact universality theorem.

7.3 Arbitrary admissible charts

Call a chart C *same-target admissible* when, before any global comparison, it has:

1. a function-valued profile on the frozen source/configuration target with proper-spatial covariance;
2. an invertible full lapse-response on a certified residual-copy-free neighborhood and unique guarded intersections there;
3. the complete clock-use anchor, readout, record, continuation, and first-failure data;
4. a nonzero chart-normal Cauchy doublet of the accepted authenticated WDW-6 solution;
5. a saturated raw restriction ideal for the actual WDW-6 successor grammar; separate invariance under local differential generators and finite proper-redescription actions; the complete WDW-5/6 label family; the common localized WDW-6 support with its Hausdorff graph topology and full common domain; a bicontinuous complete chart presentation $\mathbf{\Gamma}_{C,O}$ and independently formed inverse source marking on every germ neighborhood; a direct noncharacteristic principal-coefficient calculation and graph-smooth Cauchy estimate; the exact no-spectator conditioning range and pushed-forward form; the represented algebra; and an independently constraint-derived chart-normal connection whose restriction to the declared source lineage is proved to equal the common-normal distribution; and
6. every frozen WDW-9 non-clock, boundary, recovery, and nonclaim coordinate.

These are local antecedents. No transition or global-object equality is included.

For any two such charts C, D , their physical transition is forced by $\text{Cond}_C \text{Cond}_D^{-1}$, their graph transition on an overlap O by $\mathbf{\Gamma}_{C,O} \mathbf{\Gamma}_{D,O}^{-1}$, their label map by $\rho_{C,O} \rho_{D,O}^{-1}$, and their common-normal connection law by the derivative of the actual base map together with the two independently derived companion terms. Therefore any set-sized atlas of same-target admissible charts has the same effective descent proof wherever its domains cover the declared base.

Theorem 7.4 (W10.19 — multiple-choice closure). *Every clock/atlas alternative generated on the frozen theorem object has exactly one of the following statuses:*

1. it passes the same-target local antecedents and descends, through any lawful cover, to a neutral underlying object and enriched record canonically equivalent to $\mathfrak{R}_{10}^{\text{global}}$;
2. it changes only a complete-probe-neutral presentation label, orientation-preserving readout, frame, or refinement and is presentation skin;
3. it fails a local, overlap, coherence, or physical-domain antecedent and is fail-closed at the first loss;
4. it carries probe-visible holonomy or plurality, which is retained as CLASSIFIED – PLURAL; or
5. it changes source, matter, gauge, topology, boundary, operator family, physical form, or use target and is a richer theorem object.

At the constructed target the fourth status is unoccupied by W10.15, so no same-target global multiple-choice ambiguity remains. No preferred clock is selected.

Proof. The Physical Clock Profile Finality and Pair Normal-Form theorems first separate profile equivalence, eligibility/failure, anchor divergence, and theorem-object divergence. Passing candidates satisfy the six local antecedents above; the common-source chart maps and common-anchor factorization then construct the label-preserving transitions, and W10.12–W10.18 give descent and atlas-independent equivalence through the explicit common-source-cover comparison roofs. Probe-visible holonomy cannot be renamed skin and is assigned the fourth status. The cases are ordered and exhaustive at fixed target. W10.15 shows that the constructed atlas has no such holonomy. Selection of one local chart for calculation is governed separately by the Clock-Selection Governance theorem and adds no global authority. $\square$

Remark 7.5 (Why this is not uniqueness by definition). The admissible-chart syntax contains only local, source, domain, source-marking, and physical-use tests. Transition factorization is a theorem using the onto conditioning maps and the independently constructed chart embeddings. The evaluation-rigidity lemma, rather than W6.15, excludes a hidden graph centralizer. Coherence, descent, holonomy, refinement, and the mapping property are later consequences. A candidate with a second probe-visible gluing would survive the local syntax and force CLASSIFIED – PLURAL; it is not excluded by terminology.

8. Exact WDW-9 recovery and the complete failure boundary

8.1 Marked enriched restriction and conservative predecessor recovery

Let $i_\Phi : U_\Phi \hookrightarrow X_{10}^{\text{reg}}$ be the inherited scalar-chart inclusion and let $i_\Phi^{*,\text{enr}}$ be the restriction functor defined after (6b). The object $i_\Phi^{*,\text{enr}}\mathfrak{R}_{10}^{\text{global}}$ contains the restricted underlying descent object and every restricted coordinate of (39b), initially as an unmarked object of $\text{EnrPhysRec}_{10}(U_\Phi)$. Before any predecessor comparison, form $\text{GlueCert}_{10}^\Phi$ by replacing each restricted quotient representative $[\alpha, x, v]$ with $T_{\Phi\alpha, x}v$ and each graph/label representative with $(G_{\Phi\alpha}, \mu_{\Phi\alpha})$; W10.12 makes these formulas representative independent. This is the scalar presentation of the already constructed gluing certificate and uses no WDW-9 equality. We now spell out the $\alpha = \Phi$ instance of (38b). Put $\mathcal{B}_\Phi = \{U_\alpha \cap U_\Phi \rightarrow U_\Phi\}_\alpha$. Literal restriction gives a lawful refinement

$$r_\Phi : \mathcal{B}_\Phi \longrightarrow i_\Phi^* \mathfrak{C}_{10}^{\text{gl}}.$$

The already constructed readout transitions give a different kind of arrow

$$u_\Phi : \mathcal{B}_\Phi \longrightarrow \text{S}_{\text{clk}}(C_\Phi), \quad (u_\Phi)_\alpha = t_{\Phi\alpha}.$$

It is not called a refinement: for $\alpha = \nu, R$ it is genuine clock change. The inverse components $t_{\alpha\Phi}$ arise as the counit of the inverse 1-arrow that selects the Φ member of $\mathcal{B}_\Phi$; the unit is identity. The identity/inverse/triple laws of W10.10–W10.12 supply a two-sided inverse to u_Φ up to the transition-supported natural isomorphisms defined before W10.3. The roof (r_Φ, u_Φ) and that inverse therefore define an isomorphism

$$\zeta_\Phi : i_\Phi^* \mathfrak{C}_{10}^{\text{gl}} \xrightarrow{\cong} \text{S}_{\text{clk}}(C_\Phi) \quad \text{in } \text{ClockDesc}_{10}(U_\Phi). \quad (47a)$$

This is a comparison of clock-descent presentations on the scalar locus, not a choice of a global clock and not a relabeling of clock change as refinement; the source datum in (47a) still retains the restricted Φ, ν, R representatives and their failures.

Put $m_\Phi^{\text{loc}} = (a_\Phi, \text{id}_{\text{S}_{\text{neu}}(\mathfrak{R}_\Phi)})$ and define $\mathfrak{R}_\Phi^{10, \text{mark}}$ to be the complete marked scalar presentation

$$\begin{aligned} \mathfrak{R}_\Phi^{10, \text{mark}} := & (\text{S}_{\text{neu}}(\mathfrak{R}_\Phi(U_\Phi)), \mathcal{A}_{10}^0|_{U_\Phi}, \mathcal{I}_{10}^0|_{U_\Phi}, \\ & \Theta_{10}^0|_{U_\Phi}, \text{Coh}_{10}^0|_{U_\Phi}, \text{Hol}_{10}^{\text{raw}}|_{U_\Phi}, \\ & \{\mathcal{E}_\alpha^{\text{src}}\}|_{U_\Phi}, \{\ell_\alpha\}|_{U_\Phi}, \text{GlueCert}_{10}^\Phi; m_\Phi^{\text{loc}}) \in \text{EnrPhysRec}_{10}^{\Phi\text{-mark}}(U_\Phi). \end{aligned}$$

It has the singleton-neutral scalar object as its underlying presentation and the same restricted atlas, incidences, transitions, coherence, holonomy, markings, and gluing certificates as the source object. Thus it is neither the bare local object $\mathfrak{R}_\Phi(U_\Phi)$ nor a WDW-9 certificate.

Restricting (38) to U_Φ , retaining the scalar marking, and choosing the scalar representative gives an enriched map

$$\begin{aligned} \eta_\Phi^{10} : i_\Phi^{*,\text{enr}} \mathfrak{R}_{10}^{\text{global}} &\longrightarrow \mathcal{U}_{\text{mark}}(\mathfrak{R}_\Phi^{10,\text{mark}}), \\ \text{For}_{10}(\eta_\Phi^{10}) &= \xi_\Phi, \\ (\eta_\Phi^{10})^{\text{clk}} &= \zeta_\Phi, \\ \eta_\Phi^{10}[\alpha, x, v] &= T_{\Phi\alpha, x}v, \\ \eta_\Phi^{10,D}[\alpha, \Lambda, i] &= [G_{\Phi\alpha}^D \Lambda, \mu_{\Phi\alpha}(i)]. \end{aligned} \tag{48}$$

As an arrow (6b), (48) has underlying neutral arrow whose clock component is (47a) and whose remaining components are the displayed scalar-representative maps. It has atlas map $\sigma = \text{id}$, identity local arrows on the unchanged restricted atlas, identity incidence, transition, coherence, holonomy, source-marking, and label-marking components, and the induced gluing-certificate comparison; its underlying component is the displayed scalar-representative map on the physical, graph, record, continuation, failure, algebra, and $\mathcal{L}_*$ connection coordinates. Every square required in (6b) is respectively a transition square from W10.11, a cocycle square from W10.12, or the representative-independence square in GlueCert_{10} . Hence (48) is an arrow in the printed category, and the cocycle theorem makes its underlying formula independent of the representative. The inverse uses ζ_Φ^{-1} on the clock-descent coordinate, the scalar inclusion on the non-clock underlying coordinates, and identities on the same successor coordinates. The two composites are identity componentwise in $\text{EnrPhysRec}_{10}(U_\Phi)$.

Now define the source comparison marking from the already proved map, not conversely:

$$m_\Phi^{\text{gl}} := (a_\Phi, \text{For}_{10}(\eta_\Phi^{10})). \tag{48'}$$

This gives $(i_\Phi^{*,\text{enr}} \mathfrak{R}_{10}^{\text{global}}; m_\Phi^{\text{gl}}) \in \text{EnrPhysRec}_{10}^{\Phi\text{-mark}}(U_\Phi)$. The target marking has comparison $\text{id}_{S_{\text{neu}}(\mathfrak{R}_\Phi)}$, the local scalar component of (48) is $\text{id}_{\mathfrak{R}_\Phi}$, and hence its marking square is $\text{id}_{\text{For}_{10}(\eta_\Phi^{10})} = S_{\text{neu}}(\text{id}_{\mathfrak{R}_\Phi}) \text{For}_{10}(\eta_\Phi^{10})$. Thus (48) and its inverse upgrade to an isomorphism in $\text{EnrPhysRec}_{10}^{\Phi\text{-mark}}(U_\Phi)$. The comparison marking ties the reduct to the underlying descended object; it is not an isomorphism from bare WDW-9 data to the enriched restriction.

Applying Red_9 to $\mathfrak{R}_\Phi^{10,\text{mark}}$ forgets its successor atlas, transition, coherence, holonomy, marking, and gluing coordinates, leaving exactly the reduct of $\mathfrak{R}_\Phi^{10,\text{loc},\text{mark}}$. Consequently (26a) extends, by identity on every retained predecessor coordinate, to

$$\epsilon_\Phi : \text{Red}_9(\mathfrak{R}_\Phi^{10,\text{mark}}) \xrightarrow{\cong} \mathfrak{R}_\Phi^9(U_\Phi) = \mathfrak{R}_9^*.$$

Apply the conservative reduct to (48) and then this comparison:

$$\eta_\Phi^9 := \epsilon_\Phi \circ \text{Red}_9(\eta_\Phi^{10}) : \text{Red}_9(i_\Phi^{*,\text{enr}} \mathfrak{R}_{10}^{\text{global}}; m_\Phi^{\text{gl}}) \longrightarrow \mathfrak{R}_\Phi^9(U_\Phi) = \mathfrak{R}_9^*. \tag{48a}$$

The final equality is the W10.4 publication import on its inherited scalar locus. Equation (6e) shows that the construction is compatible with every further source restriction. No successor coordinate is silently dropped before Red_9 is applied.

Certificate 8.1 (Thirteen-role comparison). The recovery export compares the following role coordinates, not merely the Hilbert line.

ID	WDW-9 role	Value under η_{Φ}^9
F1	Kernel/admissibility	Same fixed act, report, four kernel roles, and no-repair status
F2	Spacetime transport/connection	Same Levi–Civita branch, derivative, curvature, and guards
F3	Covariant Einstein/source	Same nonvacuum EKG history, attachment, Ward/exchange data
F4	Canonical carrier/boundary	Same metric–scalar Cauchy carrier, symplectic form, constraints, reconstruction
F5	Spatial-diffeomorphism descent	Same proper action, moment zero locus, quotient, and descended data
F6	Internal gauge/Gauss	Same typed empty internal-label family
F7	Refoliation/HDA	Same HDA, characteristic plaque, finite route, continuation, and failure
F8	Configuration differential/representation	Same Sobolev chart, calculus, test algebra, CCR and regulated grammar
F9	Ordering/regulator/domain	Same ordering, heat regulator, counterterms, graph completion, and domain
F10	Anomaly/closure	Same zero-theta closure and anomaly response
F11	Physical form/observable	Same normalized state, form, radical, rank-one quotient, and algebra
F12	Clock/conditional dynamics	Same $\phi(\cdot) = \tau\mathbf{1}$ doublet, line, connection, propagator, and record use
F13	Record/continuation/reconciliation	Same lineage, RPI, exchange, cubes, five errors, failures, and nonclaims

Certificate 8.2 (Twenty-overlap comparison). For E01–E20, η_{Φ}^9 preserves respectively: connection/source; recovered connection; covariant/ canonical Cauchy data; comparator/source; spatial moment map; empty internal slot; constraint generators; quotient/characteristic compatibility; quantum configuration descent; empty internal representation; classical principal symbol; operator grammar, complete constraint labels, counterterms, graph topology, and full common domain; regulator/anomaly endpoint; joint kernel/images; physical conditioning; physical anchor/recovery separation; clock/lineage/record; characteristic continuation; source-attached clock history; and canonical recovery/source response. Each equality is the same WDW-9 base equality, because W10.4 imported it literally, (48) acts by the W10.10 source-preserving transition, and Red_9 retains every predecessor grammar/domain coordinate before ϵ_{Φ} compares it.

The restriction specification includes total assignment-typing coordinates: compatibility is defined on arbitrary coproduct-valued assignments, and the final assignment space remains the image-surjective subspace of the unchanged eligible, antecedent-faithful compatible pre-assignment space.

Theorem 8.3 (W10.20 — exact WDW-9 recovery under conservative reduct). *The map (48a) is an isomorphism of complete WDW-9 certificate objects:*

$$\boxed{\text{Red}_9(i_{\Phi}^{*,\text{enr}}\mathfrak{R}_{10}^{\text{global}}; m_{\Phi}^{\text{gl}}) \cong \mathfrak{R}_9^*}. \quad (49)$$

It preserves all thirteen roles, twenty overlaps, source lineage, response realization, assignment typing, failures, boundary data, nonclaims, and $G_9 = \text{PASS}$. Thus WDW-10 is a strict strengthening of the published endpoint rather than a parallel reconstruction. No isomorphism $i_{\Phi}^{,\text{enr}}\mathfrak{R}_{10}^{\text{global}} \cong \mathfrak{R}_9^*$ across different signatures is claimed.*

Proof. W10.13 supplies the enriched object (39b), while the explicit component calculation (48) identifies its unmarked enriched restriction with the complete singleton-neutral scalar presentation. The independent roof comparison (47a) supplies the clock-descent component of that arrow; it neither selects a global profile nor follows from the desired WDW-9 recovery. Equation (48') then forms the source comparison marking from that independent isomorphism and verifies the marking square, so (48) becomes a marked enriched isomorphism. Functoriality on arrows sends this isomorphism to an isomorphism in Cert_9 . W10.9 supplies $\epsilon_{\Phi}^{\text{loc}}$, and the displayed definition above extends it to ϵ_{Φ} after the successor-only coordinates are removed by Red_9 ; W10.4 identifies its codomain with the final WDW-9 realization, not a newly solved approximation. The two certificates above exhaust the WDW-9 standing-bearing inventory, and transition preservation fixes every listed coordinate. Their

composition is (48a), proving (49). The scalar inclusion is the inverse of η_{Φ}^{10} only; no inverse from $\mathfrak{R}_9^*$ to the enriched restriction is asserted or constructed. $\square$

The scalar chart is marked only in (48)–(49). Unmarked atlas automorphisms need not fix its label, and the global definition does not use Φ as a selector.

8.2 Ordered first-failure algebra

For any local object, overlap, atlas, or gluing, apply the first matching disposition:

Code	Status	Trigger
D0	OUT – OF – SCOPE	Source, matter, gauge, topology, boundary, operator, form, or use target changes
D1	ILL – TYPED	A profile, pullback, label map, target sort, graph, record, or cocycle is undefined on its claimed domain
D2	LOCAL – INELIGIBLE	Transversality, unique intersection, raw localization, or first-failure test fails
D3	UNPOPULATED	No authenticated source occupant exists
D4	PHYSICAL – DOMAIN – FAIL	Common-source marking, noncharacteristic coefficient, Cauchy restriction, joint kernel, graph domain, form, or algebra fails
D5	NONOVERLAP	Only clock-range overlap, or no common source/physical domain
D6	NONINTERTWINING	A carrier, label, target, domain, or complete transition law fails
D7	NONCOHERENT	A triple or higher defect is probe visible
D8	OBSTRUCTED	No global object realizes the local datum
D9	PHYSICAL – HOLONOMY	A loop changes standing-bearing global data
D10	ATLAS – DEPENDENT	Lawful refinements yield unclassified inequivalent global objects
D11	RECOVERY – MISMATCH	The reduct is ill typed, drops a predecessor coordinate, or any WDW-9 coordinate fails (49)
D12	LAWFUL – SKIN	Every earlier test passes and only complete-probe-neutral presentation differs
D13	LIVE	Every required test passes

Lemma 8.4 (Irreversible failure retention). *If a retained lineage receives D2–D11 at an event x , every extension preserving its prefix retains that status at x . A chart valid after x begins a new lineage object and does not rewrite the failed prefix.*

Proof. The complete record includes the ordered source history and first-failure coordinate. Every restriction, transition, descent, and refinement map was proved to preserve it. Removing the coordinate changes the object; replacing the prefix changes identity. Therefore no lawful continuation inside the same object can alter the earlier disposition. $\square$

Theorem 8.5 (W10.21 — no backflow and complete failure boundary). *Clock change and global descent cannot erase an earlier local failure, repair WDW-9, cross a non-clock obstruction, or authorize a richer target. The boundary of the theorem is exactly the first local, overlap, coherence, source, graph, form, recovery, continuation, or predecessor failure. D12 is applied only after D0–D11 have been passed.*

Proof. The irreversible-failure lemma handles retained lineages. W10.1 fixes predecessor coordinates; W10.11, W10.13, and W10.17 preserve them through transitions, descent, and refinement; W10.20 verifies through Red_9 that no predecessor coordinate changed on recovery. It does not classify the lawful presence of successor-only atlas data as a mismatch. The ordered disposition table prevents a later skin or uniqueness label from pre-empting an earlier failure. A non-clock change matches D0 before any atlas test. $\square$

No positive occupation of $(U_\nu \cup U_R) \setminus U_\Phi$ is established here. W10.8 proves a plural atlas with a common occupied neighborhood and global descent on its declared union, but it does not certify crossing the inherited scalar-clock failure surface. Accordingly the endpoint is PASS – GLOBAL, not PASS – GLOBAL – EXTENDED.

9. Candidate exhaustion, complete probe, and endpoint

9.1 Candidate grammar

Let Cand_{10} be the set-sized grammar generated from:

1. the exact inherited scalar chart and its lawful restrictions;
2. source-generated function-valued profiles on the frozen configuration/source target;
3. local clock-use certificates, normalized analytic normal coordinates, saturated raw-localization, the one localized WDW-6 successor germ source/domain and complete label family, bicontinuous proper-orbit-complete chart presentations with inverse common-source and label markings, directly proved noncharacteristic normal coefficients, graph-smooth chart-normal Cauchy doublets, chart-normal connections with proved $\mathcal{L}_\star$ restrictions, and no-spectator conditioning maps;
4. source pullback overlaps and complete PhysRec_{10} morphisms;
5. clock predescent data, refinement roofs, the neutral singleton embedding, and complete NeuPhysRec_{10} morphisms;
6. coherent higher-overlap data, lawful refinements, compatible global gluings, and the conservative WDW-9 reduct on a marked scalar restriction; and
7. presentation labels, orientations, frames, and normal-coordinate complements that add no standing coordinate.

The grammar does not generate new matter, gauge, boundary, topology, operator, form, physical-state, or source coordinates. Proposing one produces D0 rather than a new term of the fixed grammar.

Theorem 9.1 (W10.22 — candidate grammar completeness). *Every generated candidate has exactly one terminal normal form: inherited chart; eligible new source-generated chart; presentation change; failed local clock; failed overlap/gluing; hidden selector; or richer theorem object. Applying D0–D13 in order gives a total and exclusive terminal disposition.*

Proof. Inspect a syntax tree at its first standing-bearing constructor. A target-changing constructor is richer scope. Otherwise all expressions must first type; then a clock node either passes the complete local antecedents or fails. A passing chart is inherited or newly source generated. Atlas nodes then either have actual source overlaps, complete carrier-and-label intertwiners, coherence, descent, holonomy classification, refinement invariance, and recovery under the typed predecessor reduct, or fail at the first missing coordinate. A choice not carried by these nodes is a hidden selector. After all standing tests pass, equality under the complete probe is presentation change; otherwise the candidate is live or classified physical plurality. These tests are ordered binary partitions, so they are total and mutually exclusive. Physical Clock Profile Finality excludes an additional same-target clock-use classifier outside the declared profile; fixed-domain exhaustion excludes a second terminal test on the same coordinates. $\square$

9.2 The complete probe

Let $\mathcal{P}_9$ be the complete predecessor probe on Cert_9 , containing every coordinate listed in W10.3. Define

$$\mathcal{P}_{10} = \mathcal{P}_{\text{local}} \times \mathcal{P}_{\text{clockdesc}} \times \mathcal{P}_{\text{overlap}} \times \mathcal{P}_{\text{incidence}} \times \mathcal{P}_{\text{coherence}} \times \mathcal{P}_{\text{label}} \times \mathcal{P}_{\text{hol}} \times \mathcal{P}_{\text{mark}} \times \mathcal{P}_{\text{recovery}} \times \mathcal{P}_{\text{fail}}. \quad (50)$$

Its coordinates are:

- $\mathcal{P}_{\text{local}}$: profile definition, normalized mean/decomposition, differential, certified radius domain, anchor, readout, saturated raw preimage/ideal, differential and proper-action invariance, localized common successor source, full proper-arrow chart domain, reconstruction diffeomorphism, slice-complete chart map and inverse source marking, proper-translation action, slice-change law, grammar/counterterms, graph topology, full common domain, noncharacteristic coefficient, Cauchy doublet, joint kernel, generated range, physical form, represented algebra, $\mathcal{L}_\star$ constraint-normal connection, record, continuation, and first failure for each chart;

- $\mathcal{P}_{\text{clockdesc}}$: the representative clock cover, all local profiles, readout relations, identity/inverse/triple coherence, anchors, orientations, continuation and first-failure prefixes, lawful refinement roofs, restriction laws, and the link between the clock-descent coordinate of (5b) and the enriched atlas; it distinguishes a changed descent object while retaining presentation-equivalent refinements;
- $\mathcal{P}_{\text{overlap}}$: all seven components of every $\Theta_{\alpha\beta}$ and their domains, with $\mathbf{G} = (G, \mu)$ retained as one enriched component and both common-source marking squares;
- $\mathcal{P}_{\text{incidence}}$: source event, physical anchor, graph, record-history, boundary, and nonclaim equality on every pair and higher pullback;
- $\mathcal{P}_{\text{coherence}}$: every carrier and label cocycle, triple defect, representative 2-cell, and quadruple comparison;
- $\mathcal{P}_{\text{label}}$: $\mathfrak{I}_6$, the chart recodings ρ_α , inverse markings ℓ_α , total $\mu_{\alpha\beta} = \rho_\alpha \rho_\beta^{-1}$ actions, typed targets, literal common domains, counterterm/anomaly columns, common-family normalized-lapse relations, inverse laws, and composition laws;
- $\mathcal{P}_{\text{hol}}$: a generating loop basis and its complete carrier-and-label action;
- $\mathcal{P}_{\text{mark}}$: the inherited scalar index a_Φ , the comparison $\chi_\Phi : \mathfrak{R} \cong \mathbf{S}_{\text{neu}}(\mathfrak{R}_{a_\Phi})$ in NeuPhysRec_{10} , its inverse, its clock-descent component, its enriched restriction law, and every mark-preservation square (6c'); this successor interface remains visible and is not passed through Red_9 ;
- $\mathcal{P}_{\text{recovery}}$ is the typed composite

$$\mathcal{P}_{\text{recovery}} = \mathcal{P}_9 \circ \text{Red}_9 \circ \text{Mark}_\Phi^\chi \circ i_\Phi^{*,\text{enr}},$$

on enriched global objects equipped with a certified scalar-presentation isomorphism χ_Φ ; Mark_Φ^χ attaches (a_Φ, χ_Φ) and fails with D11 if that comparison is absent, and records thirteen WDW-9 roles, twenty overlaps, response realization, assignment typing, failures, nonclaims, and G_9 after reduction; successor-only coordinates remain visible to the other probe blocks; and

- $\mathcal{P}_{\text{fail}}$: every D0–D13 first-disposition coordinate and retained prefix.

Lemma 9.2 (Separation). *On fixed-target candidates, equality under (50) implies equality of every standing-bearing local, overlap, global, recovery, and failure coordinate.*

Proof. The local and incidence blocks determine the single-clock objects and their common bases. The clock-descent block determines the neutral first coordinate and its presentation link to the enriched atlas. The overlap, label, and coherence blocks determine the enriched descent relation. Holonomy distinguishes global carrier or label twists not visible in one patch. The empty-word evaluations at every orbit-complete point (z, g) and source markings additionally exclude an undetected graph centralizer by the rigidity lemma. The mark block fixes the comparison that ties the neutral underlying object to the singleton scalar presentation. Recovery then fixes the complete predecessor signature after Red_9 , while the local, overlap, label, coherence, holonomy, and mark blocks continue to fix the successor-only atlas structure; failure prevents erasure by restriction. These blocks exhaust the local, clock-descent, neutral, and enriched object/arrow definitions (5)–(6e), the descent definitions (37)–(41) and (39b), and the frozen target. A remaining difference is therefore only a name, local frame, orientation-preserving parameterization, or representative complement with all displayed structures fixed. $\square$

Let $\Gamma_{10}^{\text{pres}}$ be the groupoid generated by connected proper presentation arrows, lawful profile reparameterizations, source-normalized local frame changes, analytic complement changes, and cover refinements, each required to fix (50).

Theorem 9.3 (W10.23 — complete probe and maximal lawful skin). *The kernel pair of $\mathcal{P}_{10}$ is exactly the orbit relation of $\Gamma_{10}^{\text{pres}}$. Hence it is the unique maximal lawful presentation congruence on the fixed theorem object. In particular a kernel-pair pair is not merely presumed to be an arrow: a componentwise arrow is constructed from its equal probe values.*

Proof. Every generator of $\Gamma_{10}^{\text{pres}}$ fixes the corresponding block of (50) by definition and the proper-transport and common-source presentation theorems already proved, so its orbit relation lies in the kernel pair. Conversely, let two candidates have equal probe. The separation lemma fixes their source, local objects, clock-descent coordinate and refinement class, source markings, transition maps, descent relation, holonomy, recovery, and failure data. For each chart, equal conditioning ranges and forms give the unique source-normalized unitary frame map. Equal orbit-complete empty-word evaluations, Ψ , proper-translation and slice-change data, chart/source markings, $\mathfrak{I}_6$, ρ , ℓ , domains, and target data give the identity common-source automorphism by the evaluation-rigidity lemma and hence the unique graph matching. Equal profile tensors give the lawful orientation-preserving readout or

representative map; equal incidence data paste these maps on overlaps. Equal coherence and holonomy make the pasted family a global presentation arrow. Equal mark blocks fix the neutral-to-singleton scalar comparison, including its clock-descent roof, and its arrow square. Equal recovery blocks ensure equality of the predecessor reduct; the remaining successor probe blocks separately fix the enriched atlas coordinates. Equal failure blocks ensure it changes neither predecessor nor lineage identity. This constructs an arrow in $\Gamma_{10}^{\text{pres}}$. Any larger congruence would identify a pair with unequal probe and erase a standing-bearing coordinate, so maximality and uniqueness follow. $\square$

9.3 Gate ledger

Gate	Name	Proof and status
G10.0	Source/target lock	W10.1; accepted bytes, empty atlas slot; PASS
G10.1	Clock-neutral base	W10.1–W10.2; transversely germified retained lineage; PASS
G10.2	Local/neutral/enriched categories, site, and reduct	W10.2–W10.3; typed single-clock, clock-descent, neutral, and enriched objects, arrows, restrictions, S_{neu} , For_{10} , and object/arrow Red_9 ; PASS
G10.3	Inherited scalar certificate	W10.4; literal WDW-9 signature without successor prepopulation; PASS
G10.4	Intrinsic-volume chart	W10.5; multiplier inverse and certified analytic clock data; PASS
G10.5	Intrinsic-curvature chart	W10.6; elliptic inverse and certified analytic clock data; PASS
G10.6	Genuine plurality	W10.7; pairwise differential-order separation; PASS
G10.7	Cover/overlaps	W10.8; occupied pairs and triple; PASS
G10.8	Local quantum structures	W10.9; normalized profile split, saturated raw localization, one common successor graph/domain, three conjugated chart presentations and source markings, the exact evaluation-naturality square, direct intrinsic-clock noncharacteristicity, source-side graph-smooth J_{α}^{src} , separate source/evaluation/transposed joint kernels, chart-normal lines, forms, $\mathcal{L}_{\star}$ restrictions, records; PASS
G10.9	Transition population	W10.10; six ordered common-source (G^V, G^K, G^D, μ) maps; PASS
G10.10	Preservation	W10.11; typed source/label/grammar/domain, evaluation naturality, three kernel levels, normalized-lapse, and Jacobian connection laws; PASS
G10.11	Coherence	W10.12; carrier and label cancellation plus proper-presentation 2-cell coherence; PASS
G10.12	Positive descent	W10.13; explicit clock-descent coordinate, non-clock quotients, and nonzero section; PASS
G10.13	Global physical structure	W10.14; positive and separating; PASS
G10.14	Obstruction/holonomy	W10.15; directly computed identity/skin; PASS
G10.15	One gluing orbit	W10.16; source-anchor forcing and evaluation/source-marking rigidity; PASS
G10.16	Atlas independence	W10.17; literal refinements plus transition-supported comparison roofs; PASS
G10.17	Universality	W10.18; mapping bijection (47); PASS
G10.18	Multiple-choice closure	W10.19; five terminal relations; PASS
G10.19	WDW-9 recovery	W10.20; enriched marked comparison followed by conservative reduct; all roles/overlaps/repairs; PASS
G10.20	Failure/no backflow	W10.21; ordered irreversible boundary; PASS

Gate	Name	Proof and status
G10.21	Exhaustion	W10.22; total exclusive grammar; PASS
G10.22	Complete probe	W10.23; separating and maximal skin; PASS
G10.23	Endpoint	W10.24 below; exact label/nonclaims printed; PASS

9.4 Principal endpoint

For every global enriched WDW-10 object whose atlas carries the inherited scalar member on U_Φ and a certified presentation isomorphism χ_Φ , define the typed predecessor-restriction operation

$$\text{Res}_{9,\Phi}(\mathfrak{E}; \chi_\Phi) := \text{Red}_9(i_\Phi^{*,\text{enr}} \mathfrak{E}; (a_\Phi, \chi_\Phi)). \quad (50a)$$

It is not ordinary restriction alone: $i_\Phi^{*,\text{enr}}$ restricts every coordinate of (6a), χ_Φ ties the already present scalar member, through S_{neu} , to the restricted neutral underlying object, and Red_9 then projects the marked local component to the complete WDW-9 signature. For $\mathfrak{R}_{10}^{\text{global}}$ we abbreviate $\text{Res}_{9,\Phi}(\mathfrak{R}_{10}^{\text{global}}) := \text{Res}_{9,\Phi}(\mathfrak{R}_{10}^{\text{global}}; \text{For}_{10}(\eta_\Phi^{10}))$ using (48)–(48').

Theorem 9.4 (W10.24 — global clock-neutral endpoint). *On the exact frozen Branch-A WDW-9 non-clock realization, the scalar-profile, intrinsic-volume-profile, and intrinsic-curvature-profile constructions positively occupy a transversely germified clock cover of the retained homogeneous source lineage. Their common-source-derived complete transition morphisms preserve the physical form, saturated-raw localized successor graph topology and common domain, chart/source markings, total constraint-label action, typed targets, registered operator grammar and counterterms, the $\mathcal{L}_\star$ constraint-normal connection with its clock Jacobian and companion term, represented observables, records, continuation, failure typing, boundary data, and nonclaims; satisfy higher-overlap coherence; and descend to a nonempty clock-neutral realization independent of lawful atlas refinement. Every same-target global gluing preserving the independent common-source probes lies in one lawful equivalence orbit by the rigidity theorem, and the conservative reduct of marked scalar restriction recovers WDW-9 coordinatewise. Therefore*

$$\begin{aligned} \mathfrak{R}_{10}^{\text{global}} &\in \text{EnrPhysRec}_{10}(X_{10}^{\text{reg}}), \quad \text{For}_{10}(\mathfrak{R}_{10}^{\text{global}}) = \mathfrak{R}_{10}^{\text{und}} \in \text{NeuPhysRec}_{10}(X_{10}^{\text{reg}}), \\ \text{Res}_{9,\Phi}(\mathfrak{R}_{10}^{\text{global}}) &\cong \mathfrak{R}_9^*, \quad |\text{Glue}_{10}^{\text{law}} / \Gamma_{10}^{\text{law}}| = 1, \quad G_{10} = \text{PASS-GLOBAL}. \end{aligned} \quad (51)$$

The optional extension set has not been positively populated, so no PASS – GLOBAL – EXTENDED claim is made.

Proof. W10.1–W10.3 fix the theorem object and native categories. W10.4–W10.9 positively construct the three local physical clock objects and their cover. W10.10–W10.12 construct and verify complete descent data. W10.13–W10.16 construct a nondegenerate global object, compute obstructions/holonomy, and prove the singleton gluing orbit. W10.17–W10.19 prove atlas independence, the exact mapping property, and multiple-choice closure. W10.20–W10.23 prove conservative predecessor containment, no backflow, grammar completeness, and probe maximality. The gate ledger records each independent pass. These are exactly the conjuncts of (51). $\square$

Decision.

$G_{10} = \text{PASS-GLOBAL}$, $G_9 = \text{PASS unchanged}$, $\text{PASS-GLOBAL-EXTENDED}$ not claimed.

The endpoint removes global preference among admissible local clocks on the declared transversely germified lineage target; it does not create one global time or cross a non-clock failure.

10. Post-endpoint certified standing export and Schrödinger handoff

The global endpoint W10.24 is complete before this section. Only now is the separate W9.19 construction imported:

$$I_{9 \rightarrow 10}^{\text{QS,post}} := (\text{W9.19}, \text{QPred}, \text{QStand}, \text{QClose}, \Omega_9^{\text{WDW}}, H_9^{\text{QS-EXPORT}} = \text{PASS}). \quad (52)$$

It is not a coordinate of the original $I_{9 \rightarrow 10}$, $\mathfrak{R}_9^*$, Cert_9 , or Red_9 , and is not used by W10.1–W10.24.

10.1 Certified representative interface and structural standing closure

Definition 10.1 (Certified **QP/QS** boundary). Let $\mathfrak{S}$ be a nondegenerate determinate closed quantum-predictive theorem object over a domain U . A *representative predictive certificate*

$$c_{\mathfrak{S}}^{\text{QP}} = (X, \Gamma \curvearrowright X, p, \mathcal{C}, \boxtimes, \mathbf{1}, C; \zeta_{\text{prob}}, \zeta_{\text{ph}}, \zeta_{\text{ray}}, \zeta_{\text{sep}}, \zeta_{\text{comp}}, \zeta_{\text{cont}}) \quad (\text{QP-cert})$$

contains a nonempty presentation object X , the lawful scalar-redescription action $\Gamma = \mathbb{C}^\times$, a raw transition-weight family $p : X \times_U X \rightarrow [0, 1]$, a declared comparison class $\mathcal{C}$ with symmetric monoidal product, and a raw closed continuation C . Its six proof coordinates have the following exact burdens:

QP-Prob ζ_{prob} certifies that p is a standing-bearing representative-level comparison family and is stable under the declared base restrictions and lawful arrows;

QP-Phase ζ_{ph} certifies that the Γ -action is lawful redescription and that $p(\gamma x, \gamma' y) = p(x, y)$; it does *not* declare an orbit to be a standing-bearing state;

QP-Ray ζ_{ray} supplies the orbit coequalizer $q : X \rightarrow X/\Gamma$ in the declared regularity class and proves its stability under base restriction; it does *not* assert standing descent;

QP-Sep ζ_{sep} is the raw profile-completeness certificate

$$[p(x, z) = p(y, z) \text{ and } p(z, x) = p(z, y) \ \forall z] \iff q(x) = q(y);$$

QP-Comp ζ_{comp} certifies on representatives that $\boxtimes$ is independently equivariant in both inputs: for every $(\gamma, \gamma') \in \Gamma \times \Gamma$,

$$\boxtimes(\gamma x, \gamma' y) = (\gamma \gamma') \boxtimes(x, y).$$

Equivalently, it is $(\Gamma \times \Gamma)$ -equivariant when the target Γ -action is pulled back through scalar multiplication $\Gamma \times \Gamma \rightarrow \Gamma$. The same certificate proves that transition weights multiply and that the associator, unitors, and symmetry obey their coherence laws; and

QP-Cont ζ_{cont} certifies that C is Γ -equivariant, preserves p , is monoidal on the declared comparison class, and commutes with restriction and lawful arrows.

Write $\mathbf{QP}(\mathfrak{S}; c_{\mathfrak{S}}^{\text{QP}})$ when these raw data and proofs are supplied. In particular, **QP** does not assume that X/Γ carries state standing, does not supply a descended transition weight or continuation, and does not assume quotient-level composition naturality.

A *structural quantum-standing certificate*

$$c_{\mathfrak{S}}^{\text{QS}} = (P, q, \bar{p}, \bar{C}, \bar{\boxtimes}; \delta_{\text{stand}}, \delta_{\text{sep}}, \delta_{\text{cont}}, \delta_{\text{comp}}, \delta_{\text{nat}}) \quad (\text{QS-cert})$$

proves that $P = X/\Gamma$ is the standing-bearing state object; q is the universal standing descent and has exactly the scalar-redescription orbits as fibres; $\bar{p}$ is well defined and separates standing distinctions; $\bar{C}$ is the unique standing-preserving continuation with $\bar{C}q = qC$; $\bar{\boxtimes}$ is the induced strong symmetric-monoidal independent-composition law; and all four constructions are natural under lawful arrows and base restriction. Write $\mathbf{QS}(\mathfrak{S}; c_{\mathfrak{S}}^{\text{QS}})$ when this derived certificate is supplied.

The boundary contains no flat spatial carrier, carrier dimension, Laplacian, preferred time, Wigner–Stone lift, Hamiltonian-response predicate, potential, or scale anchor. Those remain projection-side burdens.

For functorial export, write $\underline{\mathbb{C}}_U := U \times \mathbb{C}$ for the explicitly defined trivial Hermitian line over U , with its canonical unit section and connection.

Let $\mathbf{QPred}(U)$ be the category of representative predictive models equipped with the complete certificate c^{QP} of Definition 10.1. An object contains the representative carrier, scalar-redescription action, raw transition weight, declared comparison groupoid, raw continuation, record and first-failure boundary, together with the six proof coordinates $\zeta_{\text{prob}}, \dots, \zeta_{\text{cont}}$. An arrow contains a physical map T , its equivariance square for the scalar action, the algebra, continuation, record and failure maps, and a strong monoidal comparison functor preserving every ζ -certificate. Its object components are $T^{\otimes n}$, with $T^{\otimes 0} = \text{id}_{\underline{\mathbb{C}}_U}$, and its action on every internal comparison-groupoid arrow $S : \mathcal{H}^{\otimes n} \rightarrow \mathcal{H}^{\otimes m}$ is

$$\mathcal{L}(T)(S) := T^{\otimes m} S (T^{\otimes n})^{-1}.$$

Restrictions retain the raw data and restrict each proof coordinate.

Let $\mathbf{QStand}(U)$ be the category of objects of $\mathbf{QPred}(U)$ equipped additionally with the complete derived certificate $c^{\mathbf{QS}}$ of Definition 10.1. An arrow must preserve the quotient map, descended transition weight, descended continuation, induced monoidal product, standing descent, separation, and every arrow/restriction naturality square. Thus membership in $\mathbf{QStand}(U)$ is certified mathematical data, not a consequence of rank, positivity, unitarity, or the types of the record fields alone. Composition and identities are componentwise, including the displayed internal-arrow action.

10.2 Certification-bearing export domains

For $U \in \mathbf{ClockSite}_{10}$, define $\mathbf{NeuPhysRec}_{10}^{\mathbf{QS}}(U)$ to have objects

$$(\mathfrak{R}, c_{\mathfrak{R}}^{\mathbf{QP}}, c_{\mathfrak{R}}^{\mathbf{QS}}, \varepsilon_{\mathfrak{R}}), \quad \varepsilon_{\mathfrak{R}} : c_{\mathfrak{R}}^{\mathbf{QS}} \xrightarrow{\cong} \mathbf{QClose}_U(c_{\mathfrak{R}}^{\mathbf{QP}}), \quad (53a)$$

where $\mathfrak{R} \in \mathbf{NeuPhysRec}_{10}(U)$, the raw physical, algebra, continuation, record, failure, scalar-action, and comparison data of $\mathfrak{R}$ carry every ζ -coordinate of a $\mathbf{QP}$ certificate, and $c_{\mathfrak{R}}^{\mathbf{QS}}$ carries every δ -coordinate of a $\mathbf{QS}$ certificate. An arrow is an arrow of $\mathbf{NeuPhysRec}_{10}(U)$ together with preservation squares for all ζ - and δ -coordinates and for ε . Thus $\mathbf{NeuPhysRec}_{10}^{\mathbf{QS}}$ is a category of certified lifts, not the full subcategory selected merely by rank and field types.

Define the other export domains as certification pullbacks:

$$\mathbf{EnrPhysRec}_{10}^{\mathbf{QS}}(U) := \mathbf{EnrPhysRec}_{10}(U) \times_{\mathbf{NeuPhysRec}_{10}(U)} \mathbf{NeuPhysRec}_{10}^{\mathbf{QS}}(U). \quad (53b)$$

$$\mathbf{Cert}_9^{\mathbf{QS}}(U) := \{(\mathfrak{C}, c_{\mathfrak{C}}) \mid \mathfrak{C} \in \mathbf{Cert}_9(U), \quad c_{\mathfrak{C}} = (c_{\mathfrak{C}}^{\mathbf{QP}}, c_{\mathfrak{C}}^{\mathbf{QS}}, \varepsilon_{\mathfrak{C}}), \quad \varepsilon_{\mathfrak{C}} : c_{\mathfrak{C}}^{\mathbf{QS}} \cong \mathbf{QClose}_U(c_{\mathfrak{C}}^{\mathbf{QP}})\}. \quad (53c)$$

The fibre product in (53b) is taken along $\mathbf{For}_{10}$ and the forgetful map from a certified lift. Arrows in (53c) preserve the complete WDW-9 signature and both certificates. All three categories are defined only after W10.24 and enlarge no core category.

Let $\mathbf{QData}(U)$ denote the category obtained from $\mathbf{QPred}(U)$ by forgetting the six ζ -proofs but retaining every representative datum and its arrows. Let $\mathbf{NeuPhysRec}_{10}^{\mathbf{raw}}(U)$ and $\mathbf{Cert}_9^{\mathbf{raw}}(U)$ be the full subcategories whose displayed physical, scalar, transition, comparison, continuation, record, and failure coordinates form such a raw datum. There are componentwise extraction functors

$$\mathbf{Raw}_{N,U} : \mathbf{NeuPhysRec}_{10}^{\mathbf{raw}}(U) \longrightarrow \mathbf{QData}(U), \quad \mathbf{Raw}_{9,U} : \mathbf{Cert}_9^{\mathbf{raw}}(U) \longrightarrow \mathbf{QData}(U).$$

They retain the representative physical carrier, scalar action, raw transition weight, comparison class, continuation, record, and failure. On an arrow they retain the physical map, its scalar-equivariance square, and the comparison-arrow action (55a). Both commute with base restriction componentwise. Their certified lifts are

$$\mathbf{QRaw}_U : \mathbf{NeuPhysRec}_{10}^{\mathbf{QS}}(U) \longrightarrow \mathbf{QPred}(U), \quad \mathbf{QRaw}_{9,U} : \mathbf{Cert}_9^{\mathbf{QS}}(U) \longrightarrow \mathbf{QPred}(U),$$

which additionally retain the six ζ -proofs. If $\theta : \mathfrak{C} \xrightarrow{\cong} \mathfrak{C}'$ is an isomorphism in $\mathbf{Cert}_9$ and c' is a $\mathbf{QP}$ certificate on $\mathbf{Raw}_9(\mathfrak{C}')$, write $(\mathbf{Raw}_9\theta)^*c'$ for the certificate transported along that raw isomorphism. Every ζ -square is transported by conjugation with $\mathbf{Raw}_9\theta$, so this operation is functorial and commutes with restriction. Define

$$\mathbf{QLoc}_U := \mathbf{QClose}_U \circ \mathbf{QRaw}_U : \mathbf{NeuPhysRec}_{10}^{\mathbf{QS}}(U) \longrightarrow \mathbf{QStand}(U). \quad (54)$$

Thus the landing in $\mathbf{QStand}$ uses W9.19; it is not inferred from rank one, positivity, unitarity, or correctly typed continuation fields.

If $f : \mathfrak{R} \rightarrow \mathfrak{R}'$ has physical unitary T_f , define on every internal comparison arrow

$$\mathcal{L}(T_f)(S) := T_f^{\widehat{\otimes}^m} S (T_f^{\widehat{\otimes}^n})^{-1}, \quad S : \mathcal{H}_{\mathfrak{R}}^{\widehat{\otimes}^n} \rightarrow \mathcal{H}_{\mathfrak{R}'}^{\widehat{\otimes}^m}, \quad (55a)$$

with $T_f^{\widehat{\otimes}^0} = \text{id}_{\mathbb{C}_U}$, and on quotient states by

$$\overline{T}_f(q_{\mathfrak{R}}u) := q_{\mathfrak{R}'}(T_f u). \quad (55b)$$

The latter is a conclusion of $\mathbf{QClose}$ from the certified equivariance of T_f .

The enriched and predecessor exports are

$$\mathbf{QExp}_{10,U} := \mathbf{QLoc}_U \circ \mathbf{For}_{10}^{\mathbf{QS}} : \mathbf{EnrPhysRec}_{10}^{\mathbf{QS}}(U) \rightarrow \mathbf{QStand}(U), \quad (56a)$$

$$\mathbf{QExp}_{9,U} := \mathbf{QClose}_U \circ \mathbf{QRaw}_{9,U} : \mathbf{Cert}_9^{\mathbf{QS}}(U) \rightarrow \mathbf{QStand}(U), \quad (56b)$$

where $\mathbf{For}_{10}^{\mathbf{QS}}$ is the pullback projection.

Lemma 10.2 (Certified functoriality and restriction). *Equations (54)–(56b) define functors on all objects, external arrows, and internal comparison-groupoid arrows. For each $j : V \hookrightarrow U$,*

$$j^* c^{\text{QP}} = c_{j^* \mathfrak{R}}^{\text{QP}}, \quad j^* c^{\text{QS}} = c_{j^* \mathfrak{R}}^{\text{QS}}, \quad j^* \text{QExp}_{k,U} = \text{QExp}_{k,V} j^*, \quad k \in \{9, 10\}, \quad (57)$$

with the same equalities on arrows.

Proof. The defining arrow of a certified lift preserves all raw data and every ζ -proof. W9.19 therefore induces (55b) and preserves every δ -proof. Formula (55a) respects identity and composition by cancellation, respects tensor products by the standard associators, and respects unit and symmetry by naturality of the Hilbert coherence maps. The zeroth component is the identity of the explicit trivial Hermitian line $\mathbb{C}_U$. For composable external arrows, $\mathcal{L}(T_{f'} T_f) = \mathcal{L}(T_{f'}) \mathcal{L}(T_f)$.

Restriction preserves the nonzero representative object, scalar action, raw weight, tensor powers, continuation, records, failures, all ζ -proofs, and formula (55a). The restriction part of W9.19 gives the corresponding δ -proofs and equalities in (57). $\square$

10.3 Lifted reduct and the exact marked pullback

On $\text{EnrPhysRec}_{10}^{\Phi\text{-mark}}(U)$ retain the core functors U_{mark} and Red_9 . Define the scalar-neutral functor, on objects and arrows, by

$$\text{ScalNeu}_{\Phi}(\mathfrak{E}; a_{\Phi}, \chi_{\Phi}) := S_{\text{neu}}(\mathfrak{R}_{a_{\Phi}}), \quad \text{ScalNeu}_{\Phi}(\Xi) := S_{\text{neu}}(f_{a_{\Phi}}).$$

Equation (6c') says precisely that χ_{Φ} is the component of a natural isomorphism

$$\text{For}_{10} \text{U}_{\text{mark}} \implies \text{ScalNeu}_{\Phi}.$$

Write U_E , U_N , and U_9 for the forgetful functors from $\text{EnrPhysRec}_{10}^{\text{qs}}$, $\text{NeuPhysRec}_{10}^{\text{qs}}$, and $\text{Cert}_9^{\text{qs}}$ to their uncertified categories.

First form the two-leg certified iso-pullback $\text{D}_{\Phi}^{\text{qs}}(U)$. Its object is

$$\mathfrak{d} = (\mathfrak{E}; \widehat{\mathfrak{E}}_{10}, \widehat{\mathfrak{R}}_{\Phi}; \lambda_{10}, \lambda_{\Phi}, \widehat{\chi}_{\Phi}),$$

where $\mathfrak{E} \in \text{EnrPhysRec}_{10}^{\Phi\text{-mark}}(U)$, $\widehat{\mathfrak{E}}_{10} \in \text{EnrPhysRec}_{10}^{\text{qs}}(U)$, $\widehat{\mathfrak{R}}_{\Phi} \in \text{NeuPhysRec}_{10}^{\text{qs}}(U)$,

$$\lambda_{10} : \text{U}_{\text{mark}}(\mathfrak{E}) \xrightarrow{\cong} \text{U}_E \widehat{\mathfrak{E}}_{10}, \quad \lambda_{\Phi} : \text{ScalNeu}_{\Phi}(\mathfrak{E}) \xrightarrow{\cong} \text{U}_N \widehat{\mathfrak{R}}_{\Phi},$$

and $\widehat{\chi}_{\Phi}$ is a certified isomorphism whose underlying arrow is

$$\lambda_{\Phi} \chi_{\Phi} (\text{For}_{10} \lambda_{10})^{-1} : \text{For}_{10} \text{U}_E \widehat{\mathfrak{E}}_{10} \xrightarrow{\cong} \text{U}_N \widehat{\mathfrak{R}}_{\Phi}.$$

Arrows contain the corresponding three certified squares. Thus certification under marking is comparison data in the pullback, not a property inferred from field types.

The certified scalar leg places $\text{ScalNeu}_{\Phi}(\mathfrak{E})$ in $\text{NeuPhysRec}_{10}^{\text{raw}}$. Since Red_9 retains exactly its predecessor raw coordinates, $\text{Red}_9(\mathfrak{E})$ lies in $\text{Cert}_9^{\text{raw}}$. The definitions of Raw_9 , Red_9 , and ScalNeu_{Φ} then give a componentwise raw natural isomorphism

$$\rho_{\Phi, \mathfrak{E}} : \text{Raw}_9 \text{Red}_9(\mathfrak{E}) \xrightarrow{\cong} \text{Raw}_N \text{ScalNeu}_{\Phi}(\mathfrak{E}). \quad (58a)$$

It is identity on the representative carrier, scalar action, transition weight, comparison arrows, continuation, record, and failure after the successor-only coordinates have been forgotten. For $\mathfrak{d} \in \text{D}_{\Phi}^{\text{qs}}(U)$ define

$$\begin{aligned} c_{\text{Red}_9 \mathfrak{d}}^{\text{QP}} &:= (\text{Raw}_N(\lambda_{\Phi}) \rho_{\Phi, \mathfrak{E}})^* c_{\Phi}^{\text{QP}}, \\ c_{\text{Red}_9 \mathfrak{d}}^{\text{QS}} &:= \text{QClose}(c_{\text{Red}_9 \mathfrak{d}}^{\text{QP}}), \\ \widetilde{\text{Red}_9}^{\text{qs}}(\mathfrak{d}) &:= (\text{Red}_9(\mathfrak{E}), c_{\text{Red}_9 \mathfrak{d}}^{\text{QP}}, c_{\text{Red}_9 \mathfrak{d}}^{\text{QS}}, \text{id}) \in \text{Cert}_9^{\text{qs}}(U). \end{aligned} \quad (58b)$$

For an arrow $\widehat{\Xi} : \mathfrak{d} \rightarrow \mathfrak{d}'$, its core component is $\text{Red}_9(\Xi)$. Its **QP** certificate component is the map induced by the commutative raw square

$$\text{Raw}_N(\lambda'_{\Phi}) \rho_{\Phi, \mathfrak{E}'} \text{Raw}_9 \text{Red}_9(\Xi) = \text{Raw}_N(\widehat{\Xi}_{\Phi}) \text{Raw}_N(\lambda_{\Phi}) \rho_{\Phi, \mathfrak{E}}, \quad (58c)$$

and its **QS** component is the image of that map under QClose . Equations (58b)–(58c) therefore define a functor on objects and arrows. Because the source certificate in (58b) is defined by pullback, the raw composite has a canonical certified lift

$$\widehat{\rho}_{\Phi, \mathfrak{d}} : \text{QRaw}_9(\widetilde{\text{Red}_9}^{\text{qs}}(\mathfrak{d})) \xrightarrow{\cong} \text{QRaw}(\widehat{\mathfrak{R}}_{\Phi}) \quad \text{in } \text{QPred}(U).$$

Its naturality square is (58c). Thus no symbol named RedCert is introduced, and QClose is never applied to a merely raw arrow.

The exact marked certified domain is the graph three-way iso-pullback

$$\text{EnrPhysRec}_{10}^{\Phi\text{-mark,qs}}(U) := \{(\mathfrak{d}, \widehat{\mathfrak{C}}_9, \lambda_9) \mid \mathfrak{d} \in D_{\Phi}^{\text{qs}}(U), \quad \widehat{\mathfrak{C}}_9 \in \text{Cert}_9^{\text{qs}}(U), \\ \lambda_9 : \widetilde{\text{Red}}_9^{\text{qs}}(\mathfrak{d}) \xrightarrow{\cong} \widehat{\mathfrak{C}}_9 \text{ in } \text{Cert}_9^{\text{qs}}(U)\}. \quad (58)$$

It is the simultaneous iso-pullback of the U_{mark} , ScalNeu_{Φ} , and Red_9 legs, with all three comparison isomorphisms $\lambda_{10}, \lambda_{\Phi}, \lambda_9$ retained. There are two different functors, and we keep them notationally distinct. Write π_D for the first projection to D_{Φ}^{qs} and π_9 for the third projection to the chosen predecessor certificate. The raw-extraction reduct is

$$\text{Red}_9^{\text{qs}} : \text{EnrPhysRec}_{10}^{\Phi\text{-mark,qs}}(U) \longrightarrow \text{Cert}_9^{\text{qs}}(U), \quad \text{Red}_9^{\text{qs}} := \widetilde{\text{Red}}_9^{\text{qs}} \circ \pi_D, \quad \pi_9(\mathfrak{d}, \widehat{\mathfrak{C}}_9, \lambda_9) := \widehat{\mathfrak{C}}_9. \quad (59)$$

Thus Red_9^{qs} is the functor constructed in (58b)–(58c), whereas π_9 is only the graph-pullback projection. On an arrow $(\widehat{\Xi}, g_9)$, the first takes $\widetilde{\text{Red}}_9^{\text{qs}}(\widehat{\Xi})$ and the second takes the certified predecessor arrow g_9 . The pullback square is exactly

$$g_9 \lambda_9 = \lambda'_9 \widetilde{\text{Red}}_9^{\text{qs}}(\widehat{\Xi}),$$

so the retained λ_9 form a natural isomorphism $\text{Red}_9^{\text{qs}} \Rightarrow \pi_9$; they do not identify the two functors by definition.

Lemma 10.3 (Lifted-reduct restriction square). *For every $j : V \hookrightarrow U \subset U_{\Phi}$, restriction preserves the three-way pullback and the square*

$$\begin{array}{ccc} \text{EnrPhysRec}_{10}^{\Phi\text{-mark,qs}}(U) & \xrightarrow{(\text{Red}_9^{\text{qs}})_U} & \text{Cert}_9^{\text{qs}}(U) \\ j^* \downarrow & & \downarrow j^* \\ \text{EnrPhysRec}_{10}^{\Phi\text{-mark,qs}}(V) & \xrightarrow{(\text{Red}_9^{\text{qs}})_V} & \text{Cert}_9^{\text{qs}}(V) \end{array} \quad \text{commutes.} \quad (60)$$

The equality holds on core objects, on every ζ - and δ -certificate, and on arrows. The same restriction equality holds for π_9 , and $j^* \lambda_9 = \lambda_{9,j^*(-)}$.

Proof. Equation (6e) gives the core-object and core-arrow square. Raw extraction, λ_{Φ} , and $\rho_{\Phi, \mathfrak{E}}$ are componentwise, so

$$j^*((\text{Raw}_N \lambda_{\Phi} \rho_{\Phi, \mathfrak{E}})^* c_{\Phi}^{\text{QP}}) = (\text{Raw}_N(j^* \lambda_{\Phi}) \rho_{\Phi, j^* \mathfrak{E}})^* j^* c_{\Phi}^{\text{QP}}.$$

This is equality of all six ζ -proofs. The restriction conclusion of W9.19 gives the equality of the five derived δ -proofs. Restricting λ_{10} , $\widehat{\chi}_{\Phi}$, and λ_9 preserves their squares, which proves (60) on the full pullback. $\square$

10.4 Positive occupation of the three-way pullback

10.4.1 The global presentation object and its comparison groupoid

Let

$$Y_{10}(x) := [\alpha, x, Y_{\alpha, x}]$$

be the authenticated unit section constructed in (40). For $U \hookrightarrow X_{10}^{\text{reg}}$ put

$$\begin{aligned} \mathcal{H}_{10,U}^{\text{gl}} &:= \mathcal{H}_{10}^{\text{gl}}|_U, & X_{10}(U) &:= \mathcal{H}_{10,U}^{\text{gl}} \setminus 0_U, \\ \Gamma_U &:= \underline{\mathbb{C}}_U^{\times}, & \gamma \cdot u &:= \gamma u, \\ q_{10,U} : X_{10}(U) &\longrightarrow P_{10}(U) := X_{10}(U)/\Gamma_U = \mathbb{P}(\mathcal{H}_{10,U}^{\text{gl}}). \end{aligned} \quad (60a)$$

where the quotient is fibrewise. The section makes every fibre nonempty and gives the unitary trivialization

$$\Phi_{10,U} : \mathcal{H}_{10,U}^{\text{gl}} \xrightarrow{\cong} \underline{\mathbb{C}}_U, \quad \Phi_{10,U}(zY_{10}(x)) = (x, z).$$

Define the representative transition weight

$$p_{10,U}(u, v) := \frac{|\langle u, v \rangle_{10,x}|^2}{\|u\|_{10,x}^2 \|v\|_{10,x}^2}, \quad (u, v) \in X_{10}(U) \times_U X_{10}(U). \quad (60b)$$

The symmetric monoidal comparison groupoid $\mathcal{L}_{10}(U)$ is the following explicit groupoid of Hermitian line bundles:

$$\text{Ob } \mathcal{L}_{10}(U) = \mathbb{N}, \quad \mathcal{L}_{10,U}(n) := (\mathcal{H}_{10,U}^{\text{gl}})^{\widehat{\otimes} n}, \quad \mathcal{L}_{10,U}(0) := \underline{\mathbb{C}}_U.$$

Put $Y_{10}^{\otimes} := 1$. In the declared comparison class, every internal arrow from n to m is uniquely of the coefficient-preserving form

$$S_s^{m,n}(zY_{10}(x)^{\otimes n}) := szY_{10}(x)^{\otimes m}, \quad s \in U(1). \quad (60c)$$

Conversely (60c) is a fibrewise unitary bundle isomorphism. This is exactly the section-trivialized global analogue of the WDW-9 class of all unitaries between its displayed one-dimensional Hilbert spaces; no larger gauge-function groupoid is silently introduced. Thus

$$S_t^{\ell,m} S_s^{m,n} = S_{ts}^{\ell,n}, \quad (S_s^{m,n})^{-1} = S_s^{n,m}, \quad \text{id}_n = S_1^{n,n}.$$

The tensor product is addition on objects and $S_s^{m,n} \widehat{\otimes} S_t^{m',n'} = S_{st}^{m+m',n+n'}$ on arrows, using the canonical tensor identification. Its tensor unit is $\underline{\mathbb{C}}_U$. The associator, left and right unitors, and symmetry are the standard Hermitian-bundle maps

$$((u \otimes v) \otimes w) \mapsto u \otimes (v \otimes w), \quad z \otimes u \mapsto zu, \quad u \otimes z \mapsto zu, \quad u \otimes v \mapsto v \otimes u.$$

They obey the pentagon, triangle, involutivity, and hexagon identities on elementary tensors, hence on every fibre and therefore as bundle maps.

If $\mu_n : \widehat{\mathbb{C}}_U^{\otimes n} \rightarrow \underline{\mathbb{C}}_U$ is canonical multiplication, define

$$m_{10,n} := \Phi_{10,U}^{-1} \mu_n \widehat{\Phi}_{10,U}^{\otimes n} : \mathcal{L}_{10,U}(n) \xrightarrow{\cong} \mathcal{H}_{10,U}^{\text{gl}}, \quad m_{10,n}(Y_{10}^{\otimes n}) = Y_{10}. \quad (60d)$$

In particular,

$$u \boxtimes_{10} v := m_{10,2}(u \widehat{\otimes} v)$$

defines the representative independent-composition law. For every n , let $p_{10,n}$ be (60b) on the n th tensor power. For nonzero representatives one has

$$\begin{aligned} (\gamma u) \boxtimes_{10} (\gamma' v) &= (\gamma \gamma')(u \boxtimes_{10} v), \\ p_{10,n+m}(u \otimes v, u' \otimes v') &= p_{10,n}(u, u') p_{10,m}(v, v'). \end{aligned} \quad (60e)$$

For every declared closed continuation arrow $a : x \rightsquigarrow y$, let $U_{10,a}$ be the physical component of Cont_{10} . Under the anchor trivialization (39), this is the transport of the accepted coefficient-preserving WDW-9 continuation $\mathfrak{K}_{\text{cont}}$ imported literally in W10.1. W10.13 descends that coordinate, and W10.20 proves that both it and the normalized authenticated class are retained by recovery. Therefore

$$U_{10,a} Y_{10}(x) = Y_{10}(y), \quad U_{10,a}(zY_{10}(x)) = zY_{10}(y).$$

Hence $U_{10,a}$ is Γ -equivariant and unitary, $p_{10,y}(U_{10,a}u, U_{10,a}v) = p_{10,x}(u, v)$, and $U_{10,a}^{\widehat{\otimes}(n+m)} = U_{10,a}^{\widehat{\otimes}n} \widehat{\otimes} U_{10,a}^{\widehat{\otimes}m}$. It commutes with $m_{10,n}$ and therefore is monoidal on $\mathcal{L}_{10}$.

10.4.2 The six global representative certificates

Proposition 10.4 (Explicit global **QP** certificate). *The data (60a)–(60e), the global continuation, the represented algebra, record, and retained first failure define a certificate c_{10}^{QP} on $\mathfrak{R}_{10}^{\text{und}}$. All six ζ -coordinates are positively supplied:*

QP-Prob ζ_{prob}^{10} is Cauchy–Schwarz for the positive form (41). It gives $0 \leq p_{10} \leq 1$ and is preserved by every physical unitary.

QP-Phase ζ_{ph}^{10} is the cancellation of $|\gamma|^2 |\gamma'|^2$ between numerator and denominator in (60b); the fibrewise $\mathbb{C}^\times$ action is the lawful scalar redescription.

QP-Ray ζ_{ray}^{10} is the fibrewise orbit coequalizer $q_{10,U}$ in (60a). The section trivialization identifies it with $U \times \mathbb{P}(\mathbb{C})$ over U .

QP-Sep ζ_{sep}^{10} is rank-one profile completeness. In one fibre any two nonzero vectors are scalar multiples, and therefore

$$[p_{10}(u, z) = p_{10}(v, z), p_{10}(z, u) = p_{10}(z, v) \ \forall z] \iff q_{10}(u) = q_{10}(v).$$

QP-Comp ζ_{comp}^{10} consists of (60c)–(60e) and the displayed Hilbert-bundle coherence diagrams. In particular it proves independent $(\Gamma \times \Gamma)$ -equivariance, multiplication of transition weights, and the associator, unitor, symmetry, pentagon, triangle, and hexagon laws.

QP-Cont ζ_{cont}^{10} is the unitary, equivariant, p -preserving and monoidal continuation calculation immediately above.

Every certificate commutes with base restriction and every lawful external arrow.

Proof. For a restriction $j : V \hookrightarrow U$, the identities

$$j^*Y_{10,U} = Y_{10,V}, \quad j^*\Phi_{10,U} = \Phi_{10,V}, \quad j^*p_{10,U} = p_{10,V}, \quad j^*\mathcal{L}_{10}(U) = \mathcal{L}_{10}(V)$$

make (60a)–(60e) restrict coordinatewise; the same is true of every standard coherence map and continuation square. Thus $j^*\zeta_{\bullet}^{10} = \zeta_{\bullet}^{10}|_V$ for all six coordinates.

Let f be a lawful external arrow between restrictions of the constructed global object and let T_f be its physical component. By (6), (6b), and the authenticated-section square, T_f is unitary, complex linear, and $T_f Y_{10} = Y'_{10}$. Consequently

$$\begin{aligned} \Phi'_{10} T_f &= \Phi_{10}, & p'_{10}(T_f u, T_f v) &= p_{10}(u, v), & \bar{T}_f q_{10} &= q'_{10} T_f, \\ T_f m_{10,n} &= m'_{10,n} T_f^{\widehat{\otimes} n}, & \mathcal{L}(T_f)(S) &= T_f^{\widehat{\otimes} m} S (T_f^{\widehat{\otimes} n})^{-1}. \end{aligned}$$

The last formula sends $S_s^{m,n}$ to $S_s^{m,n}$ on the target and preserves composition, tensor product, unit, associator, unitors, and symmetry. The continuation naturality square in (6) gives $T_f U_{10,a} = U'_{10,f(a)} T_f$. These are precisely the external-arrow squares for $\zeta_{\text{prob}}^{10}, \zeta_{\text{ph}}^{10}, \zeta_{\text{ray}}^{10}, \zeta_{\text{sep}}^{10}, \zeta_{\text{comp}}^{10}$, and ζ_{cont}^{10} . $\square$

With the certificate now constructed rather than named, put

$$\begin{aligned} c_{10}^{\text{QS}} &:= \text{QClose}(c_{10}^{\text{QP}}), & \widehat{\mathfrak{R}}_{10}^{\text{gl}} &:= (\mathfrak{R}_{10}^{\text{und}}, c_{10}^{\text{QP}}, c_{10}^{\text{QS}}, \text{id}) \in \text{NeuPhysRec}_{10}^{\text{qs}}(X_{10}^{\text{reg}}), \\ \widehat{\mathfrak{E}}_{10}^{\text{gl}} &:= (\mathfrak{R}_{10}^{\text{global}}, \widehat{\mathfrak{R}}_{10}^{\text{gl}}, \text{id}) \in \text{EnrPhysRec}_{10}^{\text{qs}}(X_{10}^{\text{reg}}). \end{aligned}$$

Restrict it to U_{Φ} . Transport its neutral certificate through $\chi_{\Phi} = \text{For}_{10}(\eta_{\Phi}^{10})$ and define

$$\begin{aligned} c_{\Phi}^{\text{QP}} &:= (\text{Raw}_N \chi_{\Phi})_* i_{\Phi}^* c_{10}^{\text{QP}}, & c_{\Phi}^{\text{QS}} &:= \text{QClose}(c_{\Phi}^{\text{QP}}), \\ \widehat{\mathfrak{R}}_{\Phi}^{\text{gl}} &:= (\text{ScalNeu}_{\Phi}(i_{\Phi}^{*,\text{enr}} \mathfrak{R}_{10}^{\text{global}}; m_{\Phi}^{\text{gl}}), c_{\Phi}^{\text{QP}}, c_{\Phi}^{\text{QS}}, \text{id}) \in \text{NeuPhysRec}_{10}^{\text{qs}}(U_{\Phi}). \end{aligned}$$

The certificate transport itself is the isomorphism $\widehat{\chi}_{\Phi} : \text{For}_{10}^{\text{qs}}(i_{\Phi}^* \widehat{\mathfrak{E}}_{10}^{\text{gl}}) \xrightarrow{\cong} \widehat{\mathfrak{R}}_{\Phi}^{\text{gl}}$.

Define the certified two-leg inhabitant

$$\mathfrak{d}_{\Phi}^{\text{gl}} := ((i_{\Phi}^{*,\text{enr}} \mathfrak{R}_{10}^{\text{global}}; m_{\Phi}^{\text{gl}}); i_{\Phi}^* \widehat{\mathfrak{E}}_{10}^{\text{gl}}, \widehat{\mathfrak{R}}_{\Phi}^{\text{gl}}; \text{id}, \text{id}, \widehat{\chi}_{\Phi}) \in \text{D}_{\Phi}^{\text{qs}}(U_{\Phi}).$$

10.4.3 Coordinatewise identity with the accepted WDW-9 certificate

Now use the core comparison η_{Φ}^9 already constructed in (48a), and write

$$\vartheta_{\Phi} := \text{Raw}_N(\lambda_{\Phi}) \rho_{\Phi, \mathfrak{E}}.$$

Lemma 10.5 (Exact certificate identity). *At $\mathfrak{d}_{\Phi}^{\text{gl}}$ there is an equality of complete representative certificates*

$$(\text{Raw}_{\eta_{\Phi}^9})^* c_{\eta_{\Phi}^9}^{\text{QP}} = (\text{Raw}_N(\lambda_{\Phi}) \rho_{\Phi, \mathfrak{E}})^* (\text{Raw}_N \chi_{\Phi})_* i_{\Phi}^* c_{10}^{\text{QP}}. \quad (60f)$$

The equality includes every coordinate of c^{QP} : the presentation carrier, scalar action, transition weight, orbit map, comparison groupoid and all its declared internal arrows, monoidal coherence, continuation, and all six ζ -proofs. The underlying raw-data isomorphism additionally matches the represented algebra, record, and failure coordinates.

Proof. W10.20 says that η_{Φ}^9 preserves the normalized authenticated physical class, form, continuation, record, and failure. Hence its physical component sends the restricted Y_{10} to Ω_9 and intertwines Φ_{10} with Φ_9 . Raw extraction of (48a), followed by the definitions of χ_{Φ} , λ_{Φ} , and $\rho_{\Phi, \mathfrak{E}}$, gives the following coordinate equalities.

Coordinate	WDW-9 pullback	Restricted WDW-10 transport
X, Γ	X_9 and scalar multiplication, pulled back by η_Φ^9	$i_\Phi^* X_{10}$ and its scalar action, transported by χ_Φ and ϑ_Φ ; linearity makes the two identical.
p	p_9 formed from the WDW-9 physical form	$i_\Phi^* p_{10}$ formed from (41); isometry of (48a) makes the normalized ratios equal.
q	$q_9 : X_9 \rightarrow \mathbb{P}\mathcal{H}_{\text{phys},9}$	$i_\Phi^* q_{10}$; equivariance and bijectivity of the physical map give the same orbit coequalizer square.
$\mathcal{L}$ and arrows	$\mathcal{L}_9(n)$ and every unitary internal arrow	$i_\Phi^* \mathcal{L}_{10}(n)$ and (60c); $T^{\widehat{\otimes} n}$ and conjugation (55a) identify them.
$\mathbf{1}$, tensor, coherence	$\mathbb{C}$, $m_{9,n}$, standard associator, unitors and flip	$\mathbb{C}$, $m_{10,n}$ and the same standard maps; $\Phi_{10} \mapsto \Phi_9$ makes every diagram equal.
C , record, failure	WDW-9 continuation and retained source package	restricted WDW-10 continuation, record and failure after successor forgetting; these are the corresponding components of (48a).
ζ_{prob}	positivity/Cauchy–Schwarz for the WDW-9 form	the same diagram for (41), transported by the physical isometry.
ζ_{ph}	scalar cancellation in p_9	scalar cancellation in p_{10} ; equivariance makes the equations literal.
ζ_{ray}	projective coequalizer of X_9	transport of the fibrewise coequalizer q_{10} ; its universal square is the same square.
ζ_{sep}	rank-one profile completeness for p_9	rank-one profile completeness for p_{10} ; equality of p and q makes the two biconditionals identical.
ζ_{comp}	$(\Gamma \times \Gamma)$ -equivariance, multiplicativity and Hilbert coherence in $\mathcal{L}_9$	(60e) and the coherence of $\mathcal{L}_{10}$; tensor conjugation identifies every component.
ζ_{cont}	unitary equivariant monoidal WDW-9 continuation	restricted global continuation; the continuation square of (48a) and its tensor powers identify the complete proof.

Every entry is preserved by further restriction because both sides use the restriction squares already proved in W10.20 and Proposition 10.4. Thus the raw data and the six proof coordinates agree, which is (60f). $\square$

Equation (60f) says exactly that the certificate defined in (58b) is $(\text{Raw}_9 \eta_\Phi^9)^* c_9^{\text{QP}}$. Applying QClose gives the corresponding identity of $\mathbf{QS}$ certificates. Hence η_Φ^9 lifts, with constructed certificate components, to

$$\widehat{\eta}_\Phi^9 : \widetilde{\text{Red}}_9^{\text{qs}}(\mathfrak{d}_\Phi^{\text{gl}}) \xrightarrow{\cong} \widehat{\mathfrak{R}}_9 := (\mathfrak{R}_9^*, c_9^{\text{QP}}, c_9^{\text{QS}}, \text{id}) \quad \text{in } \text{Cert}_9^{\text{qs}}(U_\Phi). \quad (60g)$$

Its raw certificate map is $\text{Raw}_9(\eta_\Phi^9)$ and its derived map is $\text{QClose}(\text{Raw}_9(\eta_\Phi^9))$. Finally, define

$$\mathfrak{E}_\Phi^{\text{gl}} := (\mathfrak{d}_\Phi^{\text{gl}}, \widehat{\mathfrak{R}}_9, \widehat{\eta}_\Phi^9) \in \text{EnrPhysRec}_{10}^{\Phi\text{-mark,qs}}(U_\Phi). \quad (60h)$$

Thus the actual object is an inhabitant of all three legs: its underlying comparison is id , its scalar comparison is $\widehat{\chi}_\Phi$, its reduct comparison is the nontrivial certified lift $\widehat{\eta}_\Phi^9$, and its third projection is literally the accepted $\widehat{\mathfrak{R}}_9$, not a second copy of the transported reduct. None is a membership assertion without a displayed map.

Only after (60h), let $U_{\text{mark}}^{\text{qs}}$ and $\text{ScalNeu}_\Phi^{\text{qs}}$ denote the first two projections and define

$$\text{QRes}_{\Phi,U} := \text{QLoc}_U \circ \text{ScalNeu}_\Phi^{\text{qs}} : \text{EnrPhysRec}_{10}^{\Phi\text{-mark,qs}}(U) \longrightarrow \text{QStand}(U). \quad (61)$$

The certified marking isomorphism and the image under QClose of (58a) now give, rather than assume, natural isomorphisms

$$\begin{aligned} \alpha &:= \text{QLoc}(\widehat{\chi}_\Phi) : \text{QExp}_{10} \circ U_{\text{mark}}^{\text{qs}} \Longrightarrow \text{QRes}_\Phi, \\ \beta_{(\mathfrak{d}, \widehat{\mathfrak{E}}_9, \lambda_9)} &:= \text{QClose}(\widehat{\rho}_{\Phi, \mathfrak{d}}) \circ \text{QExp}_9(\lambda_9^{-1}) : \text{QExp}_9 \circ \pi_9 \Longrightarrow \text{QRes}_\Phi. \end{aligned} \quad (62)$$

Lemma 10.6 (Marked certified recovery naturality). *The transformations in (62) are natural on every marked certified arrow and commute with every further restriction. At the positively constructed object (60h) they give*

$$\begin{aligned} \text{QExp}_{10}(U_{\text{mark}}^{\text{qs}} \mathfrak{E}_\Phi^{\text{gl}}) &\xrightarrow[\cong]{\alpha} \text{QRes}_\Phi(\mathfrak{E}_\Phi^{\text{gl}}) \\ &\xleftarrow[\cong]{\beta} \underbrace{\text{QExp}_9(\pi_9 \mathfrak{E}_\Phi^{\text{gl}})}_{\text{QExp}_9(\widehat{\mathfrak{R}}_9) = \Omega_9^{\text{WDW}}} . \end{aligned} \quad (63)$$

Proof. The marked-arrow equation (6c'), after raw extraction, is the naturality square for $\hat{\chi}_\Phi$; applying W9.19 proves the square for α . Equation (58c) is the naturality square for the certified lift $\hat{\rho}_\Phi$; applying QClose and the pullback square for λ_9 proves the square for β . Lemma 10.3 proves that both commute with restriction. At the occupied object, $\lambda_9 = \hat{\eta}_\Phi^9$ by (60h), and (60f) constructs its complete certificate map. Therefore β genuinely starts at the accepted $\hat{\mathfrak{R}}_9$ and compares it with the raw-extraction reduct; it does not start at a tautological copy of that reduct. No certificate is supplied by name or by pullback membership alone. $\square$

10.5 Clock-neutral occupied certification

Proposition 10.4 positively supplied c_{10}^{QP} and W9.19 supplied

$$c_{10}^{\text{QS}} := \text{QClose}(c_{10}^{\text{QP}}). \quad (64)$$

Use the previously constructed certified lift

$$\mathfrak{R}_{10}^{\text{global,qs}} := \hat{\mathfrak{C}}_{10}^{\text{gl}} \in \text{EnrPhysRec}_{10}^{\text{qs}}(X_{10}^{\text{reg}}).$$

Set

$$\mathfrak{Q}_{10}^{\text{gl}} := \text{QExp}_{10}(\mathfrak{R}_{10}^{\text{global,qs}}). \quad (65)$$

Theorem 10.7 (W10.25 — certified global standing export). *The object $\mathfrak{Q}_{10}^{\text{gl}}$ is a nonempty clock-neutral global model with an explicit **QP** certificate and the **QS** certificate derived by W9.19. Its quotient bundle, separated transition weight, continuation, independent composition, and every arrow/restriction naturality coordinate descend functorially. Every admissible clock chart presents the same certified export, lawful atlas replacement gives an isomorphic export, and marked scalar recovery is the certified natural isomorphism (63). Hence*

$$H_{10}^{\text{QS-GLOBAL}} = \text{PASS}.$$

Proof. Equations (60a)–(60e) and Proposition 10.4 supply every raw datum, internal arrow, coherence map, and ζ -certificate. Proposition W9.19 supplies every derived δ -certificate. Lemmas 10.2, 10.3, 10.5, and 10.6 prove functoriality, restriction, marking, reduct, coordinatewise predecessor identity, and recovery. W10.17–W10.19 give atlas independence. No preferred clock or global Stone generator is introduced. $\square$

10.6 The dependency-bearing downstream interface

Definition 10.8 (Schrödinger-side post-closure burdens). A downstream theorem object $\mathfrak{S}$ enters the handoff when:

1. it positively supplies $c_{\mathfrak{S}}^{\text{QP}}$, but assumes none of the standing-descent, quotient-separation, descended-continuation, quotient-composition, or naturality coordinates of $c_{\mathfrak{S}}^{\text{QS}}$;
2. after W9.19 constructs $(P_{\mathfrak{S}}, q_{\mathfrak{S}}, \bar{p}_{\mathfrak{S}}, \bar{C}_{\mathfrak{S}}, \bar{\boxtimes}_{\mathfrak{S}})$, the downstream paper constructs a faithful closed nonrelativistic projective-Hilbert projection

$$\Pi_{\text{NR}} : P_{\mathfrak{S}} \longrightarrow \mathbb{P}L^2(\mathbb{R}^d, dx)$$

preserving $\bar{p}_{\mathfrak{S}}$, $\bar{C}_{\mathfrak{S}}$, standing, and the declared independent-composition law;

3. its projected ray continuation satisfies Wigner implementation, phase-lift, strong-continuity, and Stone-domain hypotheses;
4. the resulting generator H satisfies $\text{SSQ}_1(H)$; and
5. M_V , $\hbar$, m , and all boundary/self-adjointness data are declared and certified downstream.

The domain and preservation laws of Π_{NR} in item 2 are unavailable from item 1: they use the five derived conclusions of W9.19.

Theorem 10.9 (W10.26 — substantive certified handoff). *For every $\mathfrak{S}$ satisfying Definition 10.8, W9.19 is used first:*

$$(\mathfrak{S}, c_{\mathfrak{S}}^{\text{QP}}) \xrightarrow{\text{W9.19}} (P_{\mathfrak{S}}, q_{\mathfrak{S}}, \bar{p}_{\mathfrak{S}}, \bar{C}_{\mathfrak{S}}, \bar{\boxtimes}_{\mathfrak{S}}; c_{\mathfrak{S}}^{\text{QS}}). \quad (66)$$

Only afterward can the faithful projection be typed and the independent Schrödinger burdens applied:

$$c_{\mathfrak{S}}^{\text{QS}} \wedge \text{Faithful}(\Pi_{\text{NR}}) \wedge \text{WignerStone}(\mathfrak{S}) \wedge \text{SSQ}_1(H) \implies H = a\Delta + M_V. \quad (67)$$

The declared scale anchors then give

$$i\hbar \partial_t \psi = \left(-\frac{\hbar^2}{2m} \Delta + M_V \right) \psi.$$

Removing W9.19 removes the standing-bearing domain of Π_{NR} and the four preservation/naturality structures it must respect; these are not premises of $c_{\mathcal{S}}^{\text{QP}}$. No converse, carrier embedding, or derivation of Δ , V , $\hbar$, or m from the occupied WDW line is claimed.

Proof. Item 1 supplies exactly the antecedent of W9.19 and none of its five δ -conclusions, so (66) is substantive. Item 2 then constructs its projection from the derived standing quotient and proves preservation of the derived structures. Wigner–Stone, domains, and SSQ_1 remain independent downstream constructions. The source-integrated *Schrödinger Dynamics* main theorem imports these W9.19 conclusions before defining its faithful projective-Hilbert projection. The occupied WDW objects witness consistency but are not identified with or enlarged into the spatial carrier. $\square$

10.7 Export and publication-status ledger

Table 2: Exact certified WDW-10 to Schrödinger handoff.

Coordinate	Disposition
Original $\mathfrak{R}_9^*$, $I_{9 \rightarrow 10}$, and W10.1–W10.24	Retained exactly; the certified export first enters after W10.24.
NeuPhysRec ₁₀ ^{qs} , EnrPhysRec ₁₀ ^{qs} , and Cert ₉ ^{qs}	Defined by explicit c^{QP} and c^{QS} data with preservation squares, not by rank or field types.
Marked domain	The graph three-way iso-pullback of the $\mathbf{U}_{\text{mark}}$, ScalNeu_Φ , and Red_9 legs in (58), with all comparison isomorphisms retained.
Lifted reduct and graph projection	Equations (58a)–(59) construct Red_9^{qs} on objects and arrows and distinguish it from the third projection π_9 ; Lemma 10.3 proves their restriction squares.
Global raw certificate	Equations (60a)–(60e) construct X_{10} , q_{10} , p_{10} , $\mathcal{L}_{10}$, its internal arrows and monoidal coherence; Proposition 10.4 proves all six ζ -coordinates with restriction and external-arrow naturality.
Actual marked occupancy	Equation (60h) supplies $(\mathfrak{d}_\Phi^{\text{gl}}, \widehat{\mathfrak{R}}_9, \widehat{\eta}_\Phi^9)$; Lemma 10.5 proves the certificate identity with the accepted WDW-9 object coordinate by coordinate.
W9.19	Derives five quotient-level δ -coordinates absent from its representative-level antecedent.
W10.25	Applies W9.19 only after the explicit global raw construction, six-certificate proof, exact predecessor identity, and genuine three-leg occupation.
Faithful flat spatial projection, Wigner–Stone, domains, and SSQ_1	Reserved to the Schrödinger manuscript and applied to the derived standing quotient.
Carrier map $\mathbb{C} \rightarrow L^2(\mathbb{R}^d)$	Not claimed.

Handoff decision.

$$G_{10} = \text{PASS-GLOBAL}, \quad H_{10}^{\text{QS-GLOBAL}} = \text{PASS}, \quad H_{10}^{\text{SCHR-HANDOFF}} = \text{PASS}.$$

The status records a certificate-bearing, semantically load-bearing interface; it does not identify the rank-one WDW carrier with the independently constructed Schrödinger carrier.

11. Audit package and frozen adversarial map

11.1 Proof audit at the load-bearing links

1. **New-clock population.** The volume response follows from $\delta \log d\mu_h = \frac{1}{2} \text{tr}_h \delta h$ and $\delta_N h = 2NK$; it is $c_{\epsilon_*} N$ at the anchor. The curvature response follows from the full variation formula (13) and is $4c_{\epsilon_*} (3 - \Delta)N$.

Coercivity and elliptic regularity prove the inverse. Neither result is a record or homogeneous-monotonicity inference.

2. **Successor graph localization.** Equation (16) localizes the actual WDW-6 smooth trace source on $\tilde{O} = \pi_{\text{pr}}^{-1}(O)$. Separate lemmas prove ideal invariance for local differential generators and finite proper actions. Equations (18)–(20) then retain the complete W5.12/W6.15 label family, typed targets, counterterms, graph topology, literal common domain, and connected proper action. No non-invariant raw function is evaluated on O , no fixed lapse is treated as basic, and no coarse refoliation quotient is used as an off-shell scalar domain.
3. **Common-source chart population.** Equation (20) is one accepted off-shell support. Each $\Gamma_{\alpha,O}$ is constructed by pullback through the full WDW-2 free-regular trivialization $\Psi_{\alpha,O}(z,g)$, not by restriction to $g = e$. Surjectivity onto $\tilde{O}$, freeness, two-sided compact-cap estimates, explicit proper translation, and the $\mathcal{R}_a^*$ slice-change law prove the inverse and its compatibility before every chart operator is defined by conjugation. The inverse source marking and label marking are retained. W5.12/W6.15 are explicitly treated as conditional and are not used to create a WDW-3 quantum refoliation lift.
4. **Quantum occupation.** The W6R.S12 retained finite-rank DeWitt calculation proves $a_{\nu,\star} \neq 0$ and $a_{R,\star} = 12a_{\nu,\star} \neq 0$, followed by a quantitative half-value neighborhood. The three-clock trace lemma proves compact-open finite-word estimates rather than extending W7.2–W7.4 by assertion. The evaluation-naturality lemma first proves $\Gamma_{\alpha}^{\mathcal{D}} \iota_6 = \Gamma_{\alpha}^{\mathcal{K}} \iota_6 = \iota_{\alpha} \hat{Q}_{\alpha}$. The localized authenticated source-kernel lemma proves that the restricted WDW-6 occupant remains in $\mathcal{Z}_6^{\text{src}}$ and retains value one. Equation (21) then defines J_{α}^{src} source-side and J_{α}^{ev} only on its evaluation image; no trace on an arbitrary dual element is invoked. Equations (15a)–(15c) fix a normalized weighted-average coordinate and make $L_{\alpha,z}^{-1} \mathbf{1}$ have unit T_{α} -speed. Direct division of the normalized Cauchy form gives the companion matrix (22b) before the connection (25). The common-normal core lemma proves rather than assumes its restriction to $\mathcal{L}_{\star}$.
5. **Transitions.** $T_{\alpha\beta}$ exists only after both conditioning maps are onto their generated ranges, while $\mathbf{G}_{\alpha\beta} = (G_{\alpha\beta}^V, G_{\alpha\beta}^{\mathcal{K}}, G_{\alpha\beta}^{\mathcal{D}}, \mu_{\alpha\beta})$ is independently defined by cancellation through the common source: $G_{\alpha\beta}^V = \hat{Q}_{\alpha} \hat{Q}_{\beta}^{-1}$, $(G_{\alpha\beta}^{\mathcal{K}}, G_{\alpha\beta}^{\mathcal{D}}) = (\Gamma_{\alpha}^{\mathcal{K}} (\Gamma_{\beta}^{\mathcal{K}})^{-1}, \Gamma_{\alpha}^{\mathcal{D}} (\Gamma_{\beta}^{\mathcal{D}})^{-1})$, and $\mu_{\alpha\beta} = \rho_{\alpha} \rho_{\beta}^{-1}$. Equations (28) and (33)–(35) include the common-family normalized-lapse relation, base derivative, clock Jacobian, and independently derived companion terms on the proved common-normal core.
6. **Coherence.** Conditioning maps cancel on their physical ranges while the middle $\hat{Q}$, Γ , and ρ maps cancel on their exact source/graph domains. The proper-arrow coordinate is transported through each map rather than erased. The μ -cocycle, Jacobian chain rule, and WDW-2 proper-presentation 2-cells are proved before strictification; Hilbert-line equality is not promoted to graph coherence.
7. **Positive descent.** The refinement-localized clock datum (38a), quotient relation (37), anchor trivialization (39), and nonzero section (40) construct $\mathfrak{R}_{10}^{\text{und}}$ in the independently defined NeuPhysRec₁₀ recipient; equation (39b) then places that object, the populated atlas, incidence, transitions, coherence, holonomy, markings, and gluing certificates in the independently defined EnrPhysRec₁₀. No distinguished global profile, stack label, or retrospective category enlargement supplies existence.
8. **Uniqueness.** Any rival live transition must commute with the onto source conditioning map and the independently constructed common-source and label markings. Empty-word evaluations separate the restriction quotient, so the rigidity lemma eliminates a hidden graph centralizer; the normalized authenticated section removes phase freedom. The proof does not attribute uniqueness to W6.15.
9. **Universality and containment.** Equation (47) prints the variance and mapping bijection on the underlying NeuPhysRec₁₀ object, with local objects entering through $\mathbf{S}_{\text{neu}}$. Equations (5a)–(6e) independently define the clock-descent, neutral, enriched object and arrow signatures, restrictions, comparison-marked category, and reduct on objects and arrows. Equation (48) first constructs an unmarked enriched isomorphism whose clock component is the independent refinement roof (47a); (48') uses it to tie the underlying neutral object to the selected scalar member, and only then do (48a)–(49) apply the typed reduct and compare the full WDW-9 signature rather than one line. No inverse from bare WDW-9 data to the enriched restriction is claimed.

11.2 Bounded re-review of W10.9–W10.12 and direct consequences

Node	Re-tested dependency	Result
W10.9	Normalized decomposition, saturated raw ideal, one common graph support, orbit-complete chart/source markings, evaluation naturality, intrinsic noncharacteristicity, source-side trace estimates, three kernel levels, and physical line	PASS: (15a)–(15c), (16)–(26); $\Psi_{\alpha,O}$ covers the full saturated preimage, g remains in the Cauchy tangential variables, X_{10}^{pt} and $\mathcal{R}_{10}^{\text{rf}}$ are separate, no quantum refoliation lift is assumed, and no fixed-lapse scalar-basic carrier remains.
W10.10	Six ordered complete transitions	PASS: (29) forms every $\mathbf{G}_{\alpha\beta} = (G_{\alpha\beta}^V, G_{\alpha\beta}^K, G_{\alpha\beta}^D, \mu_{\alpha\beta})$ from two independently formed presentations of one common support; inverse rows cancel the same chart maps.
W10.11	Typed grammar/domain, joint-kernel, and normal-label preservation	PASS: (30)–(35) use the primal/domain/dual transition square and total chart-label recoding in both directions, transport the source, evaluation, and transposed kernels separately, prove the common-family equality between i_{β}^{nor} and the $j_{\alpha\beta}$ -scaled i_{α}^{nor} , and only then apply the companion/Jacobian law.
W10.12	Enriched triple composition	PASS: (36) proves both G and μ cocycles, finite-word and target-sort composition, joint-kernel path independence, affine companion chaining, and WDW-2 proper-presentation 2-cell coherence before strictification; the $\mathcal{R}_a^*$ slice-change maps compose by proper-groupoid associativity.
W10.3	Clock-neutral recipient	PASS: (5a)–(5c) define ClockDesc_{10} , its roof arrows, transition-supported natural isomorphisms and restriction, NeuPhysRec_{10} , and the singleton embedding conservative on the complete non-clock signature before descent.
W10.13	Populated neutral object	PASS: (38a) supplies the missing clock-descent coordinate before (39b) claims enriched membership; (37)–(41) populate the non-clock coordinates.

Node	Re-tested dependency	Result
Direct down-stream	W10.14–W10.26 PASS after certified source-native reciprocal integration, following descent of complete equivariant graph packages and physical lines; holonomy, refinement, universality, recovery, exhaustion, and the complete probe retain the label action and the plural clock representative. W10.13 types the neutral first coordinate and full tuple in EnrPhysRec_{10} ; W10.20 constructs an arrow in $\text{EnrPhysRec}_{10}^{\Phi\text{-mark}}$, separates it from exact predecessor recovery under object/arrow Red_9 , and W10.23 keeps successor markings visible outside the recovery block. No downstream theorem reconstructs or enlarges a failed W10.9 map. Equations (60a)–(60e) construct the global representative datum and Proposition 10.4 proves all six ζ -coordinates before W10.25 applies QClose ; W10.26 first invokes W9.19 to construct the standing quotient and its descent/naturality certificate and then treats the spatial projection, Wigner–Stone, and SSQ_1 clauses as new downstream burdens.	

11.3 Verification scope

The source lineage, clock registry, transversality data, local quantum ledger, overlap and transition data, coherence conditions, descent population, obstruction analysis, refinement comparison, WDW-9 conservative reduct, candidate disposition, and complete-probe specification are all represented by the theorem-bearing text and tables of this manuscript.

Verification boundary. Mechanical checks cover source lineage, cross-references, overlap coverage, dependency order, gate consistency, endpoint labels, and LaTeX warnings. It does not prove the response inverses, noncharacteristic graph theorem, unitarity, descent, holonomy, universality, maximality, the downstream Hamiltonian-response exhaustion, or imported WDW-1–9 results.

11.4 Adversarial adjudication A–CY

An objection is closed only by the exact theorem or construction shown. Kernel governance is not used to populate a clock or to replace an analytic or categorical proof.

ID	Challenge	Theorem-level adjudication
A	One global clock is assumed.	CLOSED: W10.2, W10.8, and W10.13 use a cover and descent; no chart is global.
B	The clocks are reparameterizations on the homogeneous solution.	CLOSED: W10.7 uses neighborhood Fréchet differentials and differential order, not one history.
C	A changing volume record is promoted to a clock.	CLOSED: W10.5 proves only the independent profile inverse and certified intersection; full standing is deferred to W10.9.
D	Rank-one $\mathcal{H}_{\text{phys}}$ makes the atlas vacuous.	CLOSED: W10.2 and W10.9 retain base domains, profiles, normal responses, records, continuation, and failures.

ID	Challenge	Theorem-level adjudication
E	The native category is only declared.	CLOSED: W10.2–W10.3 construct objects, arrows, pullbacks, composition, automorphisms, and restrictions.
F	Transitions are defined from the desired global object.	CLOSED: W10.10 uses independently formed conditioning maps before W10.13.
G	Numerical clock-range overlap is used.	CLOSED: W10.8 and equation (27) use source pullbacks and retained quotient events.
H	Pairwise maps are treated as enough.	CLOSED: W10.12 proves complete triple and representative quadruple coherence.
I	Unitarity is inferred from one vector.	CLOSED: equation (31) holds on every vector in the exact generated fibre.
J	Constraint covariance is assumed after chart change.	CLOSED: (16)–(20), (30), and (32) construct saturated localization, total label transport, typed counterterm/target squares, graph bicontinuity, and exact common-domain intertwiners.
K	Equal expectations replace observable intertwining.	CLOSED: W10.11 intertwines the entire represented physical algebra on its domain.
L	Records are equated by clock values.	CLOSED: W10.11 compares complete records at one source event; readouts remain distinct.
M	A turning point is repaired by a new chart.	CLOSED: W10.21 retains the failed prefix; a later patch has new lineage identity.
N	Holonomy is ignored because the base is contractible.	CLOSED: W10.15 computes actual defect and loop words; contractibility is unused.
O	Every residual phase is skin.	CLOSED: only frame coboundaries fixing $\mathcal{P}_{10}$ are skin; probe-visible phases are physical holonomy.
P	Strict cocycle is notation.	CLOSED: W10.12 retains proper-presentation 2-cells and proves quadruple coherence before strictification.
Q	Atlas independence is checked only for the seed.	CLOSED: W10.17 applies to arbitrary same-target atlases whose cross-transitions have first been positively constructed; it pulls their data back separately to a common source cover and compares them by the transition-supported roofs (44a) and their W10.12 inverse. W10.19 proves that every passing arbitrary chart supplies this antecedent.
R	C_Φ is a hidden preferred clock.	CLOSED: it is marked only as the input to (48)–(49); the unmarked object uses (37)–(39).
S	Multiple choice is eliminated by definition.	CLOSED: W10.19 permits physical plurality and uses independently proved transition/descent/holonomy theorems.
T	A different clock necessarily changes the theorem object.	CLOSED: W10.5–W10.7 retain all non-clock coordinates while constructing profile-divergent atlas roles.
U	A momentum/York clock is inserted informally.	CLOSED: no C_K appears in the construction; it remains reserve-only.
V	Improper diffeomorphism or boundary time is absorbed as skin.	CLOSED: W10.1 freezes connected proper and compact-boundaryless scope; changes are D0.
W	A new clock repairs anomaly or domain failure.	CLOSED: W10.21 copies every non-clock failure into all charts.
X	One gluing orbit follows from chart count.	CLOSED: W10.16 uses onto source-conditioning squares, independent source/label markings, and evaluation rigidity.
Y	Kernel necessity populates the clocks.	CLOSED: W10.5–W10.6 use explicit multiplier/elliptic inverses and certified intersections; W10.9 constructs the quantum packages.

ID	Challenge	Theorem-level adjudication
Z	The global object repairs a missing local inhabitant.	CLOSED: W10.4–W10.9 precede all transitions and descent; the stop rule is explicit.
AA	A sheaf/stack label proves descent.	CLOSED: equations (37)–(41) and section (40) explicitly construct effective descent.
AB	Refinement drops failures.	CLOSED: W10.17 requires complete failure preservation and uses the same source base.
AC	The theorem claims full superspace or singularity resolution.	CLOSED: W10.1 fixes X_{10}^{reg} and prints the first-failure boundary/nonclaims.
AD	WDW-9 is recovered only as a Hilbert line.	CLOSED: Red_9 retains the complete predecessor grammar/domain signature and W10.20 checks thirteen roles, twenty overlaps, repairs, failures, and nonclaims.
AE	A Standard-Model/source clock is another chart.	CLOSED: changed matter, gauge, or source is D0 and a separate enrichment.
AF	Neutrality erases real local plurality.	CLOSED: profiles, domains, anchors, and failures remain in the atlas; only global preference is absent.
AG	Common $\mathcal{H}_{\text{phys}}$ makes every clock identical.	CLOSED: W10.7 retains different profile tensors, differential orders, normal responses, and failures.
AH	Cocycle cancellation occurs outside domains.	CLOSED: W10.12 works on typed triple pullbacks and exact generated ranges.
AI	Physical holonomy is declared inadmissible.	CLOSED: W10.15 would retain any probe-visible loop as physical; direct calculation makes this ledger empty.
AJ	Optional extension inflates the claim.	CLOSED: W10.21 and W10.24 explicitly withhold PASS – GLOBAL – EXTENDED.
AK	A negative obstruction result is presented as success.	CLOSED: W10.13 exhibits a nonzero global section and W10.16 a live singleton orbit.
AL	Mechanical checks certify the mathematics.	CLOSED: the verification boundary above disclaims analytic and categorical certification.
AM	The physical line is renamed as the WDW-6 graph domain.	CLOSED: (20) constructs the full localized common graph object and domain; (24) separately constructs the physical line from the joint kernel.
AN	A normal connection is used as a full base connection.	CLOSED: (23)–(25) distinguish each $\mathcal{N}_\alpha$ from its proved $\mathcal{L}_\star$ restriction; (34)–(35) apply only on that core; transverse connection is a nonclaim.
AO	Local invertibility is promoted to global injectivity.	CLOSED: Section 3 fixes quantitative injective flow boxes and treats exit as failure; no maximal-component claim remains.
AP	W10.5/W10.6 depend on W10.9 and W10.9 depends back on them.	CLOSED: W10.5–W10.6 end with analytic clock data; W10.9 alone establishes complete PhysRec_{10} membership.
AQ	P_α^{-1} is assumed although restriction kills nonzero global source functions.	CLOSED: (16) forms the saturated raw restriction quotient first; $\hat{Q}_\alpha$ is then an isomorphism of that localized object through Ψ_α , while global restriction is never inverted.
AR	The chosen common lineage is not the chart-normal direction.	CLOSED: each $\mathcal{N}_\alpha$ is constructed from the exact inverse response, and the common-normal core lemma proves equality on $\mathcal{C}_\star$ from homogeneity and uniqueness.
AS	Restricting to one lineage collapses the three clocks to reparameterizations.	CLOSED: the target retains full transverse successor germs; W10.7 separates their Fréchet differentials on every representative neighborhood. Only the descended point set is the retained lineage.

ID	Challenge	Theorem-level adjudication
AT	Smoothing or counterterm operators need not descend through the restriction quotient.	CLOSED: the local-differential ideal lemma checks their explicit WDW-6 action; spatial smoothing occurs in local coefficients/targets and does not replace the source configuration argument.
AU	The weighted integral, profile decomposition, and inverse-response normal use incompatible scales.	CLOSED: (15a)–(15c) divide by $Z_\alpha = \int w_\alpha d\mu$; (22b), (25), and (28) use that same unit-speed convention.
AV	A non-invariant raw source function is restricted to quotient O , and proper actions violate the ideal.	CLOSED: (16) uses $\tilde{O} = \pi_{\text{pr}}^{-1}(O)$; separate differential and finite-action lemmas prove ideal invariance; the common graph object is formed on the raw restriction quotient without scalar descent.
AW	A fixed lapse is incorrectly treated as basic.	CLOSED: WDW-3 fixed-lapse nonbasicness is imported explicitly; (18)–(20) retain the full abstract descriptor family and (30) re-expresses its labels between chart coordinates.
AX	Label transport is mentioned but no map is constructed.	CLOSED: (29) positively defines $\mu_{\alpha\beta} = \rho_\alpha \rho_\beta^{-1}$ from independently formed chart label recodings, with inverse and cocycle by cancellation.
AY	Proper and refoliation quotients are conflated.	CLOSED: W10.1 separates X_{10}^{pr} from $\mathcal{R}_{10}^{\text{rf}}$ and expressly denies that $\mathcal{R}_{10}^{\text{rf}}$ acts on the off-shell quantum support.
AZ	AASC is asked to manufacture the equivariant carrier.	CLOSED: WDW-5/6 populate the one common support and the WDW-2 orbit-complete trivialization populates each chart presentation before AASC exhaustion.
BA	Graph bicontinuity ignores reindexing of the complete label family.	CLOSED: the chart-presentation theorem proves both Ψ_α and Ψ_α^{-1} compact-cap estimates and retains the bijections ρ_α and their inverses.
BB	Joint-kernel preservation is proved in only one direction.	CLOSED: every source chart kernel is the bijective Q_α image of $\mathcal{Z}_6^{\text{src}}(O)$, every evaluation kernel is the Γ_α^{D} image of $\mathcal{Z}_6^{\text{ev}}(O)$, and every transposed kernel is the Γ_α^{K} image of $\mathcal{Z}_6^{\text{x}}(O)$; (29) gives all three equalities in both directions.
BC	The label inverse or cocycle is omitted.	CLOSED: W10.10 gives $\mu_{\beta\alpha} = \mu_{\alpha\beta}^{-1}$ and W10.12 proves $\mu_{\alpha\beta}\mu_{\beta\gamma} = \mu_{\alpha\gamma}$.
BD	The normal Jacobian is absent from the lapse label.	CLOSED: equation (33) proves in the common label family that i_β^{nor} equals the $j_{\alpha\beta}$ -scaled i_α^{nor} before the companion law.
BE	Physical-line descent repairs an off-shell graph failure.	CLOSED: W10.9 constructs and verifies the complete common graph object and all chart maps before the joint kernel and physical line; W10.13 descends both separately.
BF	Holonomy ignores the label action.	CLOSED: W10.15 computes loop action on both G and μ and the complete probe contains a separate label block.
BG	A presentation 2-cell is promoted to literal equality.	CLOSED: W10.12 uses literal common-chart cancellation and separately states the WDW-2 proper-slice natural isomorphism and its quadruple coherence.
BH	An arbitrary associated target is introduced without standing.	CLOSED: target sorts are definitionally determined by the common WDW-5/6 label and only re-expressed by ρ_α ; the exhaustion theorem rejects independent additions.
BI	W5.12/W6.15 are conditional matching theorems misread as existence of a quantum refoliation lift.	CLOSED: W10.9 states that boundary explicitly, constructs no such lift, and obtains every populated chart map by conjugation from one accepted support.

ID	Challenge	Theorem-level adjudication
BJ	Classical response invertibility is mistaken for quantum noncharacteristicity.	CLOSED: W10.9 separately places the pure-trace clock covectors in W6R.S12's retained finite-rank Hessian channel, contracts them with the accepted DeWitt multiplier, proves nonzero anchor coefficients, and prints the half-value neighborhood bound.
BK	W7.2–W7.4 are extended from the scalar clock to the two intrinsic clocks by analogy.	CLOSED: the three-clock trace lemma proves the compact-open finite-word estimates for each new inverse flow box and uses the newly proved coefficient bounds before division.
BL	The singleton gluing orbit assumes that the off-shell matching has trivial centralizer.	CLOSED: W10.16 proves empty-word evaluation/source-marking rigidity on the dense restriction quotient, extends it by bicontinuity, fixes labels by ℓ_α , and removes phase freedom with the normalized authenticated section.
BM	The physical-line transition is used as an endomorphism of the full Cauchy carrier.	CLOSED: equation (34) is a generated-line vector identity with Y_α printed in every term; it expressly declines to promote $T_{\alpha\beta}$ to a full-carrier map.
BN	The authenticated WDW-6 source solution is confused with the full transposed joint kernel.	CLOSED: W10.9 defines $\mathcal{Z}_6^{\text{src}}$, $\mathcal{Z}_6^{\text{ev}}$, and $\mathcal{Z}_6^\times$ separately, proves $\Gamma^{\mathcal{D}}\iota_6 = \Gamma^{\mathcal{K}}\iota_6 = \iota_\alpha\hat{Q}_\alpha$ by evaluation, proves the localized authenticated source-kernel lemma, defines J_α^{src} before J_α^{ev} , expressly denies a trace on arbitrary dual elements, and W10.10–W10.12 transport all three levels through G^V , $G^{\mathcal{D}}$, and $G^{\mathcal{K}}$.
BO	Agreement of every predecessor projection is promoted to an isomorphism of the whole enriched objects.	CLOSED: W10.20 separates the enriched isomorphism η_Φ^{10} from the predecessor comparison η_Φ^9 and claims (49) only after Red_9 .
BP	The empty WDW-9 atlas slot is treated as a coordinate that every lawful successor must keep empty.	CLOSED: W10.3 identifies it as an authorization boundary; W10.9 positively constructs successor atlas data and W10.20 recovers only the predecessor signature.
BQ	The reduct obtains recovery by discarding WDW-9's own grammar, labels, domain, counterterms, or kernel data.	CLOSED: W10.3 lists those coordinates as retained by Red_9 , W10.20 compares them, and $\mathcal{P}_{\text{recovery}}$ probes the resulting complete predecessor tuple.
BR	The scalar inclusion is used as an inverse that manufactures successor atlas coordinates from bare WDW-9 data.	CLOSED: W10.20 types the scalar inclusion only as $(\eta_\Phi^{10})^{-1}$ between enriched objects and expressly claims no inverse from $\mathfrak{R}_9^*$.
BS	Applying the recovery reduct erases the successor markings needed for gluing rigidity and complete-probe maximality.	CLOSED: Red_9 occurs only in $\mathcal{P}_{\text{recovery}}$ and $\text{Res}_{9,\Phi}$; W10.16 and the other W10.23 probe blocks continue to retain the atlas, source/label markings, transitions, coherence, and holonomy.
BT	Defining the complete marked scalar presentation from the global restriction makes predecessor recovery circular.	CLOSED: η_Φ^{10} only chooses a representative of the already constructed descent object; the load-bearing $\epsilon_\Phi^{\text{loc}}$ is proved in W10.9 from the local lift and W10.4 import before descent, and ϵ_Φ merely discards successor-only coordinates through the independently defined Red_9 .
BU	The alleged enriched global object lies outside the category on which recovery is performed.	CLOSED: W10.3 defines (6a) with every atlas, incidence, transition, coherence, holonomy, marking, and gluing coordinate; W10.13 supplies exactly that tuple in (39b), and enriched restriction preserves it.
BV	The reduct is listed on objects but has no arrow action, composition law, or restriction naturality.	CLOSED: (6b) defines enriched arrows and their composition, (6d) defines $\text{Red}_9(\Xi)$ from the marked local arrow, and (6e) proves object-and-arrow restriction naturality.

ID	Challenge	Theorem-level adjudication
BW	The local scalar lift is silently treated as the full restricted global presentation.	CLOSED: Sing_Φ creates a typed singleton-enriched local lift before descent, whereas $\mathfrak{R}_\Phi^{10,\text{mark}}$ explicitly retains the full restricted atlas and certificates; (48) is a separately constructed enriched isomorphism between the restricted global object and that complete presentation.
BX	The reduct extracts an unrelated marked local certificate rather than reducing the underlying global object.	CLOSED: W10.3 requires the comparison $\chi_\Phi : \mathfrak{R} \cong \mathfrak{R}_{a_\Phi}$ as part of every marked object; W10.20 first proves (48) unmarked, then sets $m_\Phi^{\text{gl}} = (a_\Phi, \text{For}_{10}(\eta_\Phi^{10}))$ and verifies the marking square before applying Red_g .
BY	Restriction to one proper slice is injective on the non-basic WDW-6 carrier.	CLOSED: The final theorem does not make that claim. It replaces the one-slice map by $\hat{Q}_{\alpha,O}F(z,g) = F(\Psi_{\alpha,O}(z,g))$ on the full proper-arrow coordinate; surjectivity of $\Psi_{\alpha,O}$ proves that its kernel is exactly $\mathcal{I}_{\text{raw}}(\tilde{O})$.
BZ	The defect can be repaired by silently retaining only proper-basic functions.	CLOSED: the algebraic source remains the full non-basic $\Phi_{6,O}^{\text{tr}}$, and g remains a chart and Cauchy tangential coordinate. No invariant subalgebra or narrower completion is substituted.
CA	The inverse of $\hat{Q}_\alpha$ is obtained merely by declaring the codomain to be its image.	CLOSED: W10.9 first proves that Ψ_α is the free-regular slice diffeomorphism onto the entire saturated raw preimage, prints $f \circ \Psi_\alpha^{-1}$, and proves both compact-cap seminorm estimates before completion.
CB	Proper-redescription endomorphisms disappear in the enlarged chart.	CLOSED: W10.9 prints $(\hat{U}_h f)(z,g) = f(z,h^{-1}g)$ and proves $\hat{Q}_\alpha U_h = \hat{U}_h \hat{Q}_\alpha$ for every registered composable proper arrow.
CC	The Cauchy trace reintroduces the one-slice collapse after the carrier repair.	CLOSED: equations (15d) and (21) impose the already declared physical gauge $\chi_\alpha = 0$ and fix T_α , but retain g and every non-profile tangential source variable; the trace lemma estimates compact subsets of $\mathcal{P}_{\alpha,O} \cap \{\chi_\alpha = 0\}$ at fixed g .
CD	A change of proper slice produces an untracked selector or incoherent new presentation.	CLOSED: W10.9 prints $\hat{Q}^{s'} = \mathcal{R}_a^* \hat{Q}^s$, derives the triangular Cauchy map $\mathcal{J}_a$ including $\mathcal{R}_a^*$, and proves their inverse and product laws; W10.12 derives triple and quadruple compatibility from the product rule and proper-groupoid associativity.
CE	The clock-neutral global realization is placed in a category whose objects require one clock.	CLOSED: W10.3 keeps PhysRec_{10} as the single-clock local category, defines ClockDesc_{10} and NeuPhysRec_{10} independently in (5a)–(5c), and W10.13 supplies $\mathfrak{C}_{10}^{\text{gl}}$ before asserting $\mathfrak{R}_{10}^{\text{und}} \in \text{NeuPhysRec}_{10}$.
CF	ClockDesc_{10} is only a label; its arrows, composition, or restrictions are absent.	CLOSED: W10.3 defines arrows as lawful-refinement roofs, equality by common refinement, composition by source fibre product, identities, associativity, and componentwise restriction before the populated descent.
CG	The neutral-to-scalar recovery comparison secretly chooses C_Φ as a global clock.	CLOSED: (47a) exists only after restriction to U_Φ ; its left leg is literal refinement, its right leg is the proved $t_{\Phi\alpha}$ equivalence with inverse $t_{\alpha\Phi}$, and (38a) remains unmarked globally.
CH	Passing to ClockDesc_{10} erases the real local plurality required by the endpoint.	CLOSED: (38a) retains the three profiles, readout relations, anchors, orientations, continuations, and failures; (39b) retains the chosen atlas and all successor markings; $\mathcal{P}_{\text{clockdesc}}$ and the other successor probe blocks test both layers.

ID	Challenge	Theorem-level adjudication
CI	The singleton functor narrows or reconstructs the accepted local physical objects.	CLOSED: S_{neu} changes only the first coordinate from C to $S_{\text{clk}}(C)$ and is conservative on the complete non-clock signature; every graph, domain, kernel, form, record, continuation, failure, and non-claim coordinate is retained literally.
CJ	The clock-descent coordinate is inferred from the desired global physical quotient or from WDW-9 recovery.	CLOSED: (38a) is formed from W10.8's populated cover and the readout components of W10.10–W10.12 before (37)–(41) assemble the non-clock quotient and before W10.20 constructs recovery.
CK	Genuine clock change is illicitly renamed a lawful refinement in the scalar comparison.	CLOSED: the left leg r_Φ in (47a) is literal restriction, while the right leg u_Φ is expressly a different 1-arrow with components $t_{\Phi\alpha}$; its inverse selects the Φ member and its counit has components $t_{\alpha\Phi}$. W10.17 preserves the same polarity for arbitrary atlases.
CL	The neutral global object has no coherent local singleton presentations for universal-ity or recovery.	CLOSED: (38b) constructs every ζ_α , (39a) constructs every $\xi_\alpha : j_\alpha^* \mathfrak{R}_{10}^{\text{und}} \cong S_{\text{neu}}(\mathfrak{R}_\alpha)$, and W10.12 proves their overlap compatibility before W10.18 or W10.20 uses them.
CM	A common source cover is silently promoted to a common clock-data refinement.	CLOSED: W10.17 forms $\mathcal{A}_W$ and $\mathcal{B}_W$ separately on W_{ij} ; only r_A and r_B are literal refinements. The right legs use $(t_{ji}, T_{ji}, \mathbf{G}_{ji}, \dots)$ and their W10.12 inverses, so no datum is required to carry two distinct profiles literally.
CN	The rank-one WDW line is silently enlarged to $L^2(\mathbb{R}^d)$.	CLOSED: W10.25 retains the rank-one physical bundle; Definition 10.8 types the flat spatial carrier as an independently constructed role instance; Table 2 denies a carrier map.
CO	The downstream Schrödinger theorem is used to prove the WDW endpoint.	CLOSED: W10.24 is complete before Section 10; W10.25–W10.26 are additive, and every projection, Wigner–Stone, domain, and SSQ_1 clause remains a downstream burden.
CP	Recovery-signature leakage: W9.19 or its derived coordinates are inserted into the W10.1 predecessor and then claimed recovered by W10.20.	CLOSED: W10.1 retains the original $\mathfrak{R}_0^*$ and $I_{9 \rightarrow 10}$ byte-for-byte; W9.19 first enters through the separate post-endpoint record (52). The core Cert_9 , Red_9 , W10.20, and W10.24 signatures are unchanged, while (58a)–(60h) construct the later certified reduct and occupied recovery pullback separately.
CQ	Premise irrelevance: deleting W9.19 leaves W10.26's mathematical implication unchanged.	CLOSED: Definition 10.8 assumes only $c_{\mathfrak{S}}^{\text{QP}}$. Equation (66) obtains the standing-bearing quotient, separated $\bar{p}$, descended $\bar{C}$ and $\bar{\Xi}$, and naturality by W9.19. The Schrödinger main theorem defines Π_{NR} on that quotient and requires preservation of those derived structures. Removing W9.19 removes the typed domain and four preservation obligations; none is a c^{QP} premise.
CR	Signature mismatch: a raw antecedent or derived conclusion differs across the three manuscripts.	CLOSED: all three rebuilt sources import byte-identical <code>qs_boundary.tex</code> , separating six representative ζ -proofs from five derived δ -conclusions. QP-Comp is explicitly $(\Gamma \times \Gamma)$ -equivariant through scalar multiplication, exactly the two-input law used by W9.19; Hilbert geometry, Wigner–Stone, domains, and SSQ_1 remain downstream.

ID	Challenge	Theorem-level adjudication
CS	$\mathbf{QExp}$ is notation for a coordinate deletion, not a functor.	CLOSED: (53a)–(54) define the certified lift, $\mathbf{QRaw}$, and $\mathbf{QLoc} = \mathbf{QClose} \circ \mathbf{QRaw}$; (56a) defines the enriched composite. Lemma 10.2 proves identities, composition, certificate preservation, monoidal coherence, and restriction naturality.
CT	$\mathbf{QRes}_{\Phi}$ applies an enriched-domain functor to a local $\mathbf{PhysRec}_{10}$ object.	CLOSED: (61) gives $\mathbf{QRes}_{\Phi,U} = \mathbf{QLoc}_U \circ \mathbf{ScalNeu}_{\Phi,U}^{\mathbf{qs}}$ on the simultaneous certified pullback. The local scalar object first enters $\mathbf{NeuPhysRec}_{10}^{\mathbf{qs}}$ through its certified singleton lift; (62) applies χ_{Φ} , and Lemma 10.6 proves the naturality squares.
CU	The export specifies tensor-power object components but no action on an internal comparison-groupoid arrow $S : n \rightarrow m$.	CLOSED: (55a) defines $S \mapsto T^{\otimes m} S (T^{\otimes n})^{-1}$ for every such arrow. Lemma 10.2 proves its identity, composition, tensor, unit, associator, and symmetry laws.
CV	The superscript $\mathbf{qs}$ is inferred from rank, positivity, unitarity, and typed record fields rather than explicit predictive data.	CLOSED: (53a)–(53c) define every $\mathbf{qs}$ category by an object $c^{\mathbf{QP}}$ with all six ζ -proofs, an object $c^{\mathbf{QS}}$ with all five δ -proofs, and an isomorphism to $\mathbf{QClose}(c^{\mathbf{QP}})$. Rank and types alone imply no membership.
CW	The marked domain does not ensure that the underlying, scalar, and reduct exports inhabit their claimed certified categories.	CLOSED: (58) is the graph three-way iso-pullback with explicit comparisons $\lambda_{10}, \lambda_{\Phi}, \lambda_9$; equation (60h) positively places the actual marked global object in it before (61)–(63) are formed.
CX	The reduct certificate is supplied only by an undefined name or by pullback membership.	CLOSED: (58a) constructs the raw reduct comparison; (58b) defines the $\mathbf{QP}$ and derived $\mathbf{QS}$ components on objects; (58c) defines the certificate map on arrows; (59) distinguishes the resulting raw-extraction functor $\mathbf{Red}_9^{\mathbf{qs}}$ from the graph projection π_9 . No $\mathbf{RedCert}$ symbol is used.
CY	The accepted WDW-9 certificate and the actual marked global object are never connected by a certified arrow.	CLOSED: (60f) proves the complete certificate identity along η_{Φ}^9 , (60g) constructs $\hat{\eta}_{\Phi}^9$ in $\mathbf{Cert}_9^{\mathbf{qs}}$, (60h) uses the accepted $\hat{\mathfrak{R}}_9$ as the actual third leg, and only then do (62)–(63) form β and exact marked recovery.
CZ	The global $c_{10}^{\mathbf{QP}}$ is declared without constructing its representative carrier, quotient, transition weight, comparison groupoid, internal arrows, or coherence.	CLOSED: (60a)–(60e) define $X_{10}, q_{10}, p_{10}, \mathcal{L}_{10}$, every internal phase arrow, tensor product, unit, associator, unitors, symmetry, $m_{10,n}$, independent composition, and continuation from the global Hermitian line and authenticated unit section.
DA	The six global ζ -coordinates or the equality with the accepted WDW-9 certificate are asserted only in aggregate.	CLOSED: Proposition 10.4 proves each ζ separately and displays its restriction and external-arrow squares. Lemma 10.5 proves (60f) through a coordinate table covering every raw datum and each of the six proof coordinates.

11.5 Refutation criterion and review status

A faithful defeat must preserve the frozen target and either exhibit a same-scope candidate that passes W10.1–W10.23 yet produces a second complete-probe-visible global class, or break a particular premise or inference in those theorems. A target change, failed local clock, numerical overlap, untyped map, or already classified presentation variation is not such a counterexample.

This manuscript closes the adversarial routes A–DA at theorem level, including source-native reciprocal integration for CQ and CR. The map records the germ-injectivity, chart-normal, normalization, raw-quotient, label-equivariance, conditional-matching, noncharacteristicity, trace-extension, graph-centralizer, predecessor-recovery, categorical-domain, carrier, coherence, neutrality, plurality-retention, dependency-polarity, carrier-non enlargement, recovery-signature, semantic premise-relevance, signature-equality, export-functoriality, recovery-domain, internal-arrow, certificate-membership, marked-pullback, lifted-reduct-certificate, global-certificate population, coordinate-

wise predecessor identity, and positive occupancy objections in their stated closed or pending disposition. This authored verification map is not a substitute for independent review. Such review must inspect the elliptic response, certified-neighborhood bounds, normalized profile decomposition, restriction-ideal and proper-action invariance, orbit-complete common-source charts, evaluation naturality, source/dual separation, the finite-rank DeWitt argument, graph completion, joint-kernel preservation, normal-label Jacobian, companion law, source-anchor factorization, arbitrary-atlas quantification, conservative-reduct functoriality, complete-probe maximality, and WDW-9 import fidelity.

11.6 Publication reproducibility

The publication source builds deterministically and cross-checks all theorem nodes, core gates, handoff statuses, clocks, ordered transitions, nonempty incidences, chart markings, naturality and noncharacteristic certificates, WDW-9 roles and overlaps, comparison spans, adversarial rows, the unique live gluing orbit, and the export/non-export ledger. A missing reference, dependency inversion, non-passing core gate, or LaTeX warning blocks publication.

12. References and source lineage

- [1] A. J. Maley, *Non-Degenerate Construction and the Kernel of Admissibility: An Exhaustion Theorem for Standing, Reference, and Irreversibility*, 2026.
- [2] A. J. Maley, *Fixed-Domain Exhaustion and the Complete Disposition of Admissibility-Bearing Variations*, 2026.
- [3] A. J. Maley, *Same-Scope Operator Exhaustion under AASC*, 2026.
- [4] A. J. Maley, *Bivalence of Admissibility and Terminal Same-Scope Decision*, 2026.
- [5] A. J. Maley, *The Physical Clock as a Quotient: AASC Time Non-Primitivity II — Internal Time and Clock Choice in Constrained Quantum Gravity*, 2026.
- [6] A. J. Maley, *Connection-Typed Canonical Projection of the Frozen Einstein–Source Realization*, WDW-1, 2026.
- [7] A. J. Maley, *Groupoid-Faithful Proper-Redescription Descent of the AASC Canonical Gravity Realization*, WDW-2, 2026.
- [8] A. J. Maley, *Normal Deformation as Constrained Refoliation: Groupoid-Faithful Hypersurface-Deformation Closure under AASC*, WDW-3, 2026.
- [9] A. J. Maley, *Classical Matter Occupancy and Joint-Constraint Closure under AASC*, WDW-4, 2026.
- [10] A. J. Maley, *Configuration-Space Differentiation and the Constraint-Native Wheeler–DeWitt Candidate Family*, WDW-5, 2026.
- [11] A. J. Maley, *Quantum Constraint Survival and Renormalized Physical Realization of the Wheeler–DeWitt Family*, WDW-6, 2026.
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- [15] A. J. Maley, *Complete Joint Constraint-Kernel Realization Closure*, WDW-9, 5 August 2026.
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Source-lineage rule

The accepted WDW-9 publication endpoint is the predecessor used throughout. Any changed predecessor requires revalidation of the affected import and all downstream WDW-10 nodes. W9.19 is imported separately only after

W10.24. The Schrödinger manuscript is cited only as the downstream target of W10.26. No conclusion from it is used to prove W10.1–W10.25.

End of theorem. The exact status is $G_{10} = \text{PASS-GLOBAL}$ on the declared transversely germified retained-lineage target, conditional on the accepted WDW-1–9 imports. The stronger clock-failure- crossing label and all richer targets remain outside the theorem. The additional statuses $H_{10}^{\text{QS-GLOBAL}} = \text{PASS}$ and $H_{10}^{\text{SCHR-HANDOFF}} = \text{PASS}$ are strictly downstream and retain the non-export and publication-status boundary of Table 2.